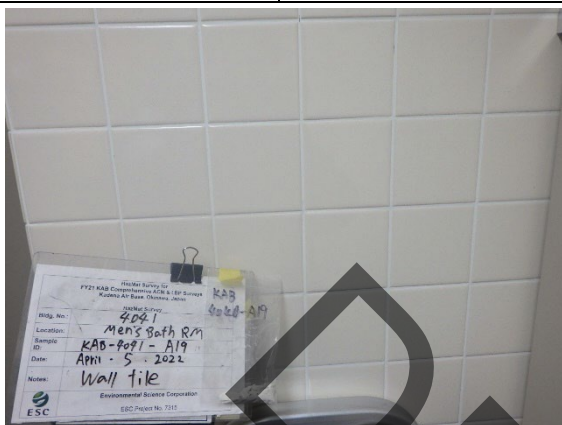


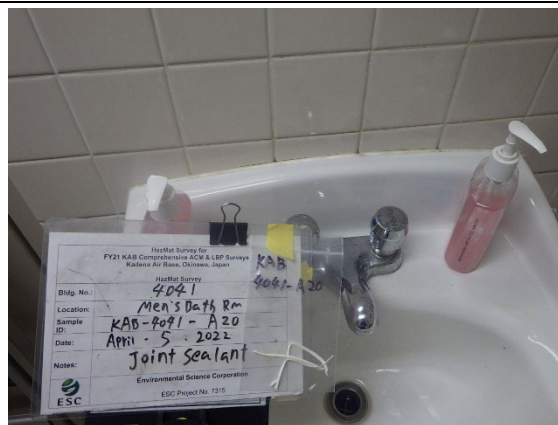


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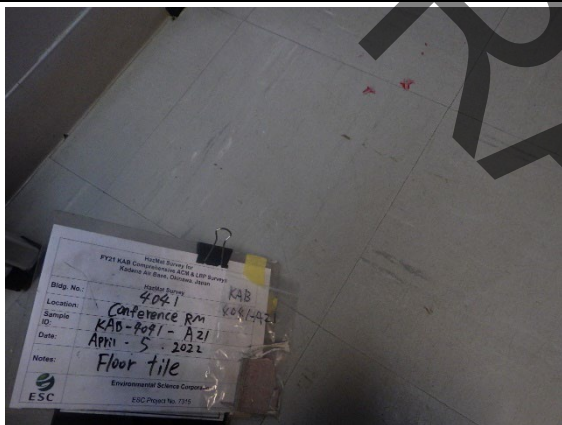
Asbestos Sampling
Kadena Air Base, Bldg.4041



KAB-4041-A19



KAB-4041-A20



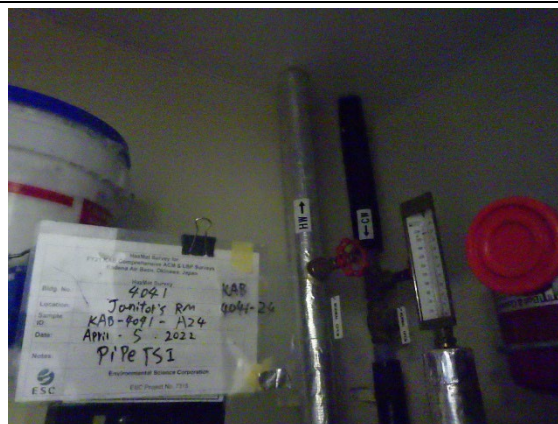
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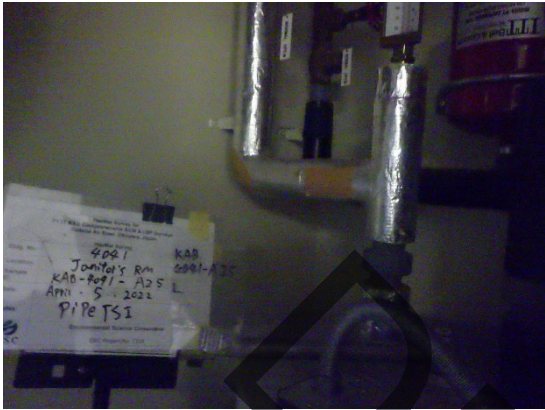
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KAB-4041-A23



KAB-4041-A24

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

HazMat Survey

Bldg. 4041



Photo Record Sheet

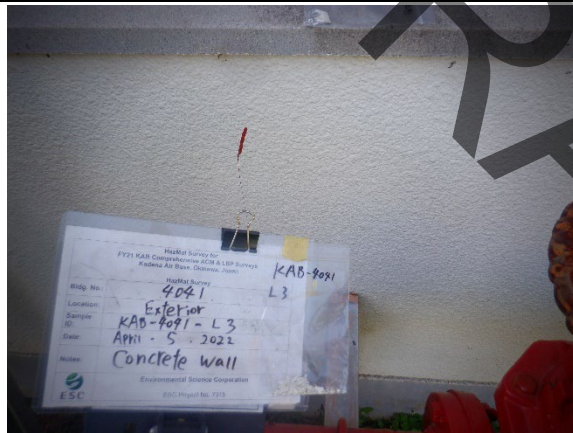
Lead Based Paint Sampling Kadena Air Base, Bldg.4041



KAB-4041-L1



KAB-4041-L2



KAB-4041-L3



KAB-4041-L4



KAB-4041-L5

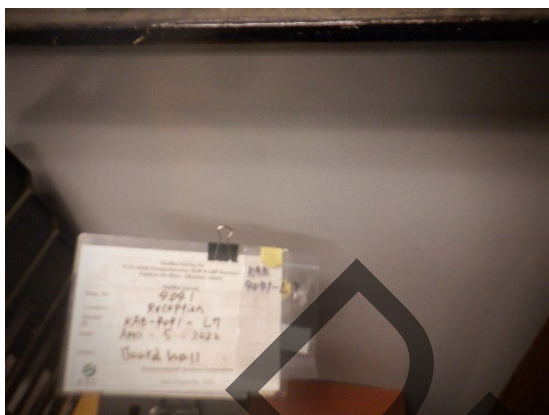


KAB-4041-L6

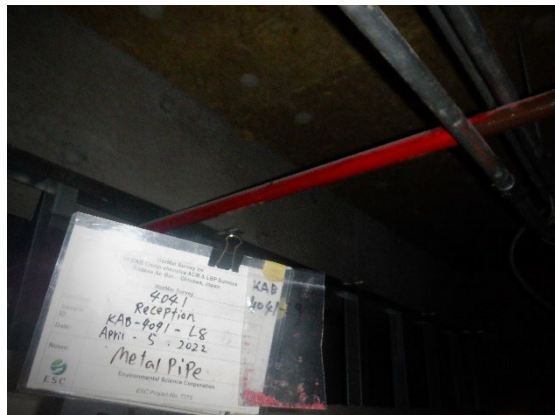


Photo Record Sheet

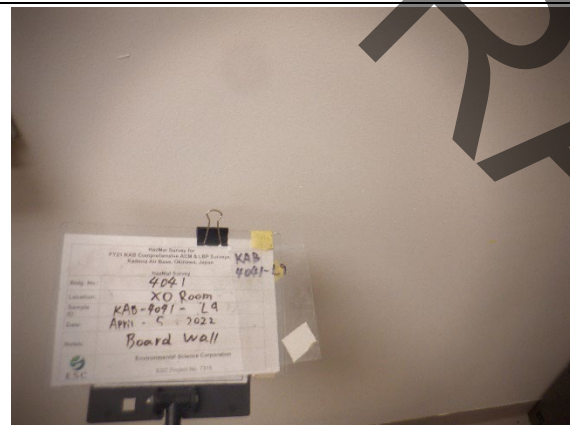
Lead Based Paint Sampling
Kadena Air Base, Bldg.4041



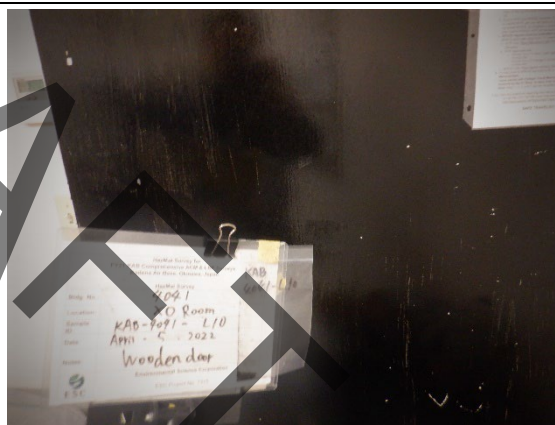
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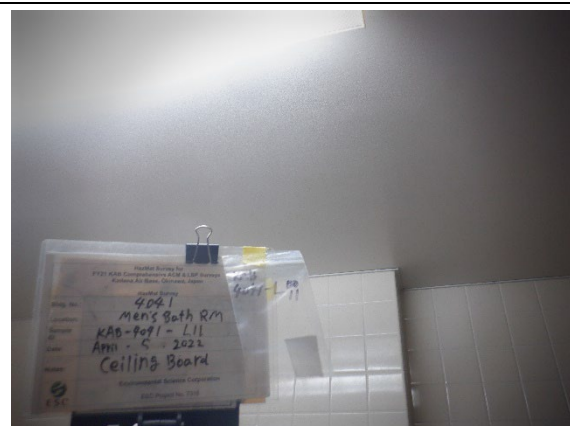
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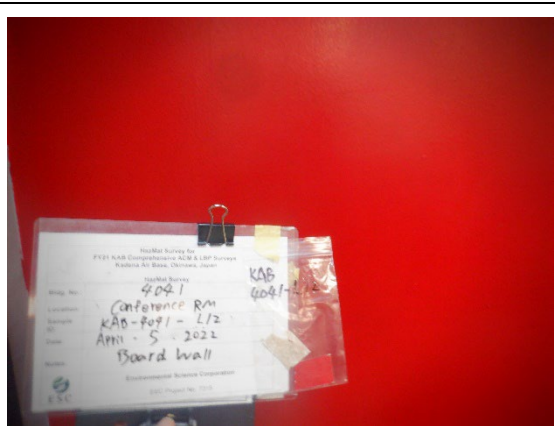
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KAB-4041-L10



KAB-4041-L11

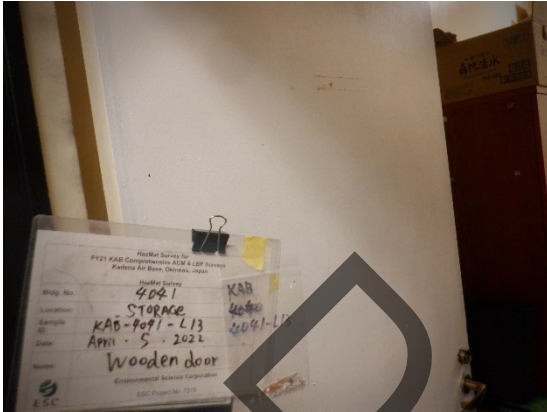


KAB-4041-L12



Photo Record Sheet

Lead Based Paint Sampling
Kadena Air Base, Bldg.4041



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Contract No. W912HV21P0027

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4042

Date: 25 February 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4042, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of twenty-eight (28) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of thirteen (13) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4042	Exterior	H1	Flange gasket	Black	N	Y	Good	N/A		Y	51553770	KAB-4042-A1	2022-10-03	ND	Valid	
4042	Exterior	H2	Concrete wall coating	Off-white, textured	N	Y	Good	250 SM		Y	51553771	KAB-4042-A2	2022-10-03	ND	Valid	
4042	Exterior	H3	Wall joint sealant	Light gray mastic	N	Y	Good	150 LM		Y	51553772	KAB-4042-A3	2022-10-03	ND	Valid	
4042	Exterior	H2	Concrete wall coating	Off-white, textured	N	Y	Good	250 SM		Y	51553773	KAB-4042-A4	2022-10-03	ND	Valid	
4042	Exterior	H3	Wall joint sealant	Light gray mastic	N	Y	Good	150 LM		Y	51553774	KAB-4042-A5	2022-10-03	ND	Valid	
4042	Exterior	H4	Window joint sealant	Dark brown mastic	N	Y	Good	30 LM		Y	51553775	KAB-4042-A6	2022-10-03	ND	Valid	
4042	Exterior	H2	Concrete wall coating	Off-white, textured	N	Y	Good	250 SM		Y	51553776	KAB-4042-A7	2022-10-03	ND	Valid	
4042	Exterior	H3	Wall joint sealant	Light gray mastic	N	Y	Good	150 LM		Y	51553777	KAB-4042-A8	2022-10-03	ND	Valid	
4042	Reception	H5	Vinyl floor tile	30x30 gray tile w/off-white mastic	N	Y	Good	220 SM		Y	51553778	KAB-4042-A9	2022-10-03	ND	Valid	
4042	Reception	H6	Gypsum board wall	Gray, smooth	N	Y	Good	350 SM		Y	51553779	KAB-4042-A10	2022-10-03	ND	Valid	
4042	Reception	H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Good	150 LM		Y	51553780	KAB-4042-A11	2022-10-03	ND	Valid	
4042	Reception	H8	Ceiling tile	60x60 off-white hole pattern	N	N	Good	220 SM		Y	51553781	KAB-4042-A12	2022-10-03	ND	Valid	
4042	Reception	H9	Duct TSI	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553782	KAB-4042-A13	2022-10-03	ND	Valid	
4042	Janitor	H10	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	Y	Y	Good	N/A		Y	51553783	KAB-4042-A14	2022-10-03	ND	Valid	
4042	Janitor	H11	Pipe TSI	Yellow fiberglass w/silver backing	Y	Y	Good	N/A		Y	51553784	KAB-4042-A15	2022-10-03	ND	Valid	
4042	Janitor	H12	Concrete floor coating	Gray	N	Y	Good	2 SM		Y	51553785	KAB-4042-A16	2022-10-03	ND	Valid	
4042	Janitor	H13	Concrete wall coating	Off-white, smooth	N	Y	Good	5 SM		Y	51553786	KAB-4042-A17	2022-10-03	ND	Valid	
4042	Men's BathRM	H14	Ceramic wall tile	10x10 off-white tile w/white grout	N	Y	Good	60 SM		Y	51553787	KAB-4042-A18	2022-10-03	ND	Valid	
4042	Men's BathRM	H15	Ceramic floor tile	20x20 beige tile w/gray tile	N	Y	Good	30 SM		Y	51553788	KAB-4042-A19	2022-10-03	ND	Valid	
4042	Men's BathRM	H16	Sink joint sealant	White mastic	N	N	Good	2 LM		Y	51553789	KAB-4042-A20	2022-10-03	ND	Valid	
4042	Men's BathRM	H17	Gypsum ceiling board	Off-white, smooth	N	N	Good	30 SM		Y	51553790	KAB-4042-A21	2022-10-03	ND	Valid	
4042	Meeting RM	H5	Vinyl floor tile	30x30 gray tile w/off-white mastic	N	Y	Good	220 SM		Y	51553791	KAB-4042-A22	2022-10-03	ND	Valid	
4042	Meeting RM	H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Good	150 LM		Y	51553792	KAB-4042-A23	2022-10-03	ND	Valid	
4042	Meeting RM	H6	Gypsum board wall	Gray, smooth	N	Y	Good	350 SM		Y	51553793	KAB-4042-A24	2022-10-03	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FR IABLE	7 AC CESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
4042	Meeting RM	H18	Wall joint sealant	White mastic	N	Y	Good	N/A		Y	51553794	KAB-4042-A25	2022-10-03	ND	Valid	
4042	FC RM	H5	Vinyl floor tile	30x30 gray tile w/off-white mastic	N	Y	Good	220 SM		Y	51553795	KAB-4042-A26	2022-10-03	ND	Valid	
4042	FC RM	H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Good	150 LM		Y	51553796	KAB-4042-A27	2022-10-03	ND	Valid	
4042	FC RM	H6	Gypsum board wall	Off-white,smooth	N	Y	Good	350 SM		Y	51553797	KAB-4042-A28	2022-10-03	ND	Valid	
LEAD PAINT SMAPLES																
4042	Exterior	S1	Metal pipe	Red		Y	Used			Y	LM220633	KAB-4042-L1	2022-06-22	<0.02	Valid	
4042	Exterior	S2	Metal flange	Light blue		Y	Used			Y	LM220634	KAB-4042-L2	2022-06-22	<0.007	Valid	
4042	Exterior	S3	Concrete wall	Off-white		Y	Used			Y	LM220635	KAB-4042-L3	2022-06-22	<0.007	Valid	
4042	Exterior	S3	Concrete wall	Off-white		Y	Used			Y	LM220636	KAB-4042-L4	2022-06-22	<0.006	Valid	
4042	Exterior	S3	Concrete wall	Off-white		Y	Used			Y	LM220637	KAB-4042-L5	2022-06-22	<0.007	Valid	
4042	Reception	S4	Gypsum board wall	Gray		Y	Used			Y	LM220638	KAB-4042-L6	2022-06-22	<0.006	Valid	
4042	Reception	S1	Metal pipe	Red		N	Used			Y	LM220639	KAB-4042-L7	2022-06-22	<0.02	Valid	
4042	Reception	S5	Wooden door frame	Red over white		Y	Used			Y	LM220640	KAB-4042-L8	2022-06-22	0.21	Valid	
4042	Reception	S6	Wooden door	Off-white over black		Y	Used			Y	LM220641	KAB-4042-L9	2022-06-22	<0.007	Valid	
4042	Janitor	S7	Concrete wall	Off-white,smooth		Y	Used			Y	LM220642	KAB-4042-L10	2022-06-22	<0.007	Valid	
4042	Men's Bathroom	S8	Gypsum ceiling board	Off-white		Y	Used			Y	LM220643	KAB-4042-L11	2022-06-22	<0.007	Valid	
4042	Meeting RM	S4	Gypsum board wall	Gray		Y	Used			Y	LM220644	KAB-4042-L12	2022-06-22	<0.006	Valid	
4042	FC RM	S9	Gypsum board wall	Off-white		Y	Used			Y	LM220645	KAB-4042-L13	2022-06-22	<0.008	Valid	

Notes;
The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS

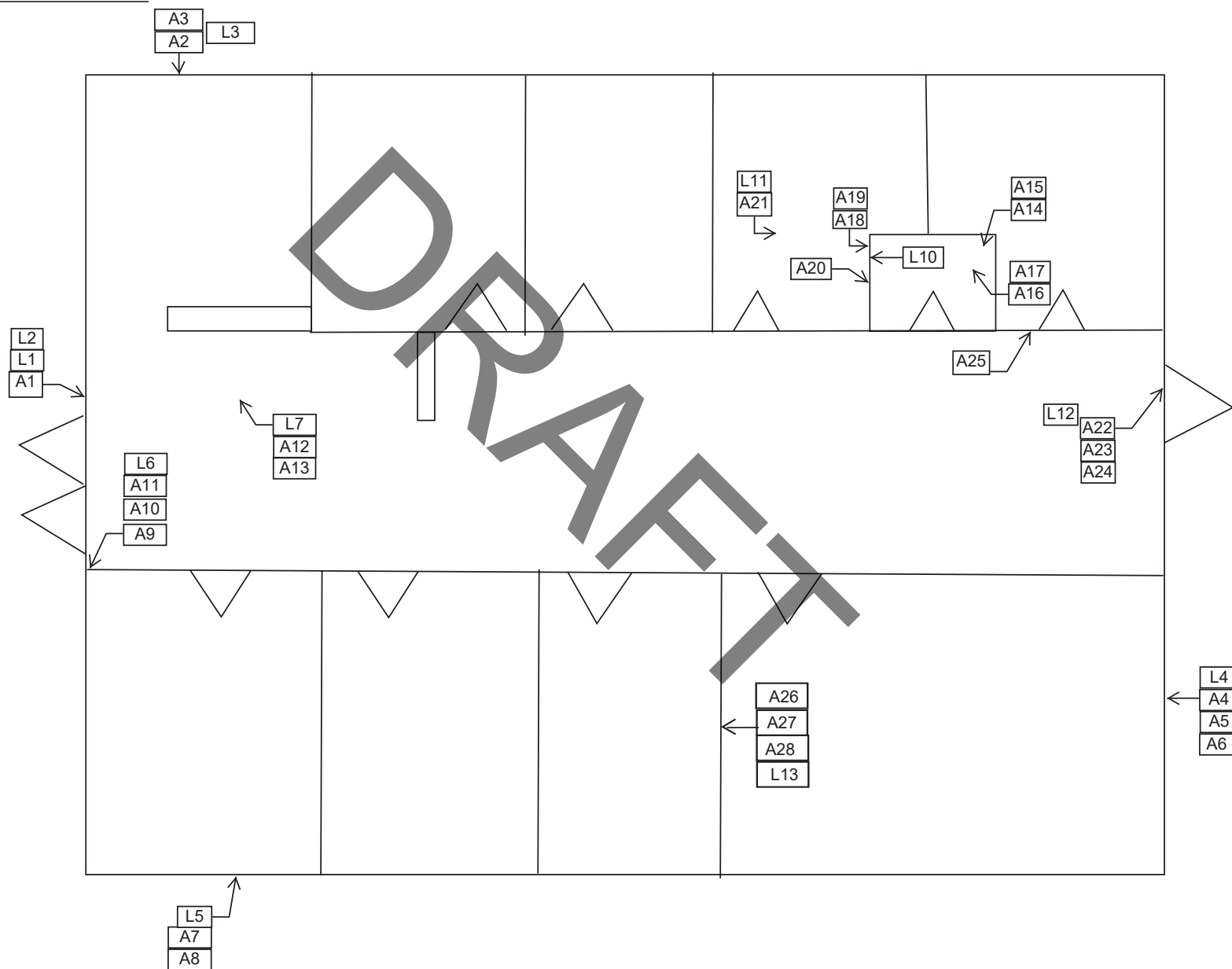
FY21 KAB COMPREHENSIVE ACM & LBP SURVEYS

KADENA AIR BASE, OKINAWA, JAPAN

BLDG NO: 4042

SAMPLING DATE: 20 Apr 2022

ASBESTOS & LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg. 4042		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project		
Comments: See attached sample logs.		☐ Metals Analysis Matrix: Method: Analytes: ☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4042-A1-A28		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				

Sampled By: Sean R Dudney Date/Time: 4/20/22 Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS

Relinquished By: Sean R Dudney	Relinquished By:	Relinquished By:
Date / Time: 26 April 2022	Date / Time:	Date / Time:
Received By: Japan Post Office	Received By: <i>Ali Wong</i>	Received By:
Date / Time: 26 April 2022	Date / Time: 5-26-22 14:00 EST	Date / Time:
Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☐ Yes ☐ No

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE					ANALYSIS RECORD DATA										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMB	ROOM OR FUNCTION	HOMOGENEOU S/SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	TEST PROC	CONDITIO N	G	C	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING	
4042	Exterior	H1	Flange gasket	Black	N	Y	Used			KAB- 4042-A1					
		H2	Concrete wall	Off-white, textured	Y	Y	Used			KAB- 4042-A2					
		H3	Wall joint sealant	Light gray mastic	Y	Y	Used			KAB- 4042-A3					
		H2	Concrete wall	Off-white, textured	Y	Y	Used			KAB- 4042-A4					
		H3	Wall joint sealant	Light gray mastic	Y	Y	Used			KAB- 4042-A5					
		H4	Window joint sealant	Dark brown mastic	N	Y	Used			KAB- 4042-A6					
		H2	Concrete wall	Off-white, textured	Y	Y	Used			KAB- 4042-A7					
		H3	Wall joint sealant	Light gray mastic	Y	Y	Used			KAB- 4042-A8					
	Reception	H5	Vinyl floor tile	30x30 gray tile w/off- white mastic	Y	Y	Used			KAB- 4042-A9					
		H6	Gypsum board wall	Gray, smooth	N	Y	Use			KAB- 4042-A10					
		H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Use			KAB- 4042-A11					
		H8	Ceiling tile	60x60 off-white hole pattern	N	N	Use			KAB- 4042-A12					
		H9	Duct TSI	Yellow fiberglass w/silver backing	N	N	Use			KAB- 4042-A13					
	Janitor	H10	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	N	Y	Use			KAB- 4042-A14					
		H11	Pipe TSI	Yellow fiberglass w/silver backing	N	Y	Use			KAB- 4042-A15					
		H12	Concrete floor coating	Gray	Y	Y	Used			KAB- 4042-A16					
	Janitor	H13	Concrete wall	Off-white, smooth	Y	Y	Used			KAB- 4042-A17					
	Men's BathRM	H14	Ceramic wall tile	10x10 off-white tile w/white grout	Y	Y	Use			KAB- 4042-A18					
		H15	Ceramic floor tile	20x20 Beige tile w/gray tile	Y	Y	Use			KAB- 4042-A19					
		H16	Sink joint sealant	White mastic	N	N	Use			KAB- 4042-A20					
	Men's BathRM	H17	Gypsum ceiling board	Off-white, smooth	N	N	Use			KAB- 4042-A21					
	Meeting RM	H5	Vinyl floor tile	30x30 gray tile w/off- white mastic	Y	Y	Used			KAB- 4042-A22					
		H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Use			KAB- 4042-A23					
		H6	Gypsum board wall	Gray, smooth	N	Y	Use			KAB- 4042-A24					
		H18	Wall joint sealant	White mastic	N	Y	Used			KAB- 4042-A25					
	PC RM	H5	Vinyl floor tile	30x30 gray tile w/off- white mastic	Y	Y	Used			KAB- 4042-A26					
		H7	Vinyl cove base	10cm gray w/tan and white mastic	N	Y	Use			KAB- 4042-A27					
		H6	Gypsum board wall	Off-white, smooth	N	Y	Use			KAB- 4042-A28					

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)

NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudney
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333884
Date Received: 05/26/22
Date Analyzed: 10/03/22
Date Printed: 10/04/22
First Reported: 10/04/22

Job ID/Site: ESC.7315; Kadena Air Base, Okinawa, Japan; Bldg. 4042

SGSFL Job ID: L1232-1
Total Samples Submitted: 28
Total Samples Analyzed: 28

Date(s) Collected: 04/20/2022

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4042-A1	51553770						
Layer: Black Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A2	51553771						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A3	51553772						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A4	51553773						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A5	51553774						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A6	51553775						
Layer: Brown Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333884

Date Printed: 10/04/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4042-A7	51553776						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A8	51553777						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A9	51553778						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A10	51553779						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)							
KAB-4042-A11	51553780						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Drywall			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A12	51553781						
Layer: Light Grey Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (35 %) Fibrous Glass (45 %)							
KAB-4042-A13	51553782						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (80 %)							
KAB-4042-A14	51553783						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (80 %)							

Client Name: Environmental Science Corporation

Report Number: B333884

Date Printed: 10/04/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4042-A15	51553784						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)		Fibrous Glass (80 %)					
KAB-4042-A16	51553785						
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A17	51553786						
Layer: Grey Cementitious Material			ND				
Layer: Off-White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A18	51553787						
Layer: Off-White Ceramic Tile			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A19	51553788						
Layer: Beige Ceramic Tile			ND				
Layer: Dark Grey Grout			ND				
Layer: Light Grey Mortar			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A20	51553789						
Layer: White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A21	51553790						
Layer: Off-White Drywall			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)		Fibrous Glass (Trace)					
KAB-4042-A22	51553791						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333884

Date Printed: 10/04/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4042-A23	51553792						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: Tan Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A24	51553793						
Layer: Off-White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)							
KAB-4042-A25	51553794						
Layer: White/Yellow Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A26	51553795						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A27	51553796						
Layer: Grey Non-Fibrous Material			ND				
Layer: Off-White Mastic			ND				
Layer: Paint			ND				
Layer: Off-White Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4042-A28	51553797						
Layer: White Drywall			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (2 %)							



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

Client Name: Environmental Science Corporation

Report Number: B333884

Date Printed: 10/04/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
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DRAFT

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4042		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4042-L1-L13		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/20/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 26 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>Ali Whang</i>		Received By:			
Date / Time: 26 April 2022		Date / Time: 5-26-22 14:00		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1 BLDG NUMBE	2 ROOM OR FUNCTIONA	3 HOMOGENEOU S /SURFACE	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIA	7 ACC	8 CONDITIO N	9 G	10 C	11 LAB/ SAMPLE ID	12 CTR COC	13 LAB ANALYSIS	14 RESULT	15 VALIDITY DETERMINATI	16 NOTES / INVALID OR UNKNOWN REASONING
4042	Exterior	S1	Metal pipe	Red		Y	Used				KAB- 4042-L1				
		S2	Metal flange	Light blue		Y	Used				KAB- 4042-L2				
		S3	Concrete wall	Off-white		Y	Used				KAB- 4042-L3				
		S3	Concrete wall	Off-white		Y	Used				KAB- 4042-L4				
		S3	Concrete wall	Off-white		Y	Used				KAB- 4042-L5				
	Reception	S4	Gypsum board wall	Gray		Y	Use				KAB- 4042-L6				
		S1	Metal pipe	Red		N	Use				KAB- 4042-L7				
		S5	Wooden door frame	Red over white		Y	Use				KAB- 4042-L8				
		S6	Wooden door	Off-white over black		Y	Use				KAB- 4042-L9				
	Janitor	S7	Concrete wall	Off-white, smooth		Y	Use				KAB- 4042- L10				
	Men's BathRM	S8	Gypsum ceiling board	Off-white		Y	Use				KAB- 4042- L11				
		S4	Gypsum board wall	Gray		Y	Use				KAB- 4042- L12				
	PC RM	S9	Gypsum board wall	Off-white		Y	Use				KAB- 4042- L13				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242209
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4042

SGSFL Job ID: L1232-1

Date(s) Collected: 04/20/22

Total Samples Submitted: 13

Total Samples Analyzed: 13

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4042-L1	LM220633	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4042-L2	LM220634	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L3	LM220635	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L4	LM220636	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4042-L5	LM220637	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L6	LM220638	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4042-L7	LM220639	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4042-L8	LM220640	Pb	0.21	wt%	0.008	EPA 3050B/7000B
KAB-4042-L9	LM220641	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L10	LM220642	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L11	LM220643	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4042-L12	LM220644	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4042-L13	LM220645	Pb	< 0.008	wt%	0.008	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

Analytical results and reports are generated by SGS Forensic Laboratories at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS Forensic Laboratories to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS Forensic Laboratories. The client is solely responsible for the use and interpretation of test results and reports requested from SGS Forensic Laboratories. SGS Forensic Laboratories is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Forensic Laboratories' Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

HazMat Survey

Bldg. 4042

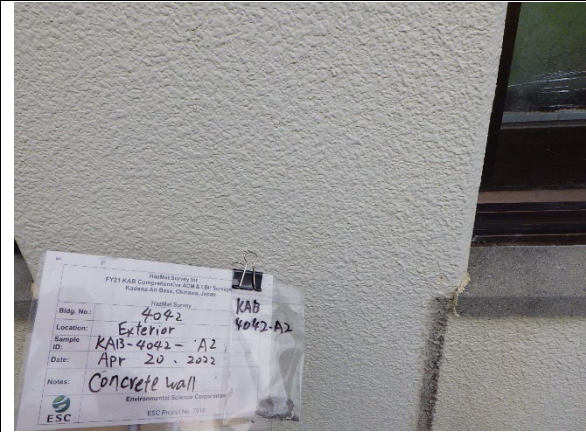


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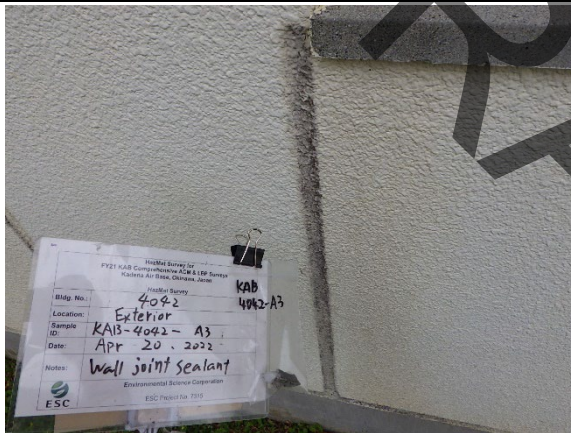
Asbestos Sampling
Kadena Air Base, Bldg.4042



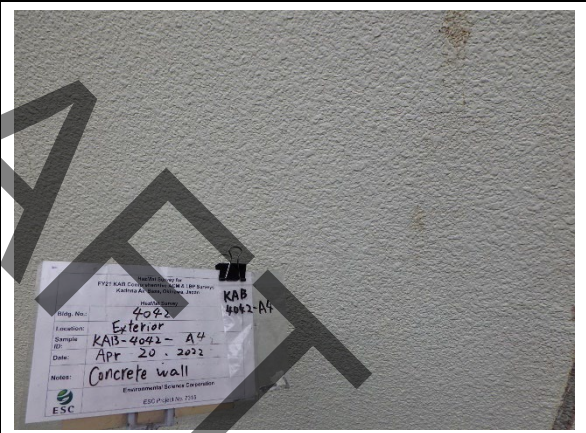
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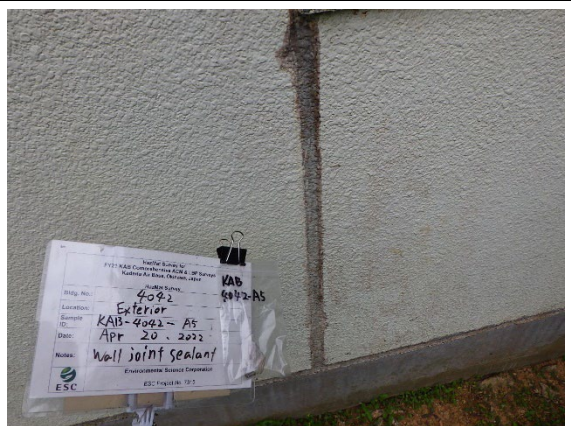
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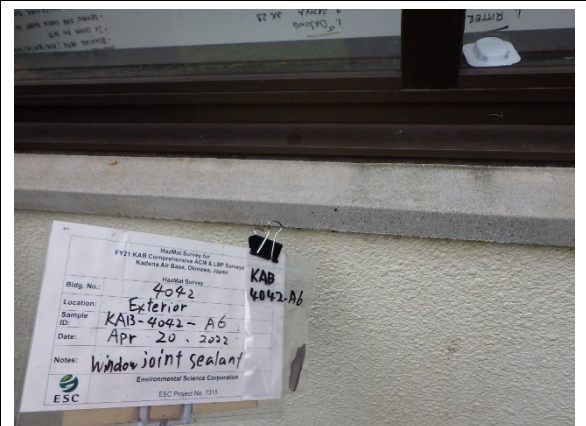
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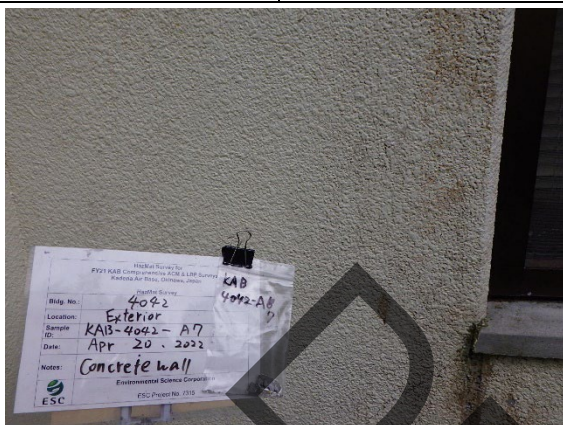


KAB-4042-A6



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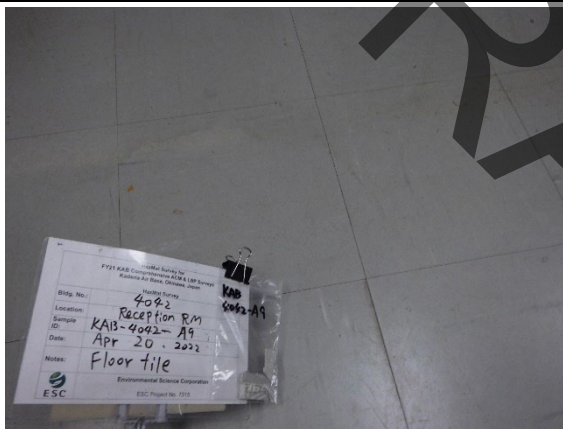
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Kadena Air Base, Bldg.4042



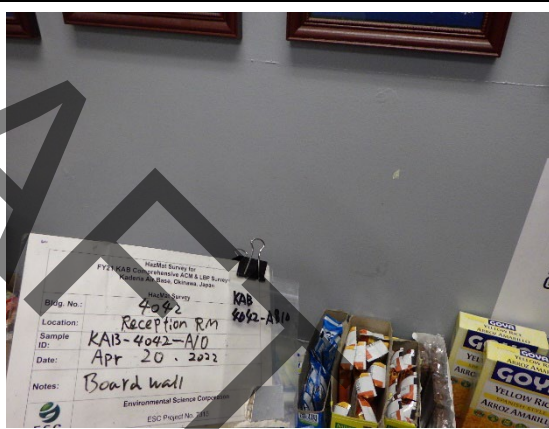
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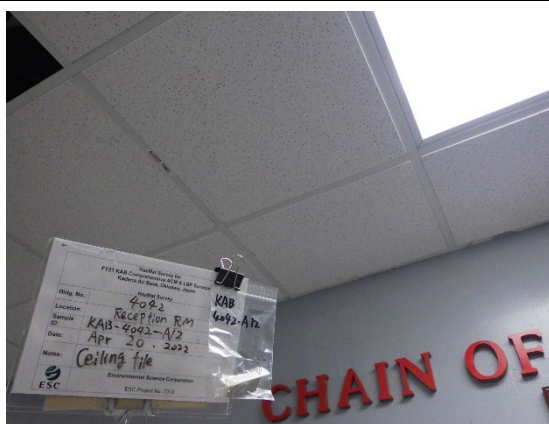
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KAB-4042-A11



KAB-4042-A12

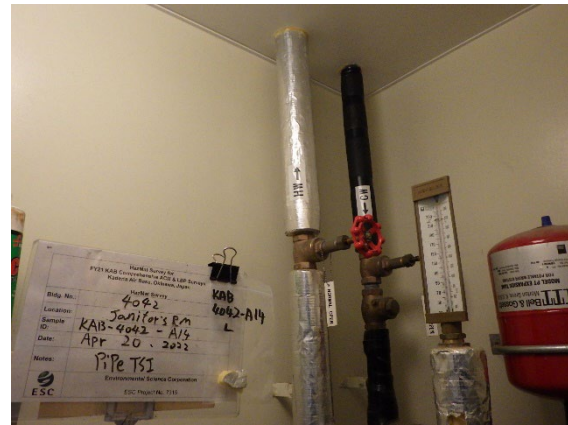


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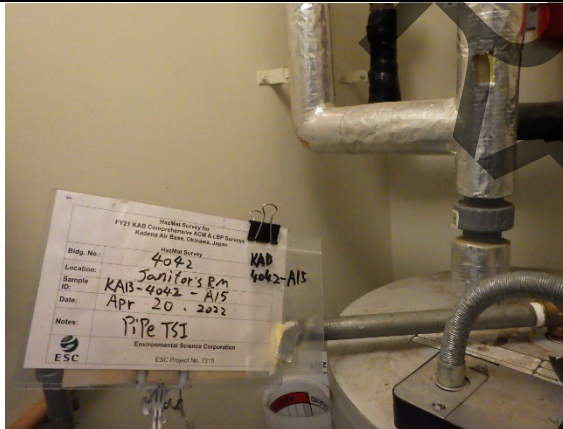
Asbestos Sampling
Kadena Air Base, Bldg.4042



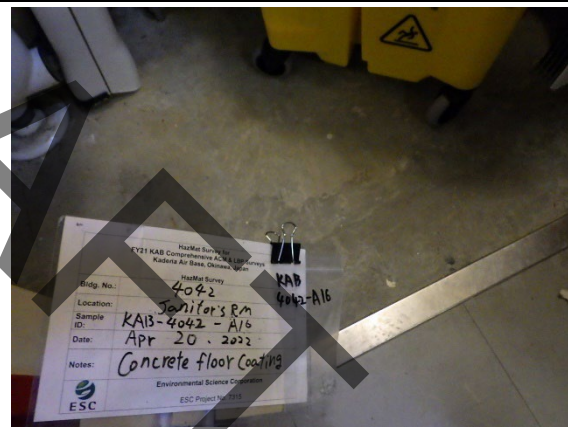
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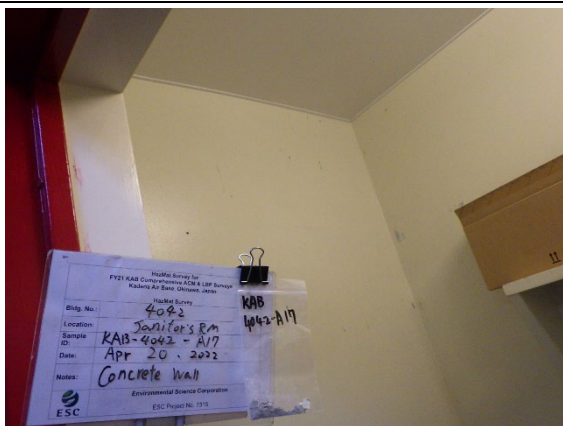
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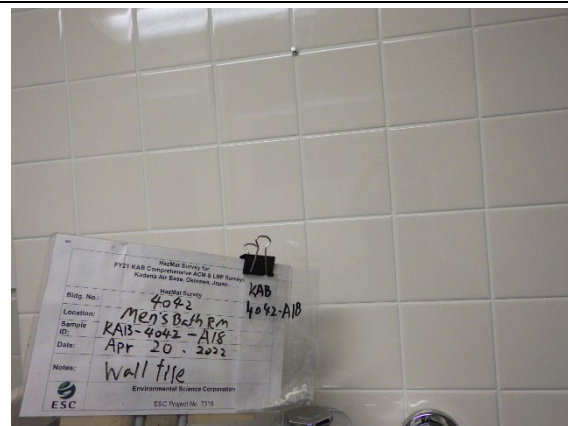
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KAB-4042-A16



KAB-4042-A17

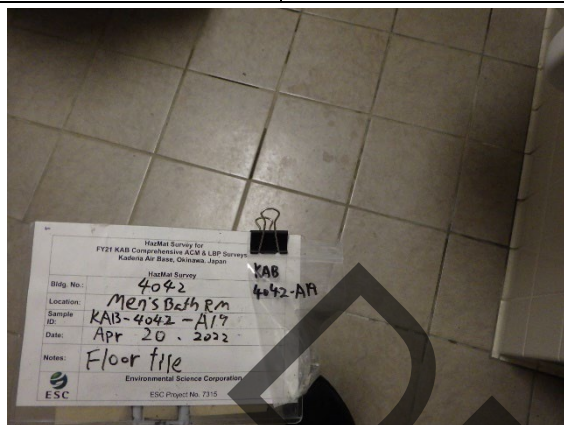


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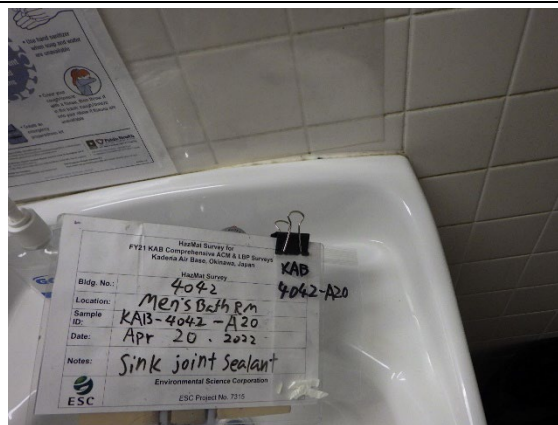


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4042



KAB-4042-A19



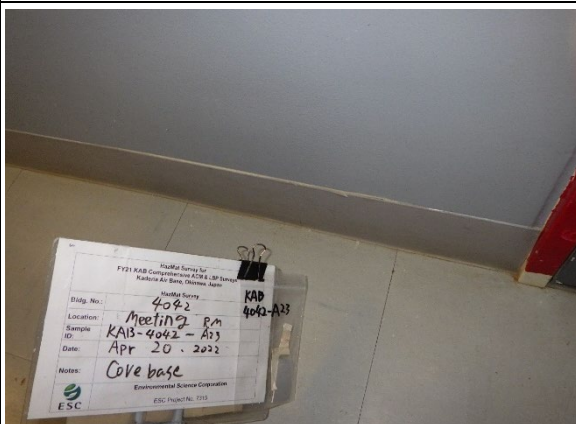
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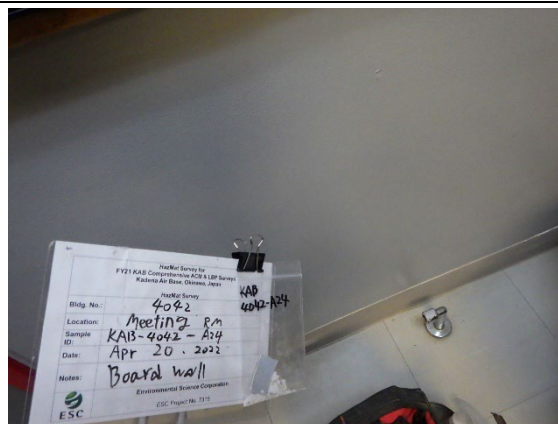
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KAB-4042-A22



KAB-4042-A23

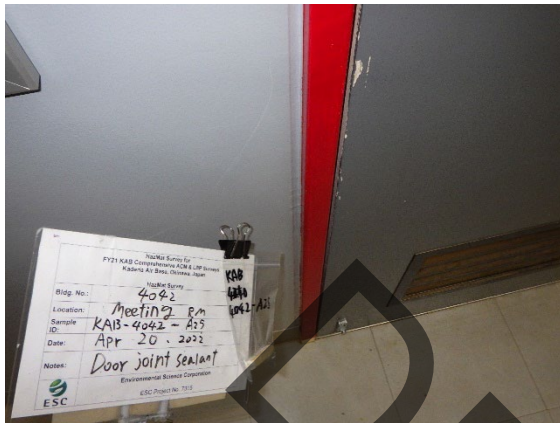


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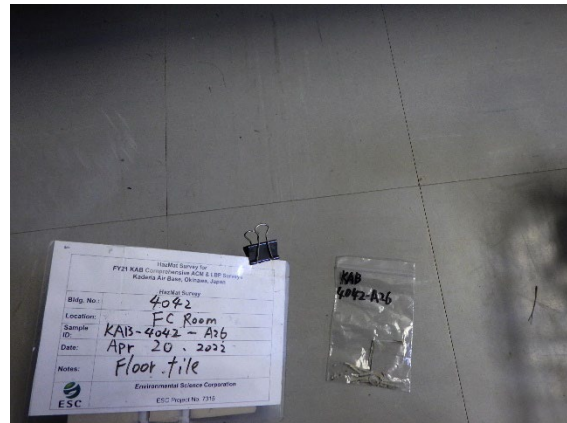


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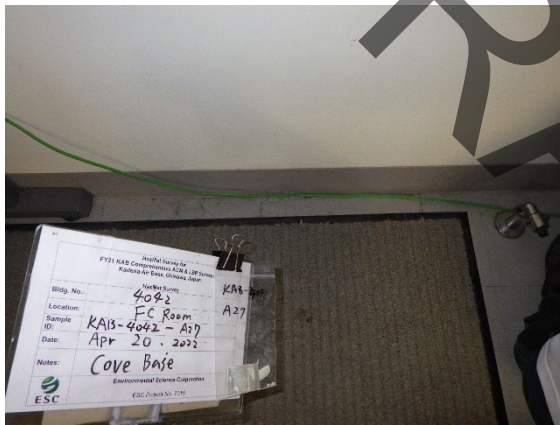
Asbestos Sampling
Kadena Air Base, Bldg.4042



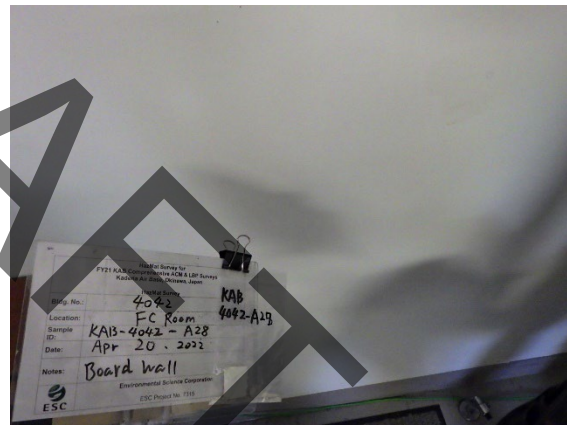
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KAB-4042-A26



KAB-4042-A27



KAB-4042-A28

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

HazMat Survey

Bldg. 4042



Photo Record Sheet

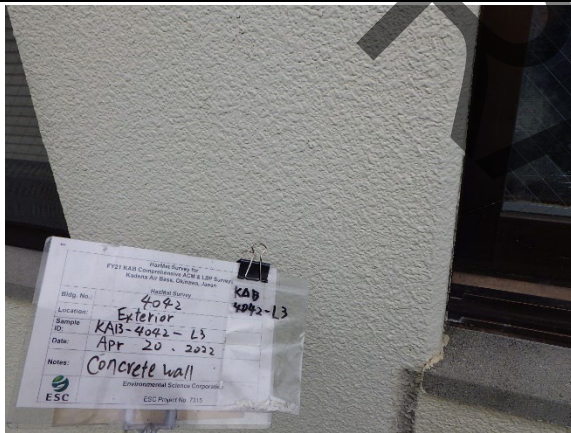
Lead Based Paint Sampling Kadena Air Base, Bldg.4042



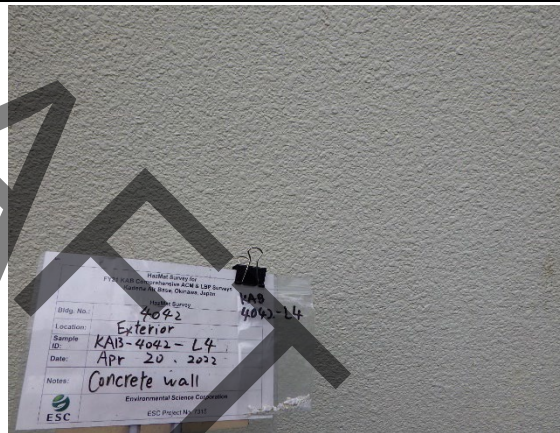
KAB-4042-L1



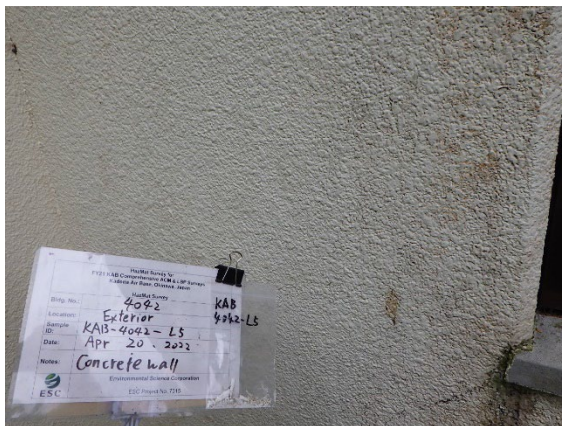
KAB-4042-L2



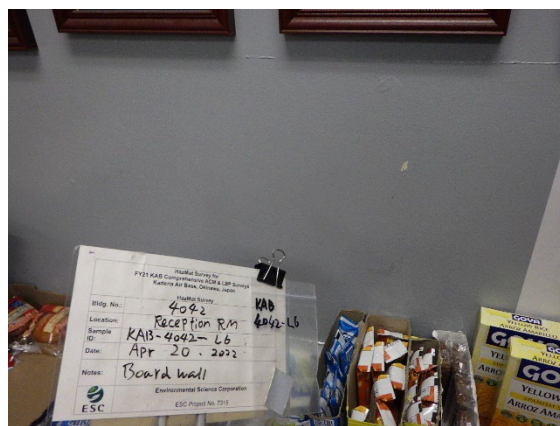
KAB-4042-L3



KAB-4042-L4



KAB-4042-L5

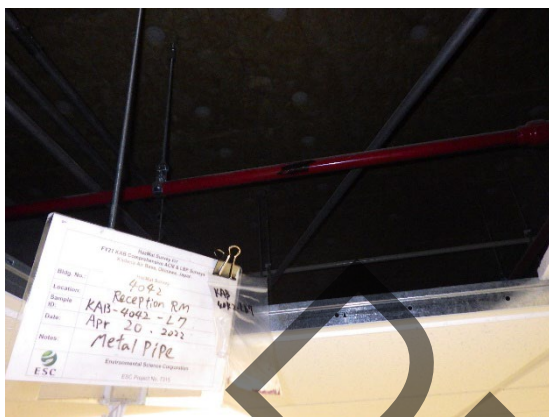


KAB-4042-L6

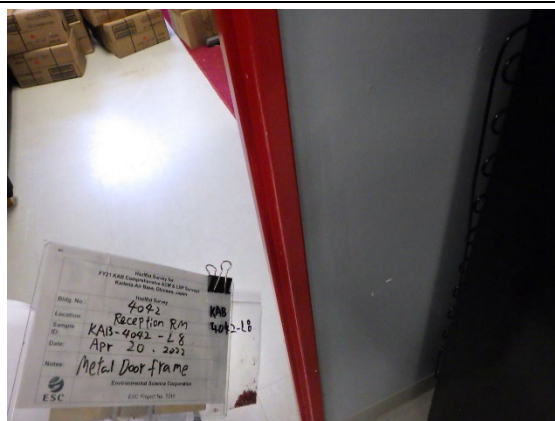


Photo Record Sheet

Lead Based Paint Sampling
Kadena Air Base, Bldg.4042



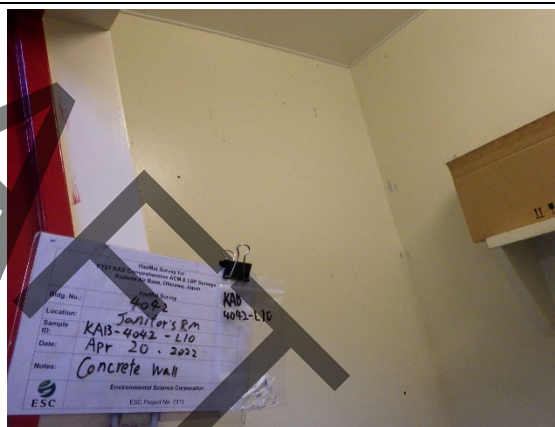
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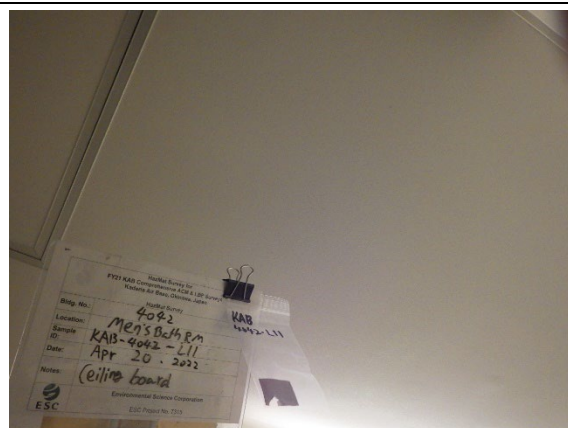
KAB-4042-L8



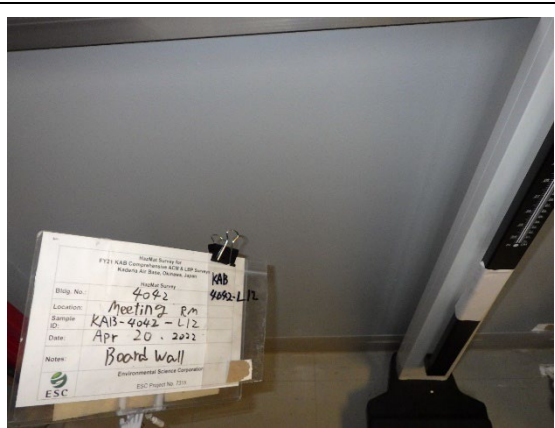
KAB-4042-L9



KAB-4042-L10



KAB-4042-L11

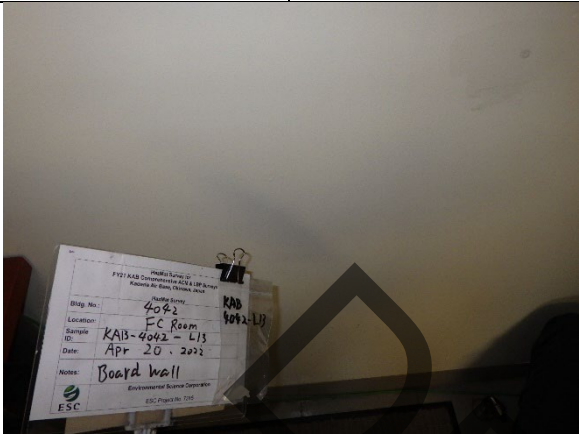


KAB-4042-L12



Photo Record Sheet

Lead Based Paint Sampling
Kadena Air Base, Bldg.4042



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KAB-4042-L13

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NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4043

Date: 25 February 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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APPENDICES

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B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4043, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of twenty-five (25) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of thirteen (13) total paint chip samples collected for this survey, one (1) paint sample was determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4043	Reception	H1	Vinyl floor tile	30x30cm gray tile w/tan and light gray mastic	N	Y	Good	220 SM		Y	51553720	KAB-4043-A1	2022-10-06	ND	Valid	
4043	Reception	H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553721	KAB-4043-A2	2022-10-06	ND	Valid	
4043	Reception	H3	Gypsum board wall	Gray	N	Y	Good	350 SM		Y	51553722	KAB-4043-A3	2022-10-06	ND	Valid	
4043	Reception	H4	Ceiling tile	60x60cm off white,all patterns	N	N	Good	220 SM		Y	51553723	KAB-4043-A4	2022-10-06	ND	Valid	
4043	Reception	H5	Duct TSI	Yellow fiberglass w/silver backing	N	N	Good	N/A		Y	51553724	KAB-4043-A5	2022-10-06	ND	Valid	
4043	Conference Rm	H1	Vinyl floor tile	30x30cm gray tile w/tan and light gray mastic	N	Y	Good	220 SM		Y	51553725	KAB-4043-A6	2022-10-06	ND	Valid	
4043	Conference Rm	H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553726	KAB-4043-A7	2022-10-06	ND	Valid	
4043	Conference Rm	H3	Gypsum board wall	Gray	N	Y	Good	350 SM		Y	51553727	KAB-4043-A8	2022-10-06	ND	Valid	
4043	Mens bathroom	H6	Sink joint sealant	Off white	N	Y	Good	2 LM		Y	51553728	KAB-4043-A9	2022-10-06	ND	Valid	
4043	Mens bathroom	H7	Ceiling board	Off white	N	Y	Good	30 SM		Y	51553729	KAB-4043-A10	2022-10-06	ND	Valid	
4043	Mens bathroom	H8	Wall tile	10x10cm off white w/white grout	N	Y	Good	60 SM		Y	51553730	KAB-4043-A11	2022-10-06	ND	Valid	
4043	Mens bathroom	H9	Ceramic floor tile	20x20cm beige w/gray grout	N	Y	Good	30 SM		Y	51553731	KAB-4043-A12	2022-10-06	ND	Valid	
4043	Janitor	H10	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	Y	Y	Good	N/A		Y	51553732	KAB-4043-A13	2022-10-06	ND	Valid	
4043	Janitor	H11	Pipe TSI (straight)	Yellow fiberglass w/silver backing	Y	Y	Good	N/A		Y	51553733	KAB-4043-A14	2022-10-06	ND	Valid	
4043	Office	H1	Vinyl floor tile	30x30cm gray tile w/tan and light gray mastic	N	Y	Good	220 SM		Y	51553734	KAB-4043-A15	2022-10-06	ND	Valid	
4043	Office	H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Good	150 LM		Y	51553735	KAB-4043-A16	2022-10-06	ND	Valid	
4043	Office	H3	Gypsum board wall	Gray	N	Y	Good	350 SM		Y	51553736	KAB-4043-A17	2022-10-06	ND	Valid	
4043	Exterior	H12	Flange gasket	Black	N	Y	Good	N/A		Y	51553737	KAB-4043-A18	2022-10-06	ND	Valid	
4043	Exterior	H13	Concrete wall coating	Off white, textured	N	Y	Good	250 SM		Y	51553738	KAB-4043-A19	2022-10-06	ND	Valid	
4043	Exterior	H14	Wall joint sealant	Off white	N	Y	Good	150 LM		Y	51553739	KAB-4043-A20	2022-10-06	ND	Valid	
4043	Exterior	H15	Window joint sealant	Brown	N	Y	Good	30 LM		Y	51553740	KAB-4043-A21	2022-10-06	ND	Valid	
4043	Exterior	H13	Concrete wall coating	Off white, textured	N	Y	Good	250 SM		Y	51553741	KAB-4043-A22	2022-10-06	ND	Valid	
4043	Exterior	H14	Wall joint sealant	Off white	N	Y	Good	150 LM		Y	51553742	KAB-4043-A23	2022-10-06	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIABLE	7 ACCESS	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
4043	Exterior	H13	Concrete wall coating	Off white, textured	N	Y	Good	250 SM		Y	51553743	KAB-4043-A24	2022-10-06	ND	Valid	
4043	Exterior	H14	Wall joint sealant	Off white	N	Y	Good	150 LM		Y	51553744	KAB-4043-A25	2022-10-06	ND	Valid	
LEAD PAINT SAMPLES																
4043	Reception	S1	Gypsum board wall	Gray		Y	Used			Y	LM220620	KAB-4043-L1	2022-06-22	<0.006	Valid	
4043	Reception	S2	Metal pipe (above ceiling)	Red		Y	Used			Y	LM220621	KAB-4043-L2	2022-06-22	0.009	Valid	
4043	Reception	S1	Gypsum board wall	Gray		Y	Used			Y	LM220622	KAB-4043-L3	2022-06-22	<0.006	Valid	
4043	Mens bathroom	S3	Ceiling board	Off white		N	Used			Y	LM220623	KAB-4043-L4	2022-06-22	<0.006	Valid	
4043	Janitor	S4	Metal door frame	Off white over red and black		Y	Used			Y	LM220624	KAB-4043-L5	2022-06-22	1.3	Valid	Positive
4043	Office	S5	Wooden door	Black over white		Y	Used			Y	LM220625	KAB-4043-L6	2022-06-22	<0.007	Valid	
4043	Office	S1	Gypsum board wall	Gray		Y	Used			Y	LM220626	KAB-4043-L7	2022-06-22	<0.007	Valid	
4043	Exterior	S6	Metal pipe	Red over white and black		Y	Used			Y	LM220627	KAB-4043-L8	2022-06-22	<0.01	Valid	
4043	Exterior	S7	Metal flange	Light blue		Y	Used			Y	LM220628	KAB-4043-L9	2022-06-22	<0.01	Valid	
4043	Exterior	S8	Concrete wall	Off white, textured		Y	Used			Y	LM220629	KAB-4043-L10	2022-06-22	<0.0006	Valid	
4043	Exterior	S8	Concrete wall	Off white, textured		Y	Used			Y	LM220630	KAB-4043-L11	2022-06-22	<0.0006	Valid	
4043	Exterior	S9	Concrete wall base	Light brown		Y	Used			Y	LM220631	KAB-4043-L12	2022-06-22	0.008	Valid	
4043	Exterior	S8	Concrete wall	Off white, textured		Y	Used			Y	LM220632	KAB-4043-L13	2022-06-22	<0.007	Valid	

Notes;
The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

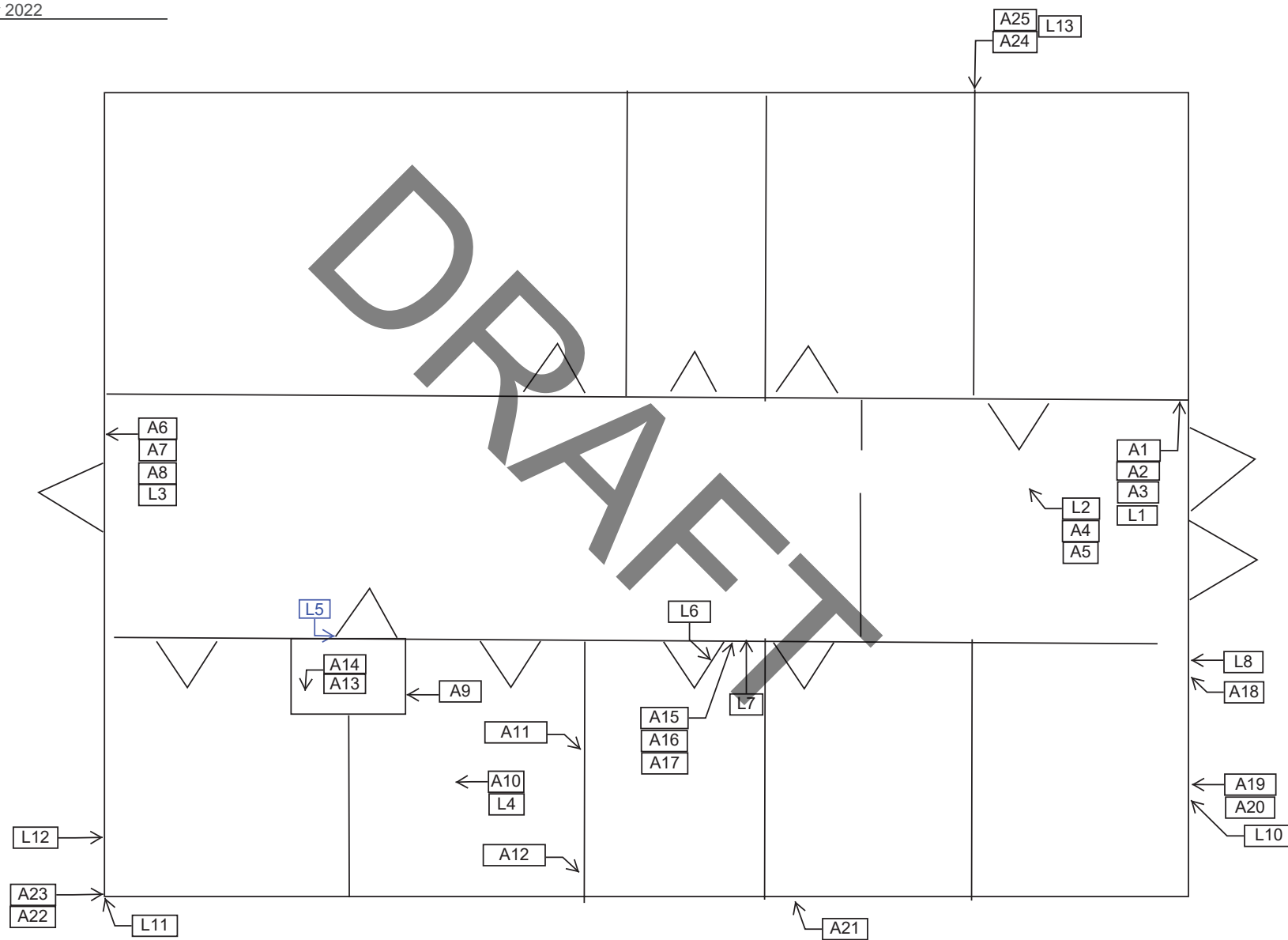
US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS

ASBESTOS & LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: ☐ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg. 4043		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project		
Comments: See attached sample logs.		☐ Metals Analysis Matrix: Method: Analytes:		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4043-A1-A25		See attached sample logs (Excel Sheet)	☐ A ☐ P ☐ C				
2.			☐ A ☐ P ☐ C				
3.			☐ A ☐ P ☐ C				
4.			☐ A ☐ P ☐ C				
5.			☐ A ☐ P ☐ C				
6.			☐ A ☐ P ☐ C				
7.			☐ A ☐ P ☐ C				
8.			☐ A ☐ P ☐ C				
9.			☐ A ☐ P ☐ C				
10.			☐ A ☐ P ☐ C				

Sampled By: Sean R Dudney		Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS	
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:
Date / Time: 25 April 2022		Date / Time:		Date / Time:
Received By: Japan Post Office		Received By: <i>Ali M...</i>		Received By:
Date / Time: 25 April 2022		Date / Time: 5-26-22 14:00		Date / Time:
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SUMMARY REPORTING SHEET (V2.0)

HOMOGENEOUS MATERIAL OR SURFACE					ANALYSIS RECORD DATA										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONAL	HOMOGENEOU S/SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	TEST	ACC	CONDITIO N	G C O T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING	
4043	Reception	H1	Floor tile	30x30 gray tile w/tan and light gray mastic	N	Y	Use			KAB- 4043-A1					
		H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB- 4043-A2					
		H3	Gypsum board wall	Gray	N	Y	Use			KAB- 4043-A3					
		H4	Ceiling tile	60x60 off white, all patterns	N	N	Use			KAB- 4043-A4					
		H5	Duct TSI	Yellow fiberglass w/silver backing	N	N	Use			KAB- 4043-A5					
	Conference Rm	H1	Floor tile	30x30 gray tile w/tan and light gray mastic	N	Y	Use			KAB- 4043-A6					
		H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB- 4043-A7					
		H3	Gypsum board wall	Gray	N	Y	Use			KAB- 4043-A8					
	Mens bathroom	H6	Sink joint sealant	Off white	N	Y	Used			KAB- 4043-A9					
		H7	Ceiling board	Off white	N	Y	Use			KAB- 4043- A10					
		H8	Wall tile	10x10 off white w/white grout	N	Y	Use			KAB- 4043- A11					
		H9	Ceramic floor tile	20x20 beige w/gray grout	N	Y	Use			KAB- 4043- A12					
	Janitor	H10	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	N	Y	Use			KAB- 4043- A13					
		H11	Pipe TSI (straight)	Yellow fiberglass w/silver backing	N	Y	Use			KAB- 4043- A14					
	Office	H1	Floor tile	30x30 gray tile w/tan and light gray mastic	N	Y	Use			KAB- 4043- A15					
		H2	Vinyl cove base	10cm gray w/white mastic	N	Y	Use			KAB- 4043- A16					
		H3	Gypsum board wall	Gray	N	Y	Use			KAB- 4043- A17					
		H12	Flange gasket	Black	N	Y	Used			KAB- 4043- A18					
		H13	Concrete wall	Off white, textured	N	Y	Used			KAB- 4043- A19					
		H14	Wall joint sealant	Off white	N	Y	Used			KAB- 4043- A20					
		H15	Window joint sealant	Brown	N	Y	Used			KAB- 4043- A21					
		H13	Concrete wall	Off white, textured	N	Y	Used			KAB- 4043- A22					
		H14	Wall joint sealant	Off white	N	Y	Used			KAB- 4043- A23					
		H13	Concrete wall	Off white, textured	N	Y	Used			KAB- 4043- A24					
		H14	Wall joint sealant	Off white	N	Y	Used			KAB- 4043- A25					

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333882
Date Received: 05/26/22
Date Analyzed: 10/06/22
Date Printed: 10/06/22
First Reported: 10/06/22

Job ID/Site: ESC.7315; Kadena Air Base, Okinawa, Japan; Bldg. 4043

SGSFL Job ID: L1232-1
Total Samples Submitted: 25
Total Samples Analyzed: 25

Date(s) Collected: 04/05/2022

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4043-A1	51553720						
Layer: Grey Tile			ND				
Layer: Light Grey Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4043-A2	51553721						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4043-A3	51553722						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4043-A4	51553723						
Layer: Off-White Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (35 %) Fibrous Glass (45 %)							
KAB-4043-A5	51553724						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (85 %)							
KAB-4043-A6	51553725						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333882

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4043-A7	51553726						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Layer: White Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (2 %)							
KAB-4043-A8	51553727						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %)							
KAB-4043-A9	51553728						
Layer: White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A10	51553729						
Layer: White Drywall			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %)							
KAB-4043-A11	51553730						
Layer: Off-White Ceramic Tile			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A12	51553731						
Layer: Beige Ceramic Tile			ND				
Layer: Dark Grey Grout			ND				
Layer: Off-White Plaster			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A13	51553732						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace) Fibrous Glass (85 %)							
KAB-4043-A14	51553733						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace) Fibrous Glass (85 %)							

Client Name: Environmental Science Corporation

Report Number: B333882

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4043-A15	51553734						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A16	51553735						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A17	51553736						
Layer: White Drywall			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4043-A18	51553737						
Layer: Black Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A19	51553738						
Layer: Grey Cementitious Material			ND				
Layer: White Skimcoat/Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A20	51553739						
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A21	51553740						
Layer: Brown Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A22	51553741						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333882

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4043-A23	51553742						
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A24	51553743						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4043-A25	51553744						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4043		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4043-L1-L13		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/5/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 25 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>alt-murphy</i>		Received By:			
Date / Time: 25 April 2022		Date / Time: 5/6/22 14:00 WPS		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.
San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SUMMARY REPORTING SHEET (V2.0)

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S /SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4043	Reception	S1	Gypsum board wall	Gray		Y	Use				KAB-4043-L1				
		S2	Metal pipe (above ceiling)	Red		Y	Use				KAB-4043-L2				
		S1	Gypsum board wall	Gray		Y	Use				KAB-4043-L3				
	Mens bathroom	S3	Ceiling board	Off white		N	Use				KAB-4043-L4				
	Janitor	S4	Metal door frame	Off white over red and black		Y	Use				KAB-4043-L5				
	Office	S5	Wooden door	Black over white		Y	Use				KAB-4043-L6				
		S1	Gypsum board wall	Gray		Y	Use				KAB-4043-L7				
		S6	Metal pipe	Red over white and black		Y	Use				KAB-4043-L8				
		S7	Metal flange	Light blue		Y	Used				KAB-4043-L9				
		S8	Concrete wall	Off white, textured		Y	Used				KAB-4043-L10				
		S8	Concrete wall	Off white, textured		Y	Used				KAB-4043-L11				
		S9	Concrete wall base	Light brown		Y	Use				KAB-4043-L12				
		S8	Concrete wall	Off white, textured		Y	Used				KAB-4043-L13				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242208
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4043

SGSFL Job ID: L1232-1

Date(s) Collected: 04/05/22

Total Samples Submitted: 13

Total Samples Analyzed: 13

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4043-L1	LM220620	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4043-L2	LM220621	Pb	0.009	wt%	0.008	EPA 3050B/7000B
KAB-4043-L3	LM220622	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4043-L4	LM220623	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4043-L5	LM220624	Pb	1.3	wt%	0.2	EPA 3050B/7000B
KAB-4043-L6	LM220625	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4043-L7	LM220626	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4043-L8	LM220627	Pb	< 0.01	wt%	0.01	EPA 3050B/7000B
KAB-4043-L9	LM220628	Pb	< 0.01	wt%	0.01	EPA 3050B/7000B
KAB-4043-L10	LM220629	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4043-L11	LM220630	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4043-L12	LM220631	Pb	0.008	wt%	0.007	EPA 3050B/7000B
KAB-4043-L13	LM220632	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

Analytical results and reports are generated by SGS Forensic Laboratories at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS Forensic Laboratories to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS Forensic Laboratories. The client is solely responsible for the use and interpretation of test results and reports requested from SGS Forensic Laboratories. SGS Forensic Laboratories is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Forensic Laboratories' Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

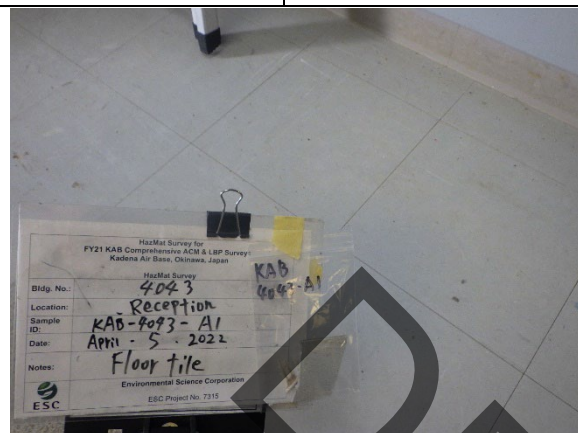
HazMat Survey

Bldg. 4043



Photo Record Sheet

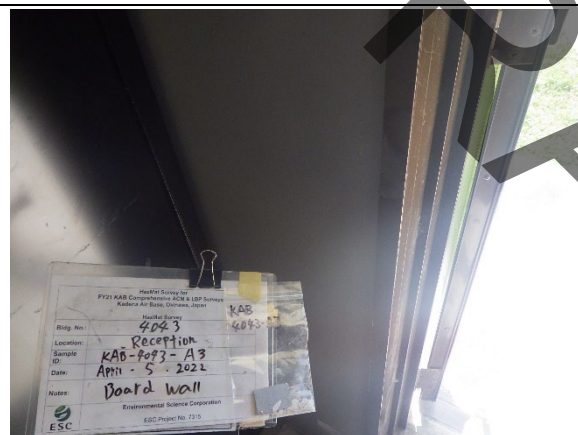
Asbestos Sampling
Kadena Air Base, Bldg.4043



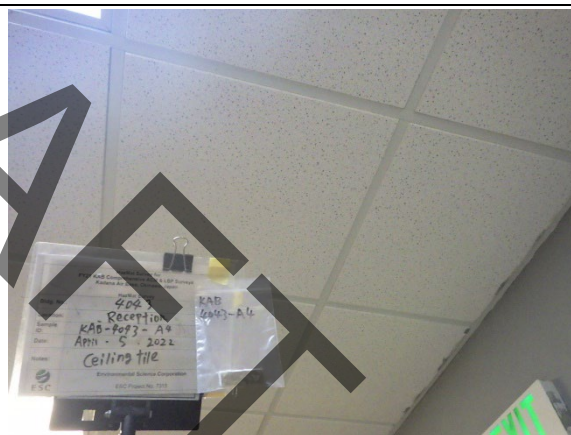
KAB-4043-A1



KAB-4043-A2



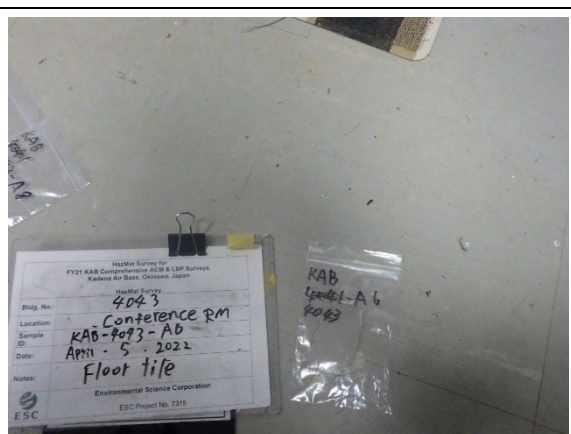
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KAB-4043-A4



KAB-4043-A5

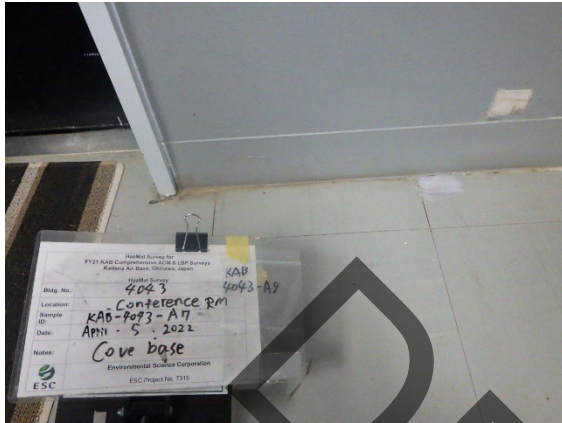


KAB-4043-A6

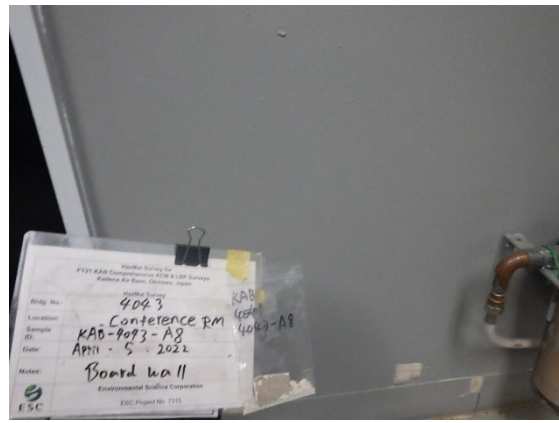


Photo Record Sheet

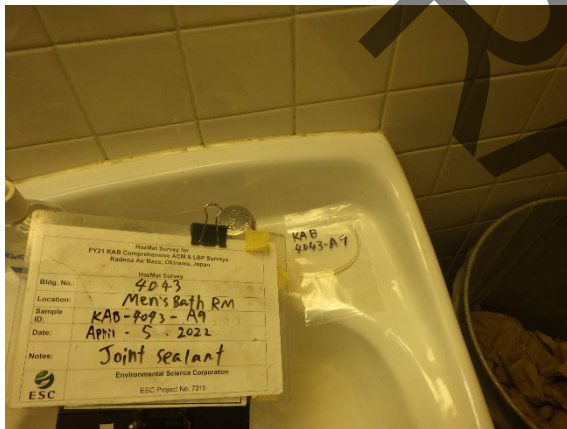
Asbestos Sampling
Kadena Air Base, Bldg.4043



KAB-4043-A7



KAB-4043-A8



KAB-4043-A9



KAB-4043-A10



KAB-4043-A11

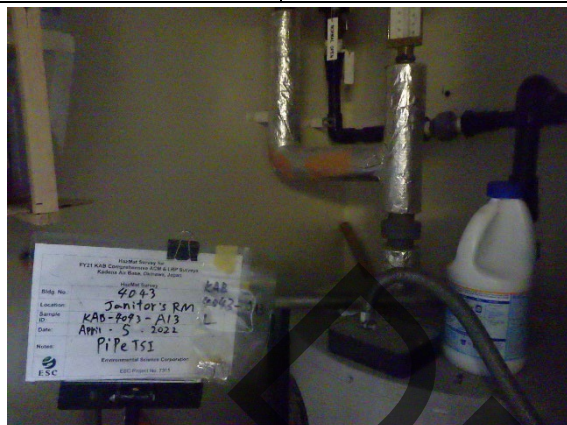


KAB-4043-A12

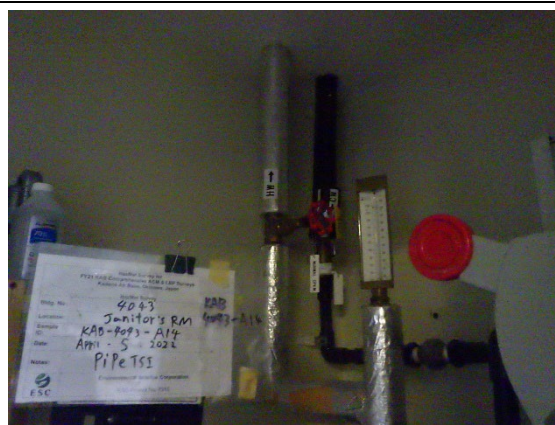


Photo Record Sheet

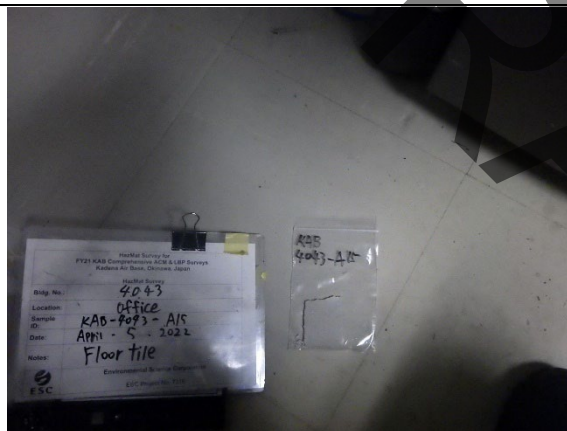
Asbestos Sampling
Kadena Air Base, Bldg.4043



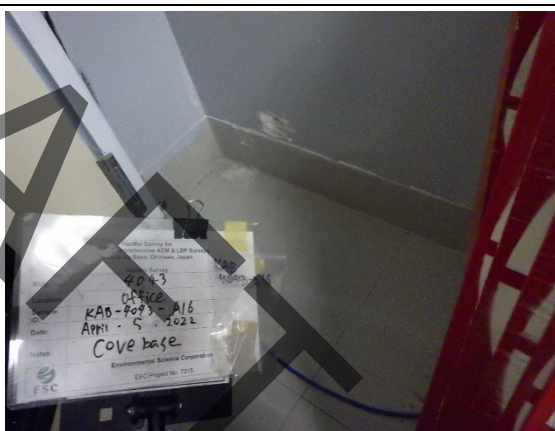
KAB-4043-A13



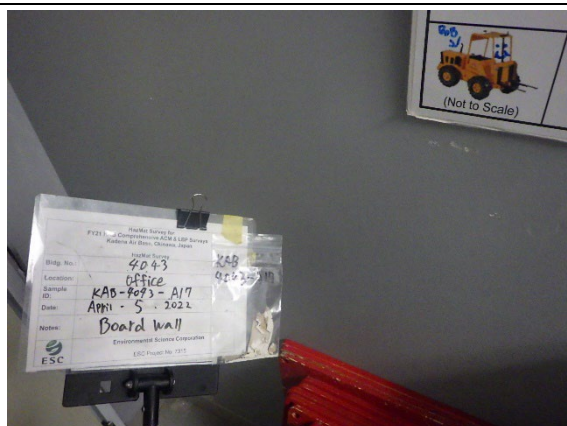
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KAB-4043-A15



KAB-4043-A16



KAB-4043-A17



KAB-4043-A18

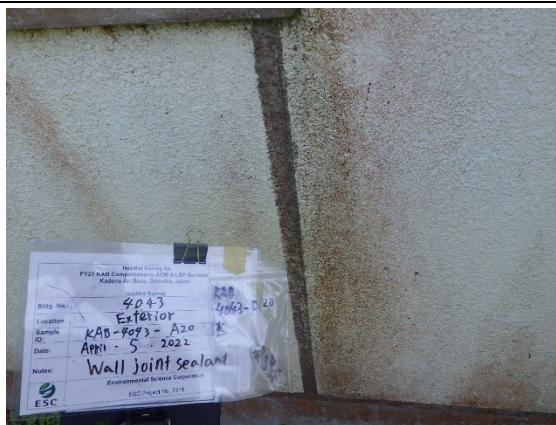


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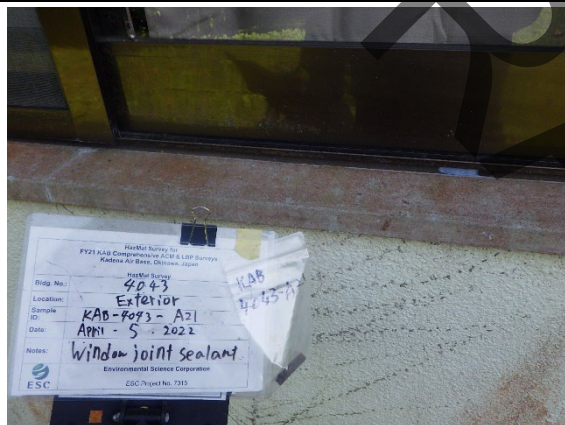
Asbestos Sampling
Kadena Air Base, Bldg.4043



KAB-4043-A19



KAB-4043-A20



KAB-4043-A21



KAB-4043-A22



KAB-4043-A23

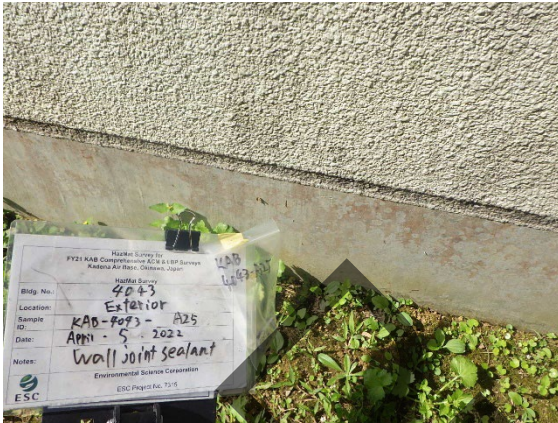


KAB-4043-A24



Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4043



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KAB-4043-A25

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

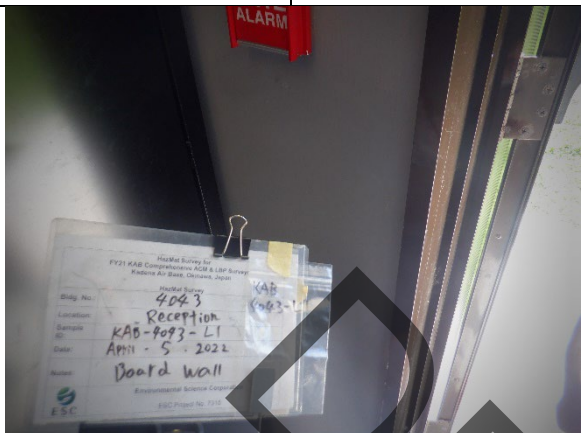
HazMat Survey

Bldg. 4043



Photo Record Sheet

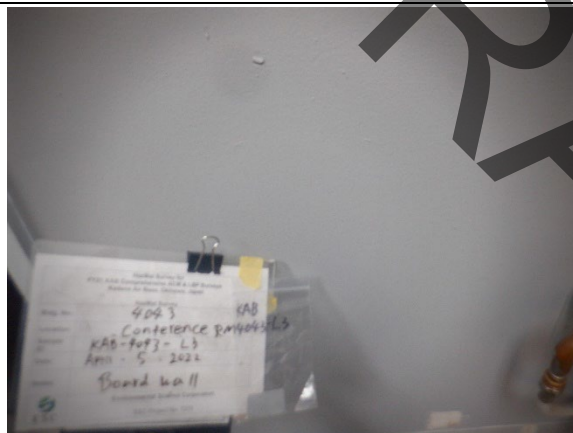
Lead Based Paint Sampling
Kadena Air Base, Bldg.4043



KAB-4043-L1



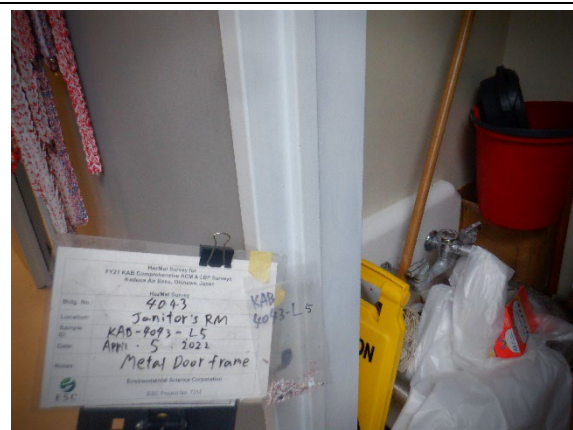
KAB-4043-L2



KAB-4043-L3



KAB-4043-L4



KAB-4043-L5

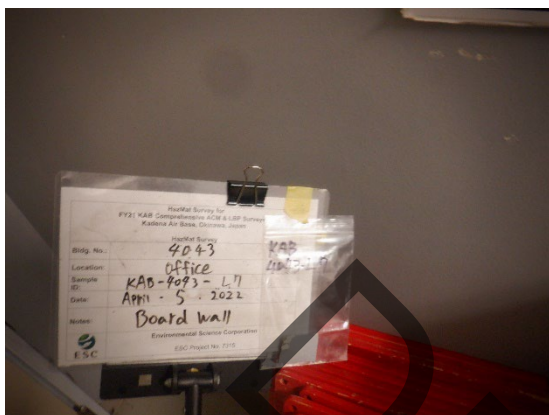


KAB-4043-L6



Photo Record Sheet

Lead Based Paint Sampling Kadena Air Base, Bldg.4043



KAB-4043-L7



KAB-4043-L8



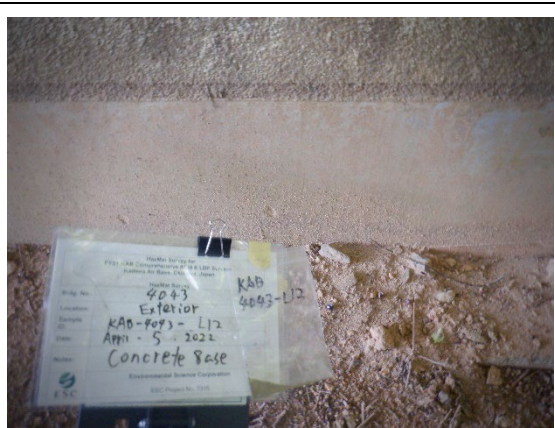
KAB-4043-L9



KAB-4043-L10



KAB-4043-L11



KAB-4043-L12



Photo Record Sheet

Lead Based Paint Sampling
Kadena Air Base, Bldg.4043



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KAB-4043-L13

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NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4044

Date: 20 March 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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2.0 ASBESTOS SAMPLING SUMMARY	1
2.1 ASBESTOS SAMPLING METHODS/ANALYSIS	2
3.0 LEAD BASED PAINT SAMPLING SUMMARY	2
3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE	2
3.2 PAINT WITH LEAD	2
4.0 REFERENCES	6

TABLES

TABLE 1 SUMMARY TABLE OF ASBESTOS AND LEAD PAINT SAMPLING	4
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APPENDICES

A	ASBESTOS / LEAD PAINT SAMPLING LOCATIONS
B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4044, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of thirty-two (32) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of fifteen (15) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
¹ BLDG NUMBER	² ROOM OR FUNCTIONAL AREA	³ HOMOGENEOUS /SURFACE NUMBER	⁴ MATERIAL NAME /APPLICATION	⁵ DESCRIPTION (Size, color, other)	⁶ FRIB AB	⁷ ACCE	⁸ CONDITION	⁹ HA QTY	¹⁰ GOV T	¹¹ C T R	¹² LAB/ SAMPLE ID NO.	¹³ CTR COC SAMPLE ID NO.	¹⁴ LAB ANALYSIS DATE	¹⁵ RESULT	¹⁶ VALIDITY DETERMINATION	¹⁷ NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4044	Reception	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	N	Y	Good	60 SM		Y	51553688	KAB-4044-A1	2022-10-04	ND	Valid	
4044	Reception	H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Good	45 LM		Y	51553689	KAB-4044-A2	2022-10-04	ND	Valid	
4044	Reception	H3	Gypsum board wall	Gray, smooth	N	Y	Good	265 SM		Y	51553690	KAB-4044-A3	2022-10-04	ND	Valid	
4044	Reception	H4	Pipe TSI (straight)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553691	KAB-4044-A4	2022-10-04	ND	Valid	
4044	Reception	H5	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553692	KAB-4044-A5	2022-10-04	ND	Valid	
4044	Reception	H6	Ceiling tile	60x60 pin hole pattern and fissured	N	N	Good	55 SM		Y	51553693	KAB-4044-A6	2022-10-04	ND	Valid	
4044	Office 1	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	N	Y	Good	60 SM		Y	51553694	KAB-4044-A7	2022-10-04	ND	Valid	
4044	Office 1	H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Good	45 LM		Y	51553695	KAB-4044-A8	2022-10-04	ND	Valid	
4044	Office 1	H3	Gypsum board wall	Gray, smooth	N	Y	Good	265 SM		Y	51553696	KAB-4044-A9	2022-10-04	ND	Valid	
4044	Office 1	H7	Concrete wall coating	Gray, smooth	N	Y	Good	20 SM		Y	51553697	KAB-4044-A10	2022-10-04	ND	Valid	
4044	Storage	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	N	Y	Good	60 SM		Y	51553698	KAB-4044-A11	2022-10-04	ND	Valid	
4044	Storage	H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Good	N/A		Y	51553699	KAB-4044-A12	2022-10-04	ND	Valid	
4044	Storage	H8	Gypsum board wall	Off-white, smooth	N	Y	Good	30 SM		Y	51553700	KAB-4044-A13	2022-10-04	ND	Valid	
4044	Storage	H4	Pipe TSI (straight)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553701	KAB-4044-A14	2022-10-04	ND	Valid	
4044	Storage	H5	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	Y	N	Good	N/A		Y	51553702	KAB-4044-A15	2022-10-04	ND	Valid	
4044	Storage	H9	Door joint sealant	Off white	N	N	Good	N/A		Y	51553703	KAB-4044-A16	2022-10-04	ND	Valid	
4044	Storage	H10	Concrete floor coating	Gray	N	Y	Good	10 SM		Y	51553704	KAB-4044-A17	2022-10-04	ND	Valid	
4044	Storage	H7	Concrete wall coating	Off white smooth	N	Y	Good	20 SM		Y	51553705	KAB-4044-A18	2022-10-04	ND	Valid	
4044	Men's bathroom	H11	Ceramic wall tile	10x10 off white w/white grout	N	Y	Good	40 SM		Y	51553706	KAB-4044-A19	2022-10-04	ND	Valid	
4044	Men's bathroom	H12	Ceramic floor tile	20x20 beige w/grey grout	N	Y	Good	20 SM		Y	51553707	KAB-4044-A20	2022-10-04	ND	Valid	
4044	Men's bathroom	H13	Sink joint sealant	White mastic	N	Y	Good	N/A		Y	51553708	KAB-4044-A21	2022-10-04	ND	Valid	
4044	Men's bathroom	H14	Ceiling board	Off white smooth	N	N	Good	20 SM		Y	51553709	KAB-4044-A22	2022-10-04	ND	Valid	
4044	Exterior	H15	Flange gasket	Black	N	Y	Good	N/A		Y	51553710	KAB-4044-A23	2022-10-04	ND	Valid	
4044	Exterior	H16	Concrete wall coating	Off white textured	N	Y	Good	220 SM		Y	51553711	KAB-4044-A24	2022-10-04	ND	Valid	
4044	Exterior	H17	Wall joint sealant	Gray	N	Y	Good	N/A		Y	51553712	KAB-4044-A25	2022-10-04	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1 BLDG NUMBER	2 ROOM OR FUNCTIONAL AREA	3 HOMOGENEOUS /SURFACE NUMBER	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIAB LE	7 ACCE SC	8 CONDITION	9 HA QTY	10 GOV T	11 C T R	12 LAB/ SAMPLE ID NO.	13 CTR COC SAMPLE ID NO.	14 LAB ANALYSIS DATE	15 RESULT	16 VALIDITY DETERMINATION	17 NOTES / INVALID OR UNKNOWN REASONING
4044	Exterior	H18	Window joint sealant	Brown	N	Y	Good	N/A		Y	51553713	KAB-4044-A26	2022-10-04	ND	Valid	
4044	Exterior	H17	Wall joint sealant	Gray	N	Y	Good	N/A		Y	51553714	KAB-4044-A27	2022-10-04	ND	Valid	
4044	Exterior	H16	Concrete wall coating	Off white textured	N	Y	Good	220 SM		Y	51553715	KAB-4044-A28	2022-10-04	ND	Valid	
4044	Exterior	H19	Floor joint sealant	Light gray	N	Y	Good	N/A		Y	51553716	KAB-4044-A29	2022-10-04	ND	Valid	
4044	Exterior	H20	Door joint sealant	Off white	N	Y	Good	N/A		Y	51553717	KAB-4044-A30	2022-10-04	ND	Valid	
4044	Exterior	H17	Wall joint sealant	Gray	N	Y	Good	N/A		Y	51553718	KAB-4044-A31	2022-10-04	ND	Valid	
4044	Exterior	H16	Concrete wall coating	Off-white, textured	N	Y	Good	220 SM		Y	51553719	KAB-4044-A32	2022-10-04	ND	Valid	
LEAD PAINT SAMPLES																
4044	Reception	S1	Gypsum board wall	Gray		Y	Used			Y	LM220579	KAB-4044-L1	2022-06-22	<0.007	Valid	
4044	Reception	S2	Metal pipe	Red over white		N	Used			Y	LM220580	KAB-4044-L2	2022-06-22	<0.007	Valid	
4044	Office 1	S3	Wooden door	Green over off white		Y	Used			Y	LM220581	KAB-4044-L3	2022-06-22	0.008	Valid	
4044	Office 1	S4	Wooden door frame	Black over red and off-white		Y	Used			Y	LM220582	KAB-4044-L4	2022-06-22	0.008	Valid	
4044	Office 1	S1	Gypsum board wall	Gray		Y	Used			Y	LM220583	KAB-4044-L5	2022-06-22	<0.006	Valid	
4044	Office 1	S5	Gypsum board wall	Off-white		Y	Used			Y	LM220584	KAB-4044-L6	2022-06-22	<0.007	Valid	
4044	Office 1	S6	Concrete wall	Off-white		Y	Used			Y	LM220585	KAB-4044-L7	2022-06-22	<0.006	Valid	
4044	Men's bathroom	S7	Ceiling board	Off-white		N	Used			Y	LM220586	KAB-4044-L8	2022-06-22	<0.007	Valid	
4044	Exterior	S8	Metal pipe	Red over off-white		Y	Used			Y	LM220587	KAB-4044-L9	2022-06-22	<0.007	Valid	
4044	Exterior	S9	Metal flange	Light blue		Y	Used			Y	LM220588	KAB-4044-L10	2022-06-22	<0.02	Valid	
4044	Exterior	S10	Concrete wall	Off-white		Y	Used			Y	LM220589	KAB-4044-L11	2022-06-22	<0.006	Valid	
4044	Exterior	S11	Concrete wall base	Light brown		Y	Used			Y	LM220590	KAB-4044-L12	2022-06-22	0.010	Valid	
4044	Exterior	S10	Concrete wall	Off-white		Y	Used			Y	LM220591	KAB-4044-L13	2022-06-22	<0.006	Valid	
4044	Exterior	S12	Metal pipe	Red		Y	Used			Y	LM220592	KAB-4044-L14	2022-06-22	<0.006	Valid	
4044	Exterior	S10	Concrete wall	Off-white		Y	Used			Y	LM220593	KAB-4044-L15	2022-06-22	<0.006	Valid	

Notes;
The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

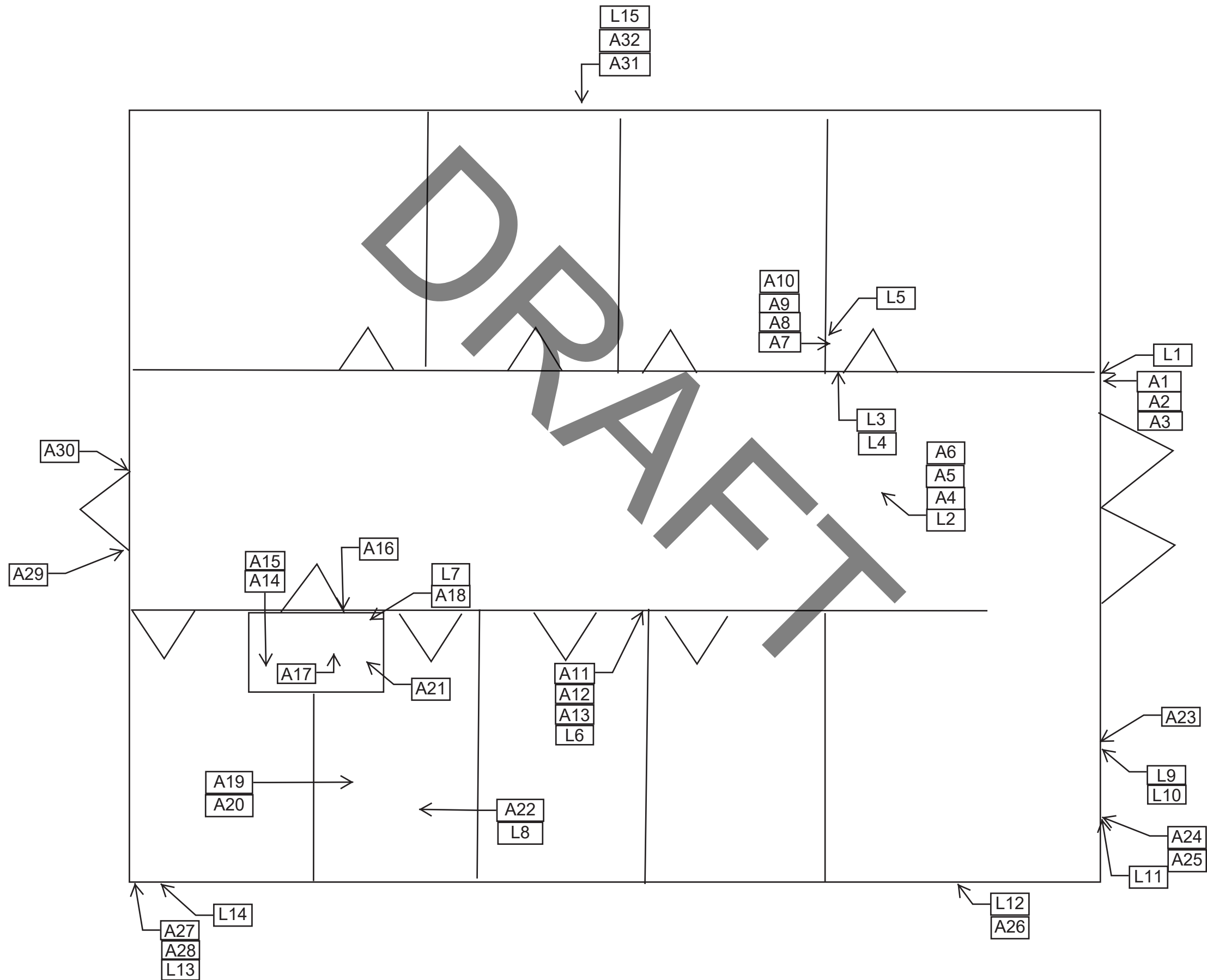
US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS



ASBESTOS AND LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: ☐ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg. 4044		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project ☐ Metals Analysis Matrix: Method:		
Comments: See attached sample logs.		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4044A1-A32		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				

Sampled By: Sean R Dudney Date/Time: 4/8/22		Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS	
Relinquished By: Sean R Dudney		Relinquished By:	
Date / Time: 25 April 2022		Date / Time:	
Received By: Japan Post Office		Received By: <i>Alvin Whiting</i>	
Date / Time: 25 April 2022		Date / Time: 5-26-22 14:00 WPS	
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No	

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

ANALYSIS RECORD DATA															
+C30+A2:AC20+A2:AC22+A2:AC24+C30+A2:A+A2:AC25															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S/SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G C O T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING	
4044	Reception	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	Y	Y	Use		KAB-4044-A1						
4044		H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Use		KAB-4044-A2						
4044		H3	Gypsum board wall	Gray smooth	N	Y	Use		KAB-4044-A3						
4044		H4	Pipe TSI (straight)	Yellow fiberglass w/silver backing	N	N	Use		KAB-4044-A4						
4044		H5	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	N	N	Use		KAB-4044-A5						
4044		H6	Ceiling tile	60x60 pin hole pattern and fisherd	N	N	Use		KAB-4044-A6						
4044	Office 1	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	Y	Y	Use		KAB-4044-A7						
4044		H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Use		KAB-4044-A8						
4044		H3	Gypsum board wall	Gray smooth	N	Y	Use		KAB-4044-A9						
4044		H7	Concrete wall	Gray smooth	N	Y	Use		KAB-4044-A10						
4044	Storage	H1	Vinyl floor tile	30x30 gray tile w/light gray mastic	Y	Y	Use		KAB-4044-A11						
4044		H2	Vinyl cove base	10cm gray w/light brown mastic	N	Y	Use		KAB-4044-A12						
4044		H8	Gypsum board wall	Off white smooth	N	Y	Use		KAB-4044-A13						
4044		H4	Pipe TSI (straight)	Yellow fiberglass w/silver backing	N	N	Use		KAB-4044-A14						
4044		H5	Pipe TSI (elbow)	Yellow fiberglass w/silver backing	N	N	Use		KAB-4044-A15						
4044		H9	Door joint sealant	Off white	N	N	Use		KAB-4044-A16						
4044		H10	Concrete floor coating	Gray	Y	Y	Use		KAB-4044-A17						
4044		H7	Concrete wall	Off white smooth	N	Y	Use		KAB-4044-A18						

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333881
Date Received: 05/26/22
Date Analyzed: 10/04/22
Date Printed: 10/06/22
First Reported: 10/06/22

Job ID/Site: ESC.7315; Kadena Air Base, Okinawa, Japan; Bldg. 4044

SGSFL Job ID: L1232-1
Total Samples Submitted: 32
Total Samples Analyzed: 32

Date(s) Collected: 04/08/2022

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4044-A1	51553688						
Layer: Grey Tile			ND				
Layer: Off-White Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4044-A2	51553689						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4044-A3	51553690						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)							
KAB-4044-A4	51553691						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (90 %)							
KAB-4044-A5	51553692						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (90 %)							
KAB-4044-A6	51553693						
Layer: Light Grey Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (35 %) Fibrous Glass (45 %)							

Client Name: Environmental Science Corporation

Report Number: B333881

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4044-A7	51553694						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A8	51553695						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: White Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A9	51553696						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4044-A10	51553697						
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A11	51553698						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A12	51553699						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Layer: White Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A13	51553700						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %)							
KAB-4044-A14	51553701						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Fibrous Glass (90 %)							

Client Name: Environmental Science Corporation

Report Number: B333881

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4044-A15	51553702						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Fibrous Glass (90 %)							
KAB-4044-A16	51553703						
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A17	51553704						
Layer: Grey Cementitious Material with Debris			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A18	51553705						
Layer: Grey Cementitious Material			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A19	51553706						
Layer: White Ceramic Tile			ND				
Layer: Grey Plaster			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A20	51553707						
Layer: Beige Ceramic Tile			ND				
Layer: Dark Grey Grout			ND				
Layer: Light Grey Mortar			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A21	51553708						
Layer: White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A22	51553709						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (20 %)							

Client Name: Environmental Science Corporation

Report Number: B333881

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4044-A23	51553710						
Layer: Black Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A24	51553711						
Layer: Off-White Plaster			ND				
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A25	51553712						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A26	51553713						
Layer: Brown Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A27	51553714						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A28	51553715						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A29	51553716						
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A30	51553717						
Layer: Off-White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333881

Date Printed: 10/06/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4044-A31	51553718						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4044-A32	51553719						
Layer: Grey Cementitious Material			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

DRAFT



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 25 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4044		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4044-L1-L15		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/11/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:	Relinquished By:				
Date / Time: 25 April 2022		Date / Time:	Date / Time:				
Received By: Japan Post Office		Received By: <i>Chen Ming</i>	Received By:				
Date / Time: 25 April 2022		Date / Time: 5-26-22 14:00 USPS	Date / Time:				
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☐ Yes ☐ No				

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1 BLDG NUMBE	2 ROOM OR FUNCTIONA	3 HOMOGENEOU S /SURFACE	4 MATERIAL NAME /APPLICATION	5 DESCRIPTION (Size, color, other)	6 FRIA	7 ACC	8 CONDITIO N	9 G	10 C	11 LAB/ SAMPLE ID	12 CTR COC	13 LAB ANALYSIS	14 RESULT	15 VALIDITY DETERMINATI	16 NOTES / INVALID OR UNKNOWN REASONING
4044	Reception	S1	Gypsum board wall	Gray		Y	Use			KAB-4044-L1					
4044		S2	Metal pipe	Red over white		N	Use			KAB-4044-L2					
4044	Office 1	S3	Wooden door	Green over off white		Y	Use			KAB-4044-L3					
4044		S4	Wooden door frame	Black over red and off white		Y	Use			KAB-4044-L4					
4044		S1	Gypsum board wall	Gray		Y	Use			KAB-4044-L5					
4044		S5	Gypsum board wall	Off white		Y	Use			KAB-4044-L6					
4044		S6	Concrete wall	Off white		Y	Used			KAB-4044-L7					
4044	Mens bathroom	S7	Ceiling board	Off white		N	Use			KAB-4044-L8					
4044	Exterior	S8	Metal pipe	Red over off white		Y	Use			KAB-4044-L9					
4044		S9	Metal flange	Light blue		Y	Use			KAB-4044-L10					
4044		S10	Concrete wall	Off white		Y	Use			KAB-4044-L11					
4044		S11	Concrete wall base	Light brown		Y	Used			KAB-4044-L12					
4044		S10	Concrete wall	Off white		Y	Use			KAB-4044-L13					
4044		S12	Metal pipe	Red		Y	Use			KAB-4044-L14					
4044		S10	Concrete wall	Off white		Y	Use			KAB-4044-L15					

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242186
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4044

SGSFL Job ID: L1232-1

Date(s) Collected: 04/11/22

Total Samples Submitted: 15

Total Samples Analyzed: 15

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4044-L1	LM220579	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4044-L2	LM220580	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4044-L3	LM220581	Pb	< 0.008	wt%	0.008	EPA 3050B/7000B
KAB-4044-L4	LM220582	Pb	< 0.008	wt%	0.008	EPA 3050B/7000B
KAB-4044-L5	LM220583	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4044-L6	LM220584	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4044-L7	LM220585	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4044-L8	LM220586	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4044-L9	LM220587	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4044-L10	LM220588	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4044-L11	LM220589	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4044-L12	LM220590	Pb	0.010	wt%	0.006	EPA 3050B/7000B
KAB-4044-L13	LM220591	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4044-L14	LM220592	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4044-L15	LM220593	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

Analytical results and reports are generated by SGS Forensic Laboratories at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS Forensic Laboratories to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS Forensic Laboratories. The client is solely responsible for the use and interpretation of test results and reports requested from SGS Forensic Laboratories. SGS Forensic Laboratories is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Forensic Laboratories' Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

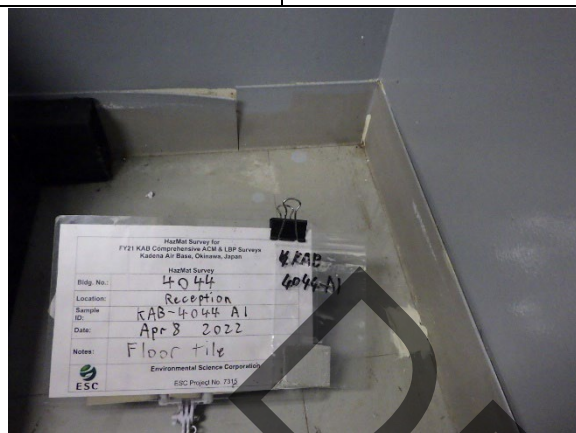
HazMat Survey

Bldg. 4044

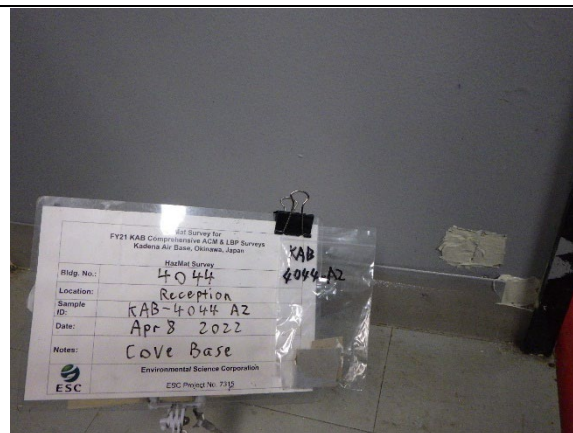


Photo Record Sheet

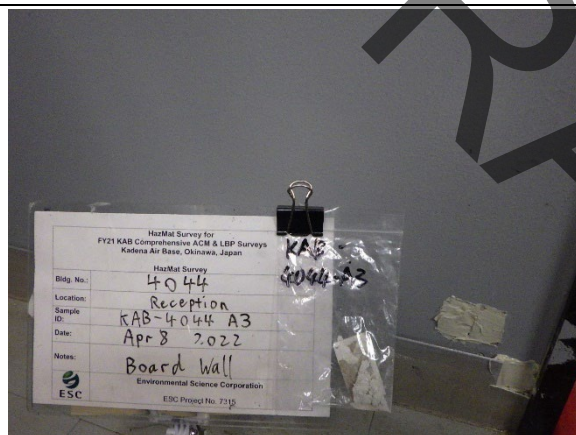
Asbestos Sampling
Kadena Air Base, Bldg.4044



KAB-4044-A1



KAB-4044-A2



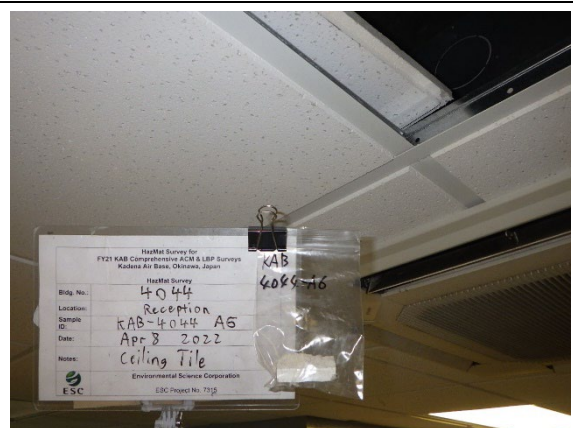
KAB-4044-A3



KAB-4044-A4



KAB-4044-A5

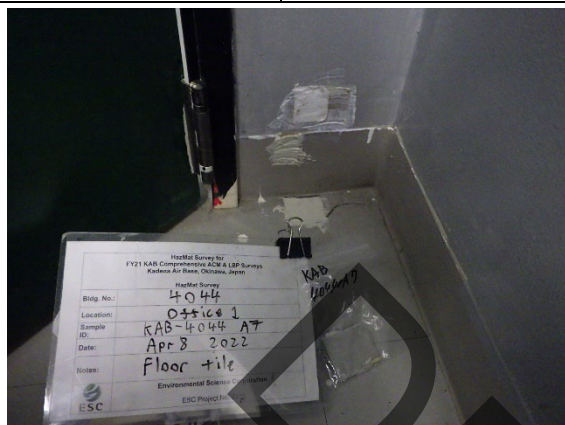


KAB-4044-A6



Photo Record Sheet

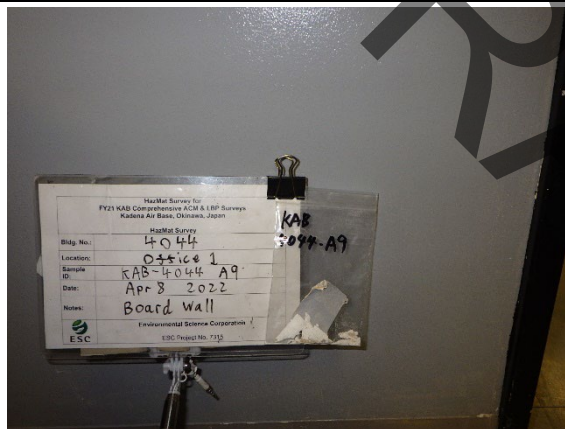
Asbestos Sampling
Kadena Air Base, Bldg.4044



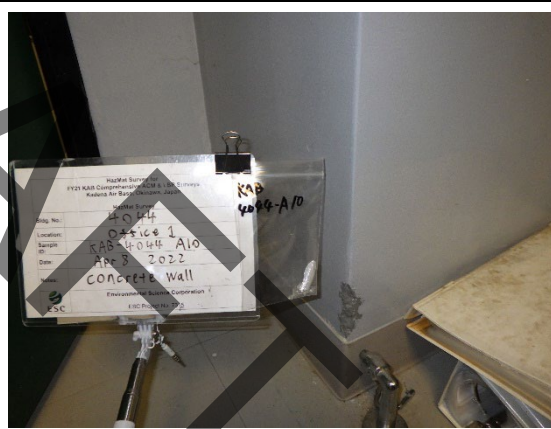
KAB-4044-A7



KAB-4044-A8



KAB-4044-A9



KAB-4044-A10



KAB-4044-A11



KAB-4044-A12

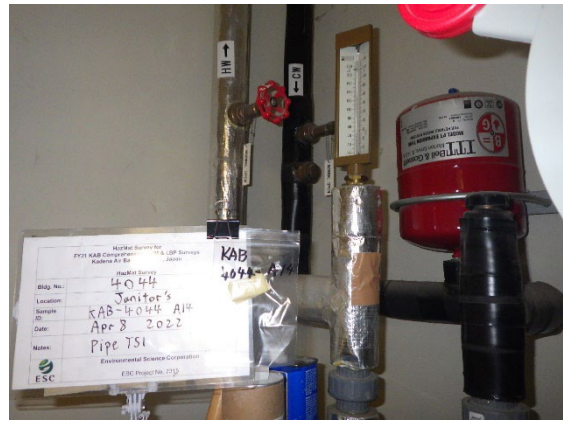


Photo Record Sheet

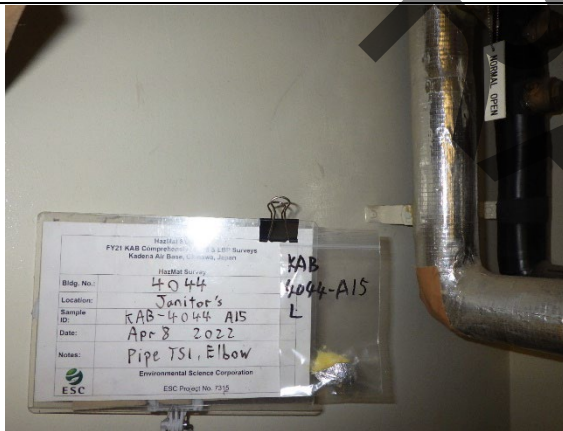
Asbestos Sampling Kadena Air Base, Bldg.4044



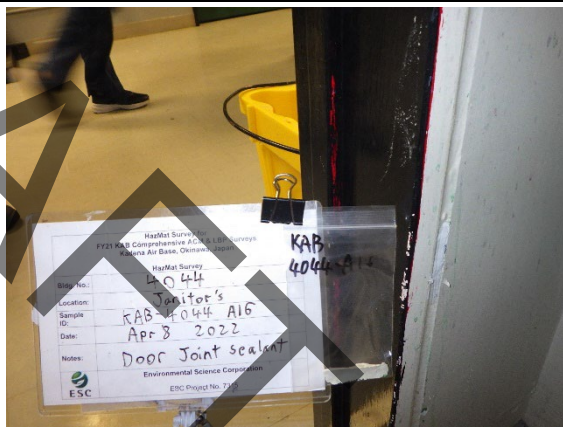
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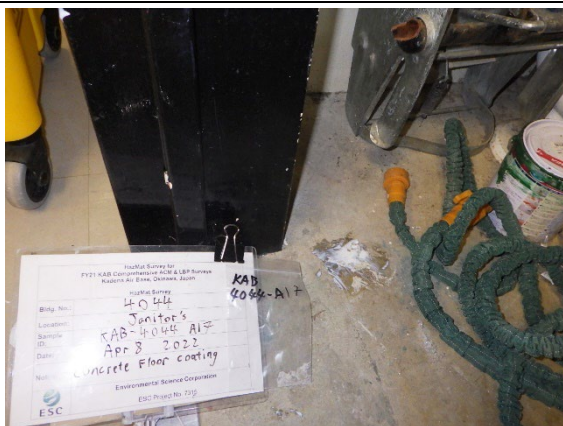
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KAB-4044-A15



KAB-4044-A16



KAB-4044-A17

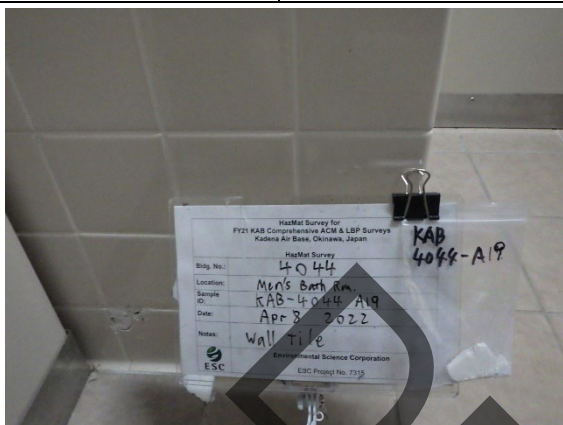


KAB-4044-A18

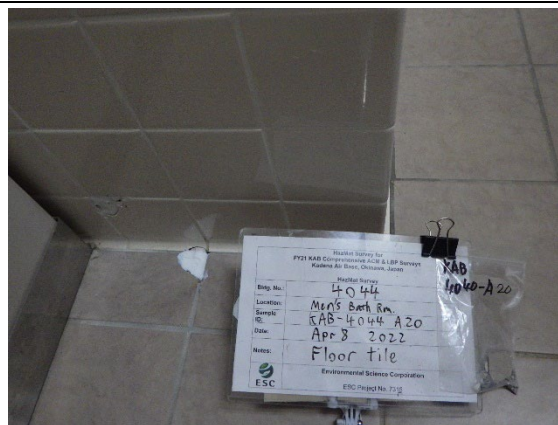


Photo Record Sheet

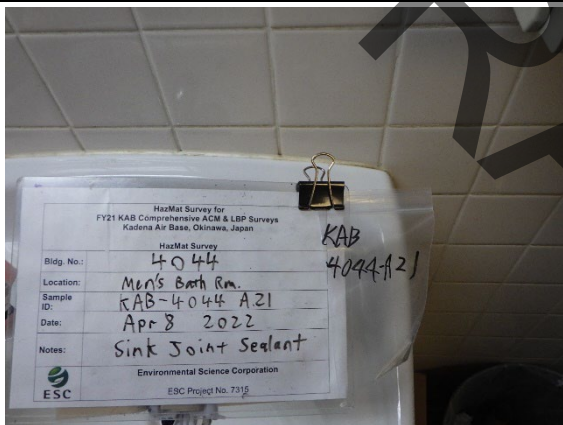
Asbestos Sampling
Kadena Air Base, Bldg.4044



KAB-4044-A19



KAB-4044-A20



KAB-4044-A21



KAB-4044-A22



KAB-4044-A23



KAB-4044-A24

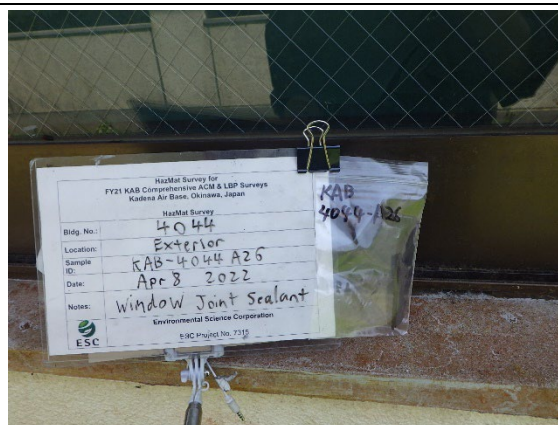


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4044



KAB-4044-A25



KAB-4044-A26



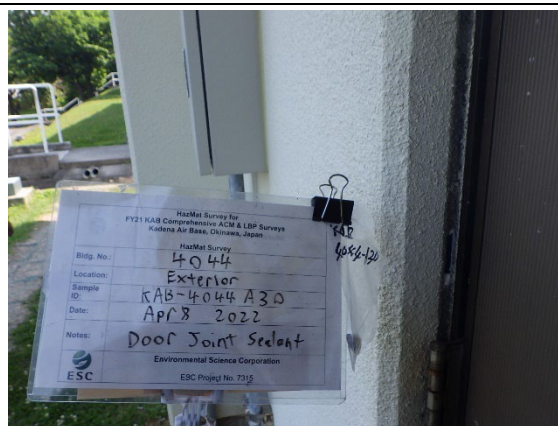
KAB-4044-A27



KAB-4044-A28



KAB-4044-A29

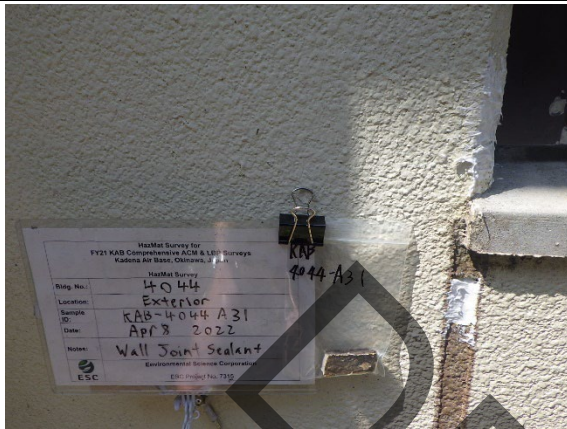


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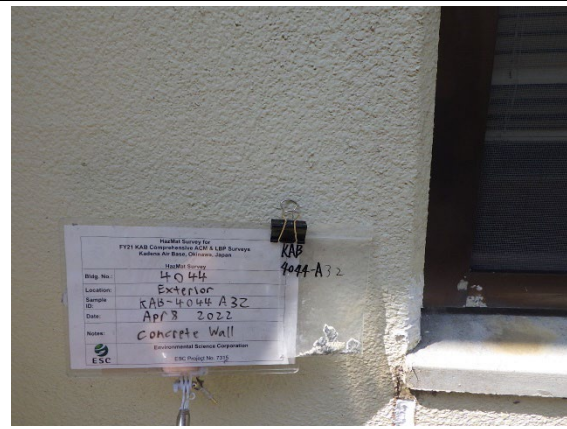


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4044



KAB-4044-A31



KAB-4044-A32

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

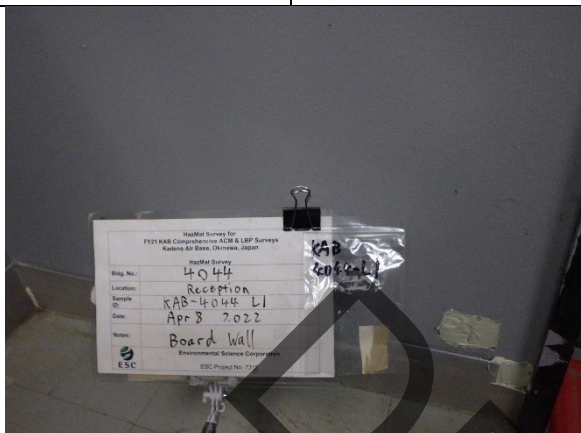
HazMat Survey

Bldg. 4044



Photo Record Sheet

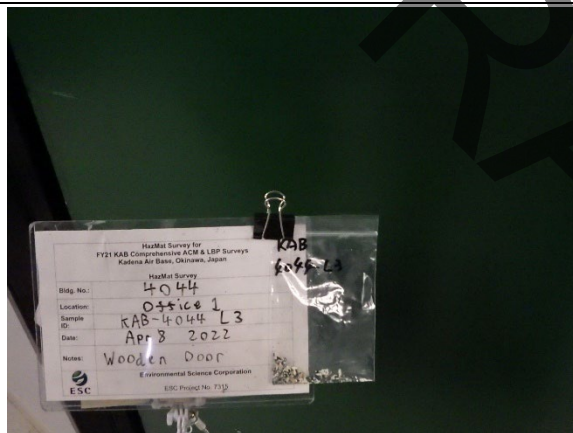
Lead Based Paint Sampling Kadena Air Base, Bldg.4044



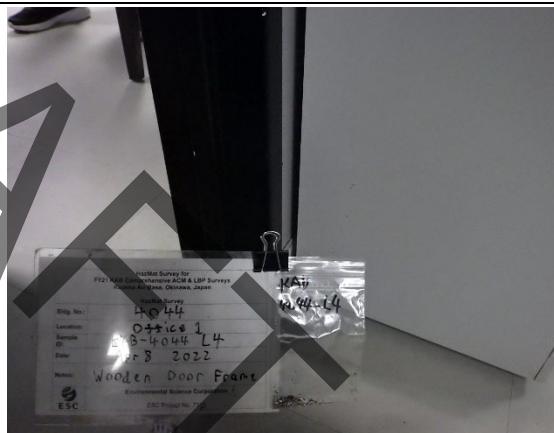
KAB-4044-L1



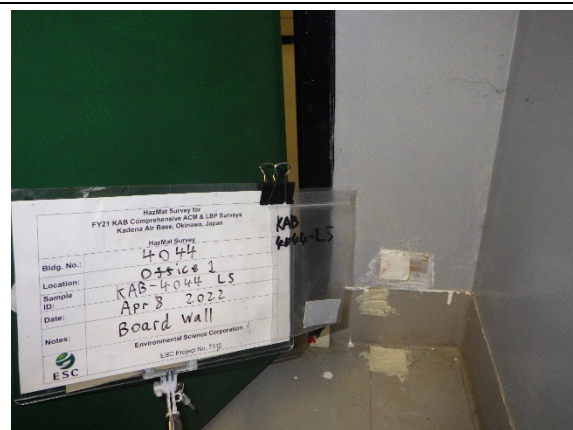
KAB-4044-L2



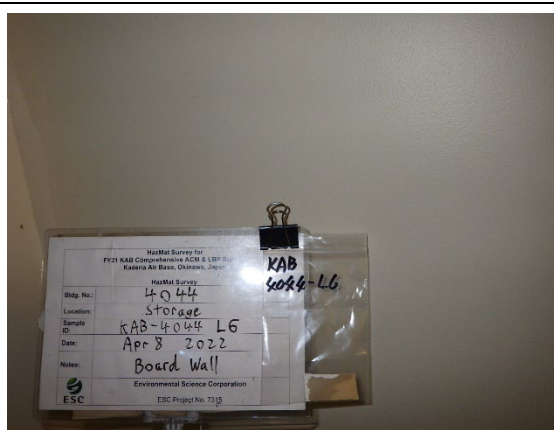
KAB-4044-L3



KAB-4044-L4



KAB-4044-L5



KAB-4044-L6



Photo Record Sheet

Lead Based Paint Sampling Kadena Air Base, Bldg.4044



KAB-4044-L7



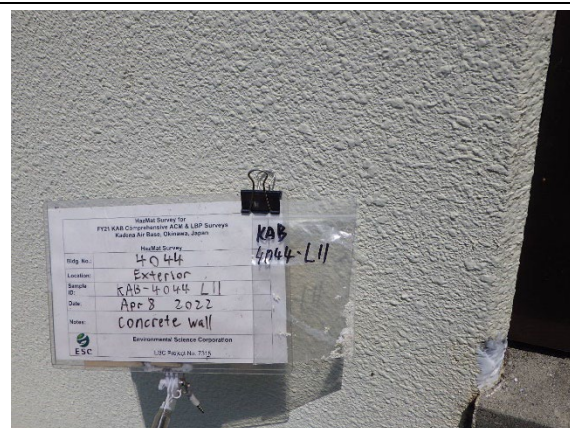
KAB-4044-L8



KAB-4044-L9



KAB-4044-L10



KAB-4044-L11

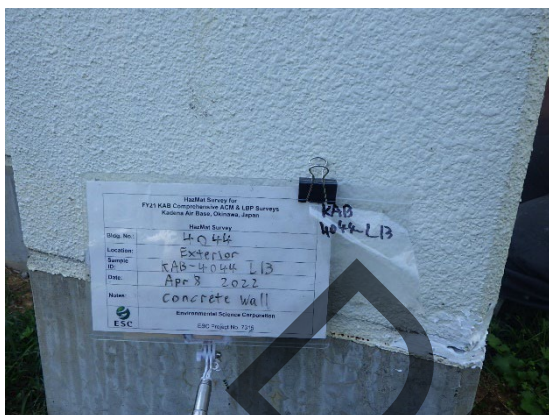


KAB-4044-L12

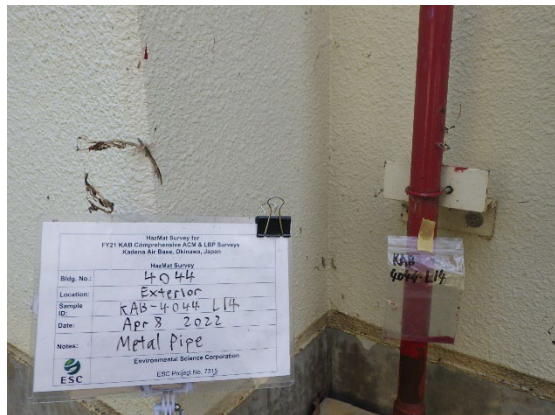


Photo Record Sheet

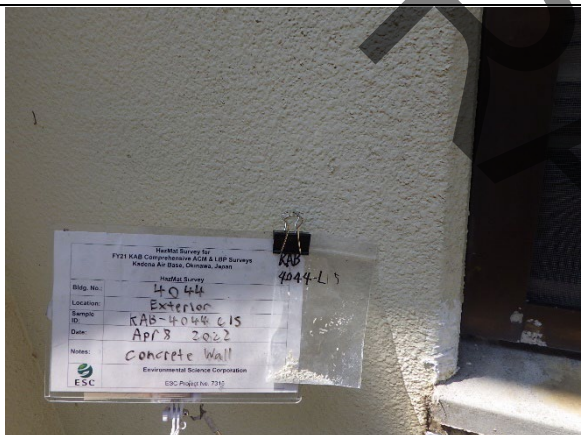
Lead Based Paint Sampling
Kadena Air Base, Bldg.4044



KAB-4044-L13



KAB-4044-L14



KAB-4044-L15

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NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4045

Date: 6 March 2023
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE	2
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APPENDICES

A	ASBESTOS / LEAD PAINT SAMPLING LOCATIONS
B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4045, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of thirty-five (35) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of thirteen (13) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

3.2 PAINT WITH LEAD

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter or more than 0.5 percent by weight. Lead-containing paint is paint containing detectable concentrations of lead, but not at sufficient concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2022).

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIABLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4045	Classroom	H1	Vinyl floor tile	30x30cm grey w/ grey and tan mastic	N	Y	Good	250 SM		Y	51553647	KAB-4045-A1	2022-10-10	ND	Valid	
4045	Classroom	H2	Vinyl cove base	10cm light grey w/ grey and tan mastic	N	Y	Good	100 LM		Y	51553648	KAB-4045-A2	2022-10-10	ND	Valid	
4045	Classroom	H3	Concrete wall coating	Off-white, smooth	N	Y	Good	20 SM		Y	51553649	KAB-4045-A3	2022-10-10	ND	Valid	
4045	Classroom	H4	Gypsum board wall	Off-white, smooth	N	Y	Good	200 SM		Y	51553650	KAB-4045-A4	2022-10-10	ND	Valid	
4045	Classroom	H5	Duct TSI	Yellow fiberglass w/ silver backing	Y	N	Good	N/A		Y	51553651	KAB-4045-A5	2022-10-10	ND	Valid	
4045	Classroom	H6	Ceiling insulation	Fiberglass w/ silver backing	Y	N	Good	N/A		Y	51553652	KAB-4045-A6	2022-10-10	ND	Valid	
4045	Classroom	H7	Ceiling tile	60x60cm off-white fissured	N	N	Good	280 SM		Y	51553653	KAB-4045-A7	2022-10-10	ND	Valid	
4045	Classroom	H8	Concrete wall coating	Above ceiling, black coating	N	N	Good	N/A		Y	51553654	KAB-4045-A8	2022-10-10	ND	Valid	
4045	Classroom	H1	Vinyl floor tile	30x30cm grey w/ grey and tan mastic	N	Y	Good	250 SM		Y	51553655	KAB-4045-A9	2022-10-10	ND	Valid	
4045	Classroom	H2	Vinyl cove base	10cm light grey w/ grey and tan mastic	N	Y	Good	100 LM		Y	51553656	KAB-4045-A10	2022-10-10	ND	Valid	
4045	Storage C	H9	Concrete block wall coating	Grey painted	N	N	Good	80 SM		Y	51553657	KAB-4045-A11	2022-10-10	ND	Valid	
4045	Storage C	H10	Pipe TSI straight	Yellow fiberglass w/ silver backing	Y	N	Good	N/A		Y	51553658	KAB-4045-A12	2022-10-10	ND	Valid	
4045	Storage C	H11	Pipe TSI elbow	Yellow fiberglass w/ silver backing	Y	N	Good	N/A		Y	51553659	KAB-4045-A13	2022-10-10	ND	Valid	
4045	Men's bathroom	H12	Ceramic floor tile	20x20cm grey w/ grey grout	N	Y	Good	30 SM		Y	51553660	KAB-4045-A14	2022-10-10	ND	Valid	
4045	Men's bathroom	H13	Ceramic wall tile	10x10cm off-white tile w/ white grout and grey cement	N	Y	Good	50 SM		Y	51553661	KAB-4045-A15	2022-10-10	ND	Valid	
4045	Men's bathroom	H14	Gypsum ceiling board	Off-white	N	N	Good	30 SM		Y	51553662	KAB-4045-A16	2022-10-10	ND	Valid	
4045	Men's bathroom	H15	Joint sealant (door)	White mastic	N	Y	Good	3 LM		Y	51553663	KAB-4045-A17	2022-10-10	ND	Valid	
4045	Men's bathroom	H15	Joint sealant (sink)	White mastic	N	Y	Good	3 LM		Y	51553664	KAB-4045-A18	2022-10-10	ND	Valid	
4045	Office	H1	Vinyl floor tile	30x30cm grey w/ grey and tan mastic	N	Y	Good	280 SM		Y	51553665	KAB-4045-A19	2022-10-10	ND	Valid	
4045	Office	H2	Vinyl cove base	10cm light grey w/ grey and tan mastic	N	Y	Good	100 LM		Y	51553666	KAB-4045-A20	2022-10-10	ND	Valid	
4045	Corridor	H16	Door joint sealant	White mastic	N	Y	Good	15 LM		Y	51553667	KAB-4045-A21	2022-10-10	ND	Valid	
4045	Office	H4	Gypsum board wall	Off-white, smooth	N	Y	Good	200 SM		Y	51553668	KAB-4045-A22	2022-10-10	ND	Valid	
4045	Exterior	H17	Wall joint sealant	White mastic	N	Y	Good	150 LM		Y	51553669	KAB-4045-A23	2022-10-10	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIBLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
4045	Exterior	H18	Concrete block wall coating	Off-white, textured	N	Y	Good	250 SM		Y	51553670	KAB-4045-A24	2022-10-10	ND	Valid	
4045	Exterior	H19	Concrete wall base coating	Red	N	Y	Good	30 SM		Y	51553671	KAB-4045-A25	2022-10-10	ND	Valid	
4045	Exterior	H17	Wall joint sealant	White mastic	N	Y	Good	150 LM		Y	51553672	KAB-4045-A26	2022-10-10	ND	Valid	
4045	Exterior	H20	Door joint sealant	Grey mastic	N	Y	Good	6 LM		Y	51553673	KAB-4045-A27	2022-10-10	ND	Valid	
4045	Exterior	H18	Concrete block wall coating	Off-white, textured	N	Y	Good	250 SM		Y	51553674	KAB-4045-A28	2022-10-10	ND	Valid	
4045	Exterior	H17	Wall joint sealant	White mastic	N	Y	Good	150 LM		Y	51553675	KAB-4045-A29	2022-10-10	ND	Valid	
4045	Exterior	H21	Flex sheet	Black sheet	N	Y	Damaged	N/A		Y	51553676	KAB-4045-A30	2022-10-10	ND	Valid	
4045	Exterior	H22	Duct TSI	Black foam w/ tan mastic	Y	Y	Damaged	N/A		Y	51553677	KAB-4045-A31	2022-10-10	ND	Valid	
4045	Exterior	H23	Duct mastic	Grey mastic on metal duct	N	Y	Good	N/A		Y	51553678	KAB-4045-A32	2022-10-10	ND	Valid	
4045	Exterior	H24	Concrete base coating	AHU, grey	N	Y	Good	5 SM		Y	51553679	KAB-4045-A33	2022-10-10	ND	Valid	
4045	Exterior	H25	Floor joint sealant	Solid black	N	Y	Good	50 LM		Y	51553680	KAB-4045-A34	2022-10-10	ND	Valid	
4045	Exterior	H26	Window joint sealant	Black mastic	N	Y	Good	20 LM		Y	51553681	KAB-4045-A35	2022-10-10	ND	Valid	
LEAD PAINT SAMPLES																
4045	Classroom	S1	Concrete wall	Off-white		Y	Used			Y	LM220362	KAB-4045-L1	2022-06-22	0.011	Valid	
4045	Classroom	S2	Gypsum board wall	Off-white		Y	Used			Y	LM220363	KAB-4045-L2	2022-06-22	<0.007	Valid	
4045	Classroom	S3	Concrete wall (above ceiling)	Black		N	New			Y	LM220364	KAB-4045-L3	2022-06-22	<0.007	Valid	
4045	Storage C	S4	Concrete block wall	Grey		Y	New			Y	LM220365	KAB-4045-L4	2022-06-22	0.008	Valid	
4045	Corridor	S5	Concrete block wall	Grey and off-white		Y	New			Y	LM220366	KAB-4045-L5	2022-06-22	<0.007	Valid	
4045	Men's bathroom	S6	Gypsum ceiling board	Off-white		N	New			Y	LM220367	KAB-4045-L6	2022-06-22	<0.006	Valid	
4045	Corridor	S7	Metal pipe (electrical)	Off-white		Y	New			Y	LM220368	KAB-4045-L7	2022-06-22	<0.02	Valid	
4045	Office	S8	Gypsum board wall	Off-white		Y	New			Y	LM220369	KAB-4045-L8	2022-06-22	<0.007	Valid	
4045	Exterior	S4	Concrete block wall	Off-white		Y	New			Y	LM220370	KAB-4045-L9	2022-06-22	0.022	Valid	
4045	Exterior	S9	Concrete wall base	Red		Y	Used			Y	LM220371	KAB-4045-L10	2022-06-22	<0.009	Valid	
4045	Exterior	S4	Concrete block wall	Off-white		Y	New			Y	LM220372	KAB-4045-L11	2022-06-22	0.012	Valid	
4045	Exterior	S10	Metal pipe (electrical)	Off-white		Y	Damaged			Y	LM220373	KAB-4045-L12	2022-06-22	0.053	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIABLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
4045	Exterior	S11	Plastic drainpipe	Off-white		Y	Used			Y	LM220374	KAB-4045-L13	2022-06-22	<0.02	Valid	

Notes;

The quantities provided in this table represents rough order quantities for the purpose to aid in identifying the homogeneous material identified by the CBI at the time of the survey. This estimate should not be utilized as a basis to develop an estimate for the purpose of formulating a bid for any future work involving all or part of such material.

RED HIGHLIGHT – Asbestos Positive
BLUE HIGHLIGHT – Lead Based Paint Positive
GREEN HIGHLIGHT – Unknown, Trace result with Insufficient Material for Additional Analysis

DRAFT

4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

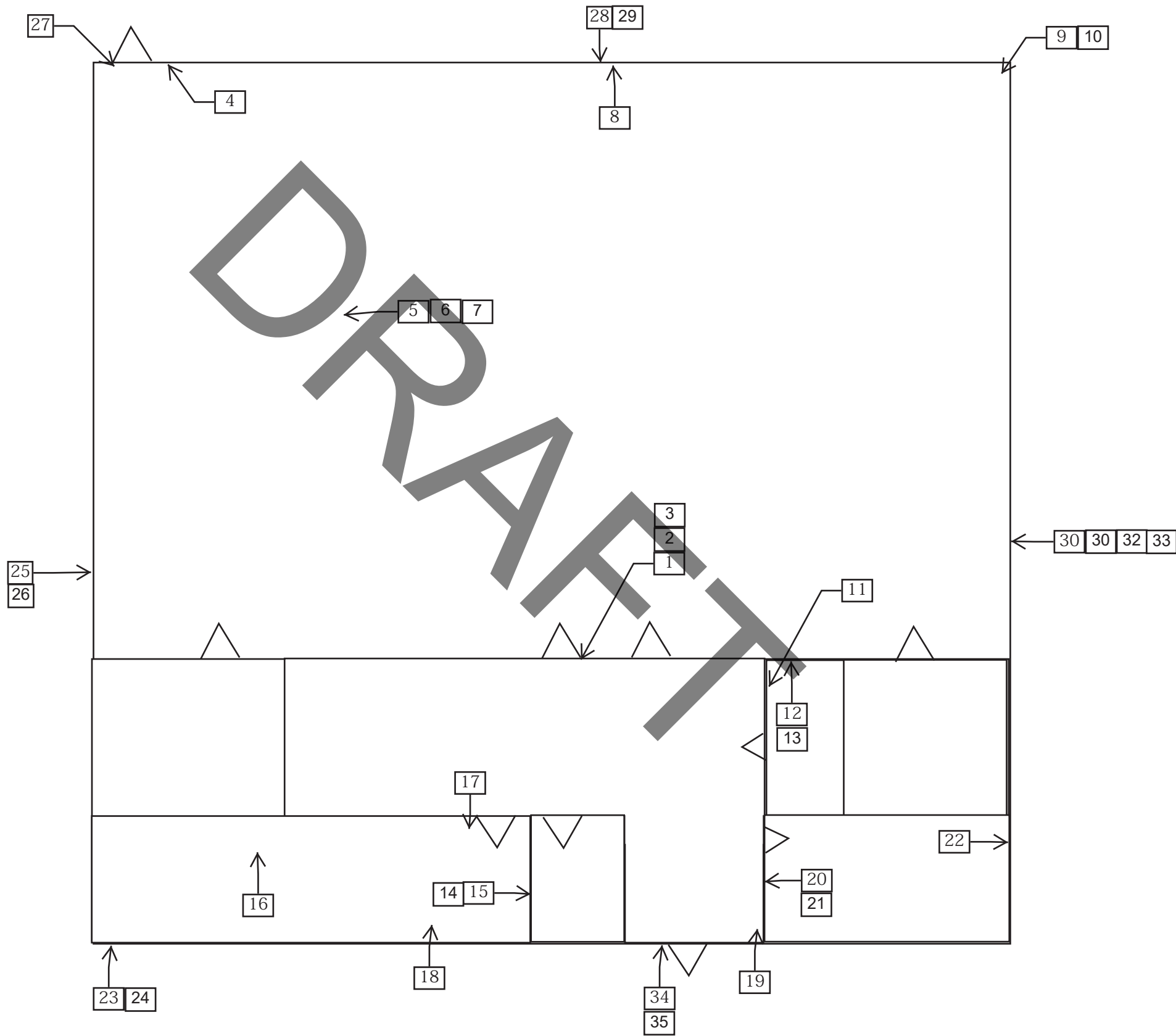
US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

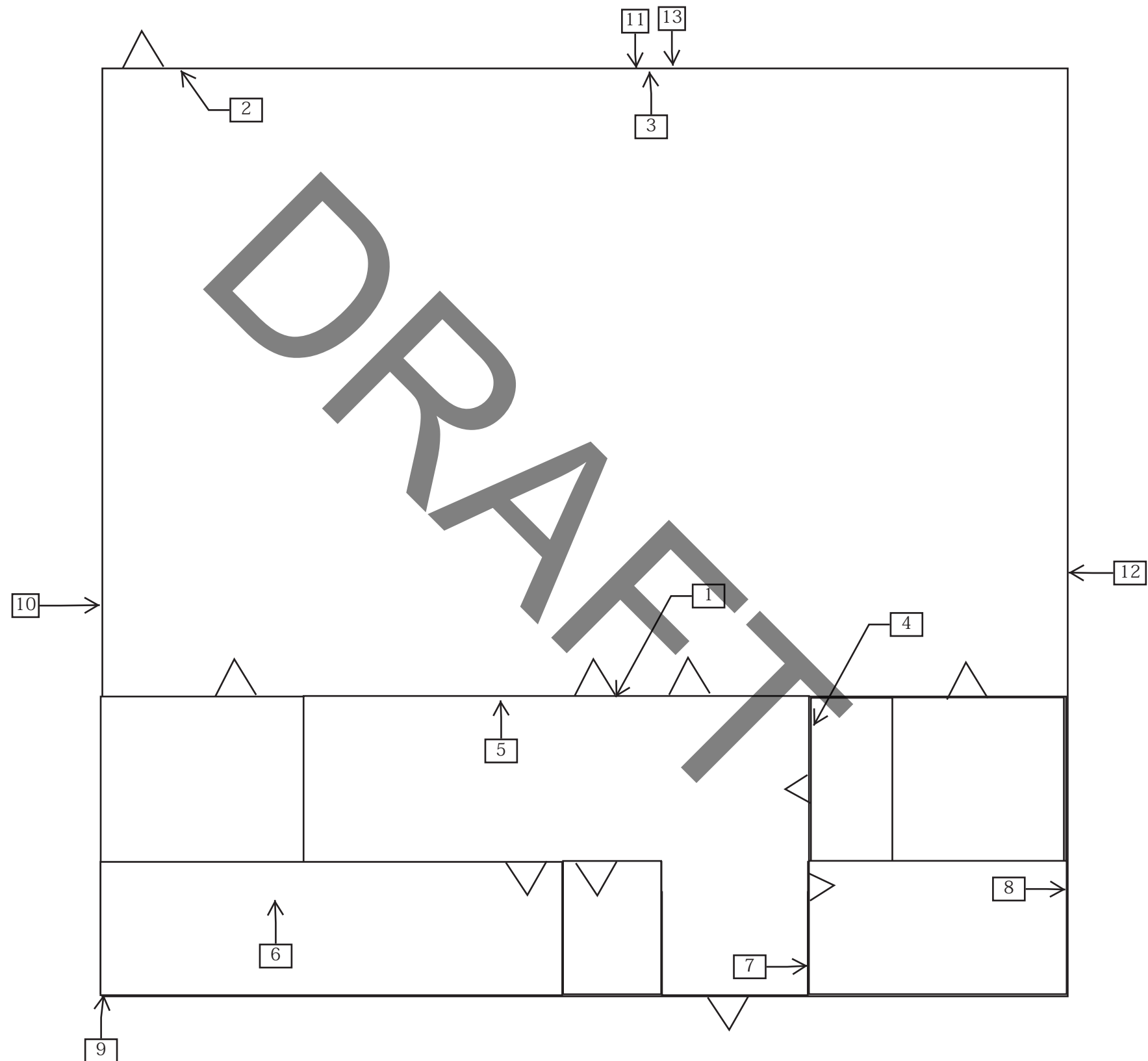
APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS



ASBESTOS SAMPLES



LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: ☐ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg. 4045		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project ☐ Metals Analysis Matrix: Method:		
Comments: See attached sample logs.		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4045-A1-A35		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.		7	A P C				

Sampled By: Sean R Dudney		Date/Time: 3/8/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS	
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:
Date / Time: 26 April 2022		Date / Time:		Date / Time:
Received By: Japan Post Office		Received By: <i>all m...</i>		Received By:
Date / Time: 26 April 2022		Date / Time: 5/26/22 14:00 USPT		Date / Time:
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S/SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4045	Classroom	H1	Vinyl floor tile	30x30cm grey w/ grey and tan mastic	N	Y	Used				KAB- 4045-A1				
		H2	Vinyl cove base	10cm light grey w/ grey and tan mastic	N	Y	Used				KAB- 4045-A2				
		H3	Concrete wall	Off-white, smooth	N	Y	Used				KAB- 4045-A3				
		H4	Gypsum board wall	Off-white, smooth	N	Y	Used				KAB- 4045-A4				
		H5	Duct TSI	Yellow fiberglass w/ silver backing	Y	N	New				KAB- 4045-A5				
		H6	Ceiling insulation	Fiberglass w/ silver backing	Y	N	Used				KAB- 4045-A6				
		H7	Ceiling tile	60x60cm off-white fissured	N	N	Used				KAB- 4045-A7				
		H8	Concrete wall coating	Above ceiling, black caoting	N	N	New				KAB- 4045-A8				
		H1	Vinyl floor tile	30x30cm grey w/ grey and tan mastic	N	Y	Used				KAB- 4045-A9				
		H2	Vinyl cove base	10cm light grey w/ grey and tan mastic	N	Y	Used				KAB- 4045- A10				
	Storage C	H9	Concrete block wall	Grey painted	N	N	Used				4045- A11				
		H10	Pipe TSI straight	Yellow fiberglass w/ silver backing	Y	N	New				4045- A12				
		H11	Pipe TSI elbow	Yellow fiberglass w/ silver backing	Y	N	New				4045- A13				
	Mens bathroom	H12	Ceramic floor tile	20x20cm grey w/ grey grout	N	Y	Used				4045- A14				
		H13	Ceramic wall tile	10x10cm off-white tile w/ white grout and grey cement	N	Y	New				KAB- 4045- A15				
		H14	Gypsum ceiling board	Off-white	N	N	New				KAB- 4045- A16				
		H15	Joint sealant (door)	White mastic	N	Y	Used				4045- A17				

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333880
Date Received: 05/26/22
Date Analyzed: 10/10/22
Date Printed: 10/11/22
First Reported: 10/11/22

Job ID/Site: ESC.7315; KAdena Air Base, Okinawa, Japan; Bldg. 4045

SGSFL Job ID: L1232-1
Total Samples Submitted: 35
Total Samples Analyzed: 35

Date(s) Collected: 03/08/2022

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4045-A1	51553647						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Layer: Light Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A2	51553648						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: White Mastic			ND				
Layer: Paint			ND				
Layer: White Joint Compound			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A3	51553649						
Layer: Grey Cementitious Material			ND				
Layer: Off-White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A4	51553650						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4045-A5	51553651						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace) Fibrous Glass (90 %)							

Client Name: Environmental Science Corporation

Report Number: B333880

Date Printed: 10/11/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4045-A6	51553652						
Layer: Light Yellow Fibrous Material			ND				
Layer: Tan Fibrous Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)	Fibrous Glass (90 %)						
KAB-4045-A7	51553653						
Layer: Off-White Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (35 %)	Fibrous Glass (45 %)						
KAB-4045-A8	51553654						
Layer: Grey Cementitious Material			ND				
Layer: Black Coating			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A9	51553655						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A10	51553656						
Layer: Grey Non-Fibrous Material			ND				
Layer: Clear Mastic			ND				
Layer: Paint			ND				
Layer: Off-White Joint Compound			ND				
Layer: Tan Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A11	51553657						
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A12	51553658						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)	Fibrous Glass (90 %)						

Client Name: Environmental Science Corporation

Report Number: B333880

Date Printed: 10/11/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4045-A13	51553659						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material with Foil			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)		Fibrous Glass (75 %)					
KAB-4045-A14	51553660						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A15	51553661						
Layer: Off-White Ceramic Tile			ND				
Layer: Grey Grout			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A16	51553662						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)		Fibrous Glass (Trace)					
KAB-4045-A17	51553663						
Layer: White Non-Fibrous Material			ND				
Layer: Yellow Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A18	51553664						
Layer: White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A19	51553665						
Layer: Off-White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A20	51553666						
Layer: Grey Tile			ND				
Layer: Tan Mastic			ND				
Layer: Light Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333880

Date Printed: 10/11/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4045-A21	51553667						
Layer: Grey Non-Fibrous Material			ND				
Layer: Clear Mastic			ND				
Layer: Paint			ND				
Layer: Off-White Skimcoat/Joint Compound			ND				
Layer: Grey Cementitious Material			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A22	51553668						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)	Fibrous Glass (Trace)						
KAB-4045-A23	51553669						
Layer: Off-White Non-Fibrous Material			ND				
Layer: Paint			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A24	51553670						
Layer: Grey Cementitious Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A25	51553671						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A26	51553672						
Layer: Off-White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4045-A27	51553673						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333880

Date Printed: 10/11/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4045-A28	51553674						
Layer: Grey Non-Fibrous Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A29	51553675						
Layer: Off-White Non-Fibrous Material			ND				
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A30	51553676						
Layer: Black Woven Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace) Fibrous Glass (75 %)							
KAB-4045-A31	51553677						
Layer: Black Foam			ND				
Layer: Tan Adhesive			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A32	51553678						
Layer: Tan/ Silver Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A33	51553679						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A34	51553680						
Layer: Black Cementitious Material			ND				
Layer: Tan Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4045-A35	51553681						
Layer: Brown Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333880

Date Printed: 10/11/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
-----------	------------	------------------	---------------------	------------------	---------------------	------------------	---------------------

DRAFT



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400-1000 / ☐ CARB 435					
Site Location: Bldg. 4045		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4045-L1-L13		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 3/8/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 26 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>Chen Shun</i>		Received By:			
Date / Time: 26 April 2022		Date / Time: 5-26-22 14:00 JPS		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S /SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4045	Class room	S1	Concrete wall	Off-white			Used				KAB- 4045-L1				
		S2	Gypsum board wall	Off-white			Used				KAB- 4045-L2				
		S3	Concrete wall (above ceiling)	Black			New				KAB- 4045-L3				
	Storage C		Concrete block wall	Grey			New				KAB- 4045-L4				
	Corridor		Concrete block wall	Grey and off-white			New				KAB- 4045-L5				
	Mens bathroom		Gypsum ceiling board	Off-white			New				KAB- 4045-L6				
	Corridor		Metal pipe (electrical)	Off-white			New				KAB- 4045-L7				
	Office		Gypsum board wall	Off-white			New				KAB- 4045-L8				
	Exterior	S4	Concrete block wall	Off-white			New				KAB- 4045-L9				
			Concrete wall base	Red			Used				KAB- 4045- L10				
		S4	Concrete block wall	Off-white			New				KAB- 4045- L11				
			Metal pipe (electrical)	Off-white			Damaged				KAB- 4045- L12				
			Plastic drain pipe	Off-white			Used				KAB- 4045- L13				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242122
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4045

SGSFL Job ID: L1232-1

Date(s) Collected: 03/08/22

Total Samples Submitted: 13

Total Samples Analyzed: 13

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4045-L1	LM220362	Pb	0.011	wt%	0.006	EPA 3050B/7000B
KAB-4045-L2	LM220363	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4045-L3	LM220364	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4045-L4	LM220365	Pb	0.008	wt%	0.006	EPA 3050B/7000B
KAB-4045-L5	LM220366	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4045-L6	LM220367	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4045-L7	LM220368	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4045-L8	LM220369	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4045-L9	LM220370	Pb	0.022	wt%	0.006	EPA 3050B/7000B
KAB-4045-L10	LM220371	Pb	< 0.009	wt%	0.009	EPA 3050B/7000B
KAB-4045-L11	LM220372	Pb	0.012	wt%	0.007	EPA 3050B/7000B
KAB-4045-L12	LM220373	Pb	0.053	wt%	0.007	EPA 3050B/7000B
KAB-4045-L13	LM220374	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

Analytical results and reports are generated by SGS Forensic Laboratories at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS Forensic Laboratories to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS Forensic Laboratories. The client is solely responsible for the use and interpretation of test results and reports requested from SGS Forensic Laboratories. SGS Forensic Laboratories is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Forensic Laboratories' Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

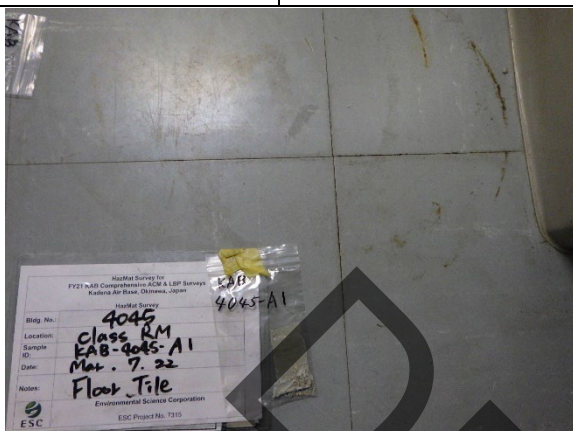
HazMat Survey

Bldg. 4045

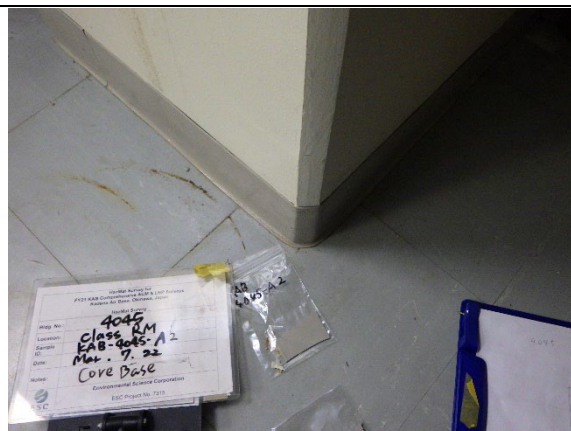


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4045



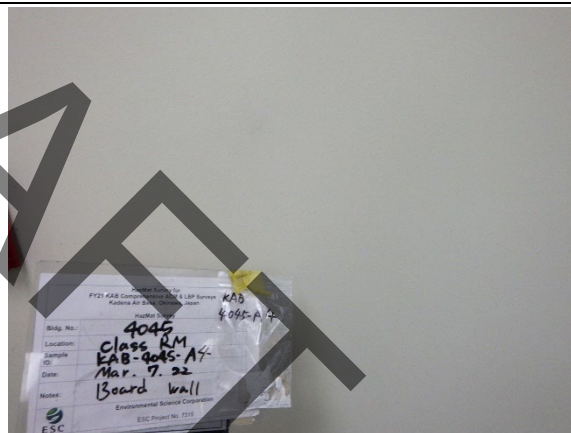
KAB-4045-A1



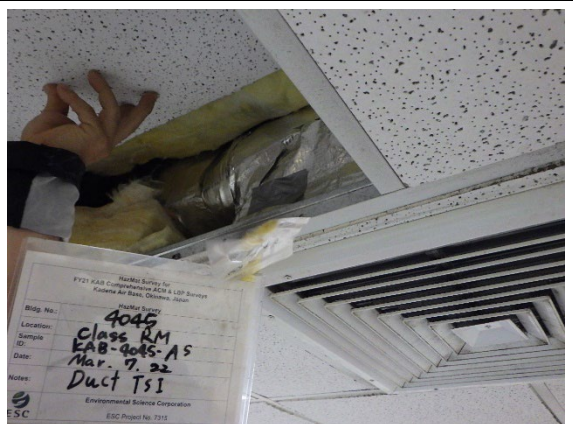
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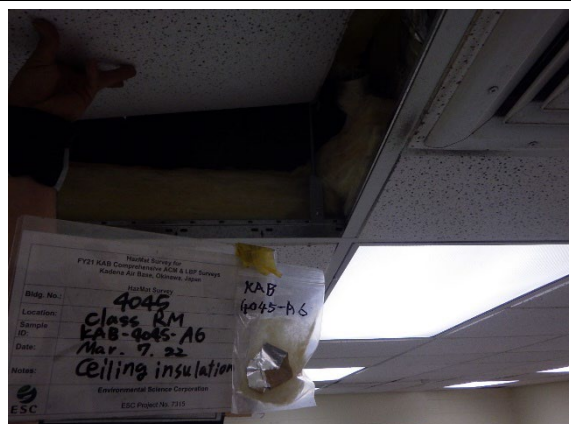
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KAB-4045-A4



KAB-4045-A5



KAB-4045-A6



Photo Record Sheet

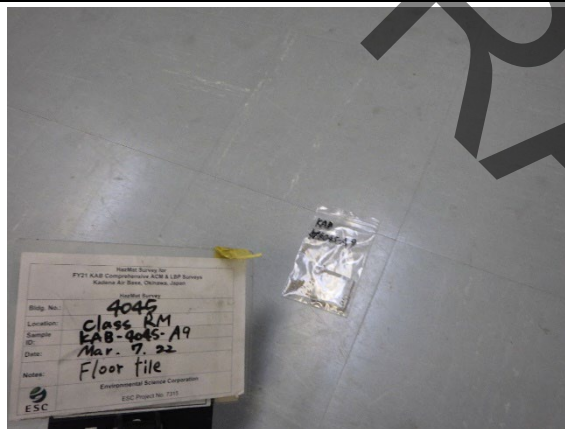
Asbestos Sampling
Kadena Air Base, Bldg.4045



KAB-4045-A7



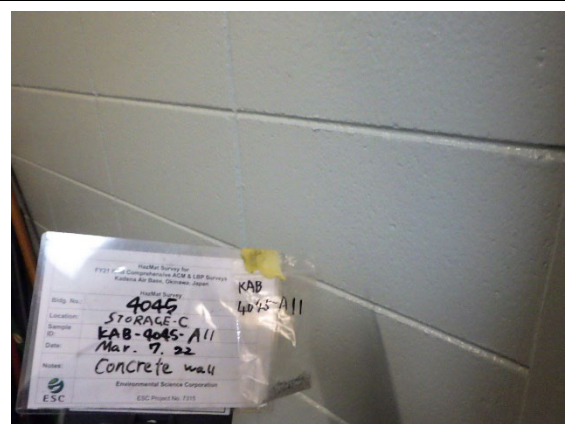
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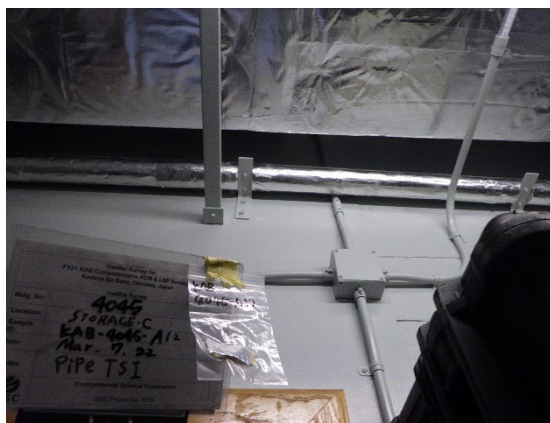
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KAB-4045-A10



KAB-4045-A11

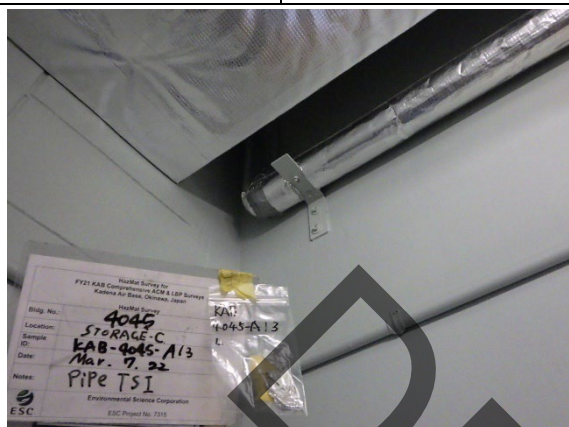


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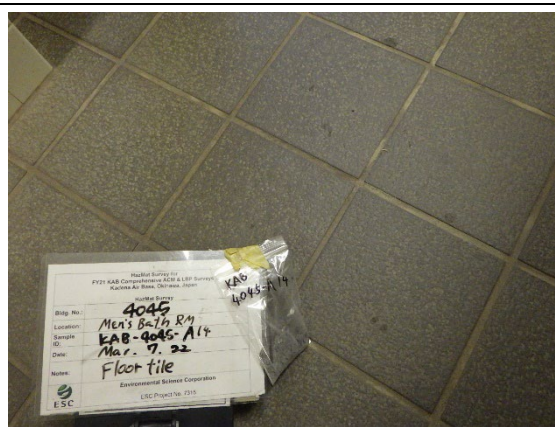


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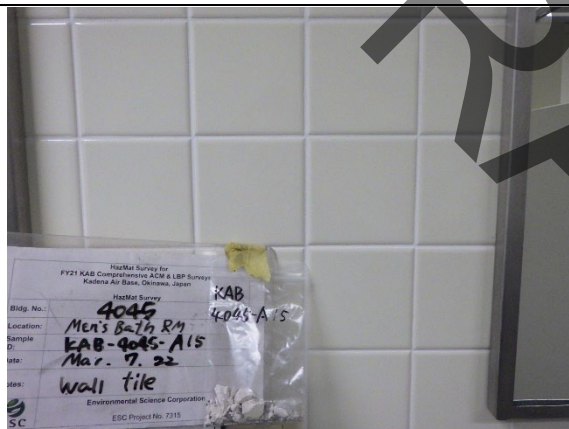
Asbestos Sampling
Kadena Air Base, Bldg.4045



KAB-4045-A13



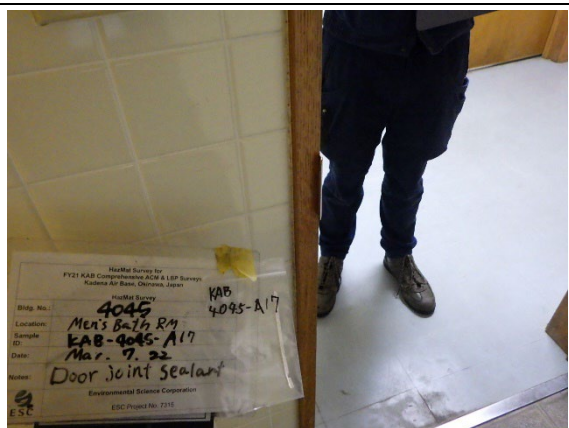
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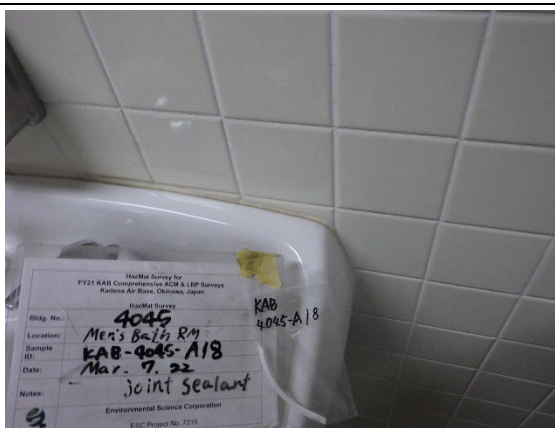
KAB-4045-A15



KAB-4045-A16



KAB-4045-A17

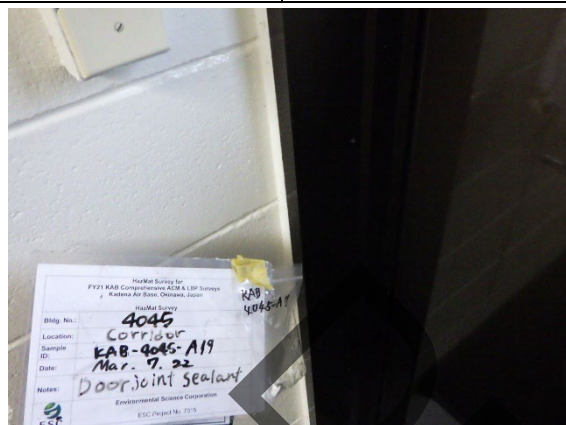


KAB-4045-A18



Photo Record Sheet

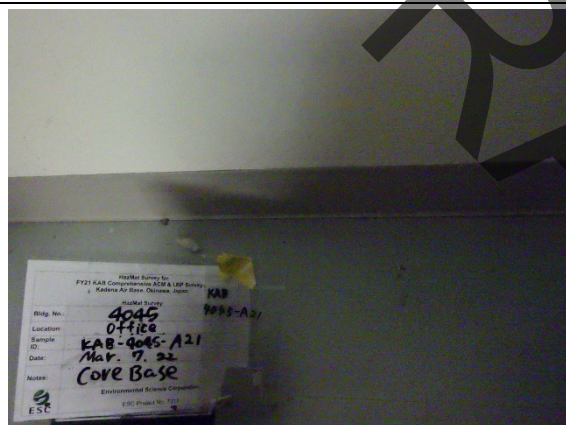
Asbestos Sampling
Kadena Air Base, Bldg.4045



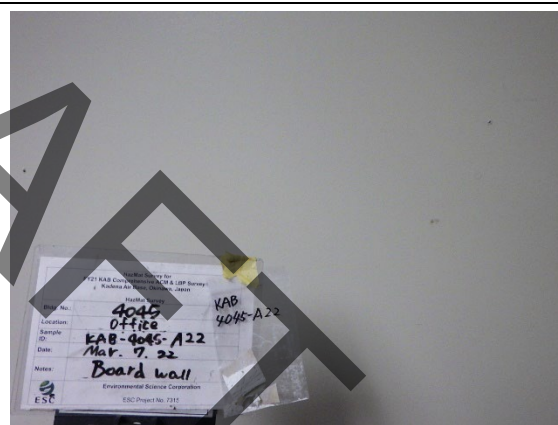
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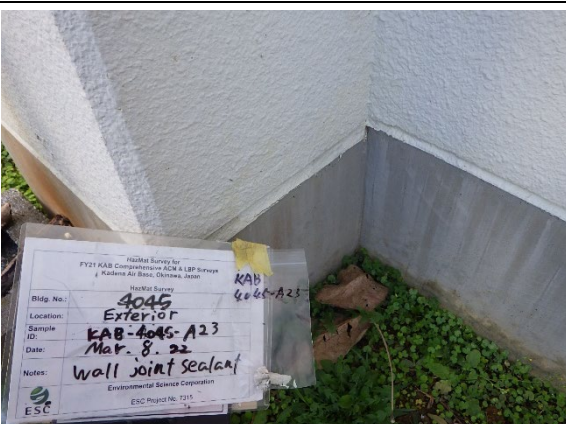
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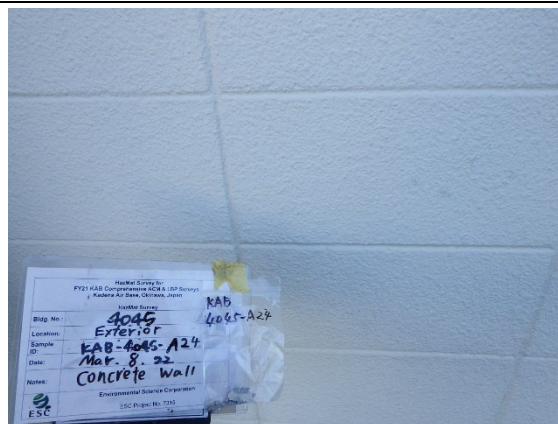
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KAB-4045-A22



KAB-4045-A23



KAB-4045-A24

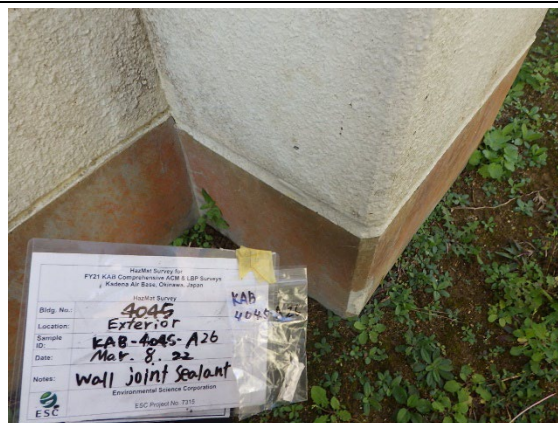


Photo Record Sheet

Asbestos Sampling
Kadena Air Base, Bldg.4045



KAB-4045-A25



KAB-4045-A26



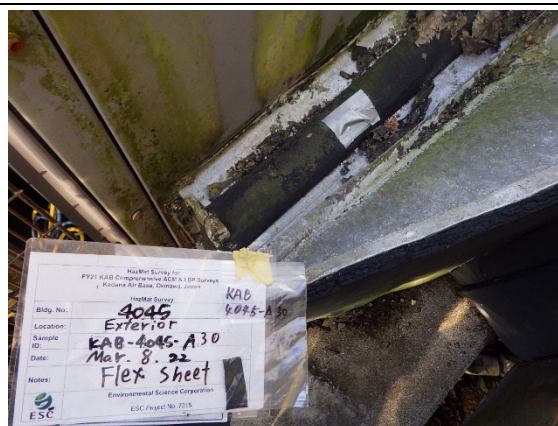
KAB-4045-A27



KAB-4045-A28



KAB-4045-A29

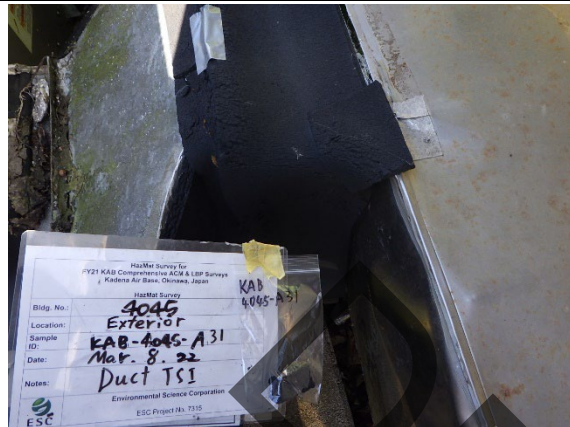


KAB-4045-A30



Photo Record Sheet

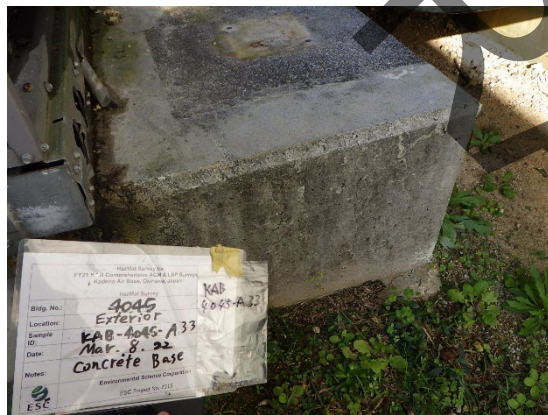
Asbestos Sampling
Kadena Air Base, Bldg.4045



KAB-4045-A31



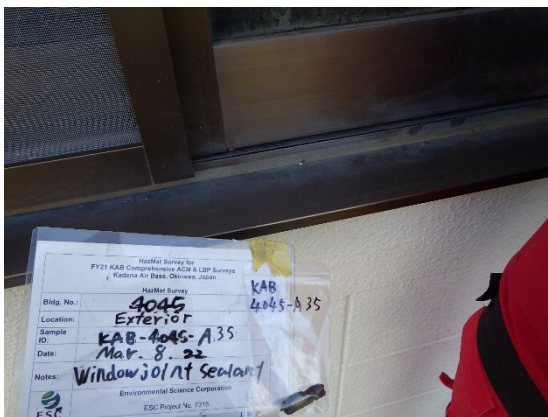
KAB-4045-A32



KAB-4045-A33



KAB-4045-A34



KAB-4045-A35

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LEAD BASED PAINT SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

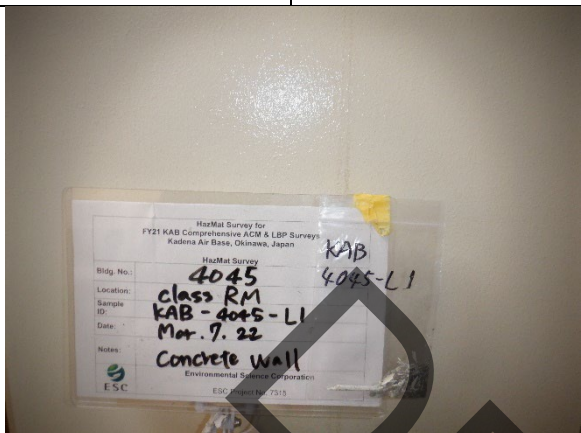
HazMat Survey

Bldg. 4045

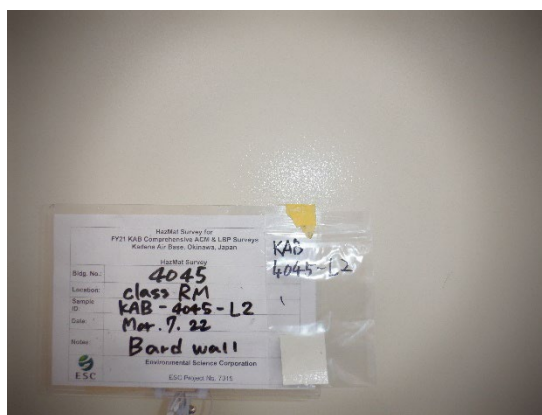


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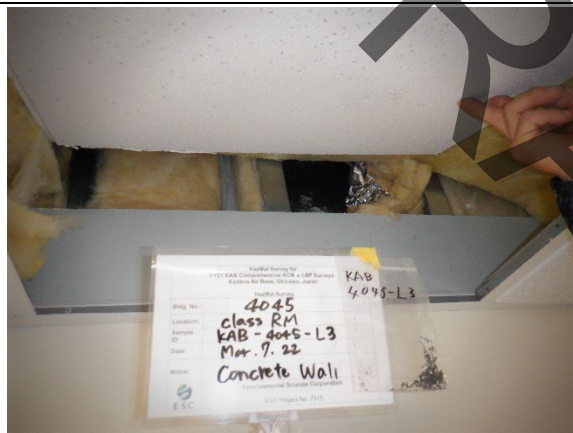
Lead Based Paint Sampling Kadena Air Base, Bldg.4045



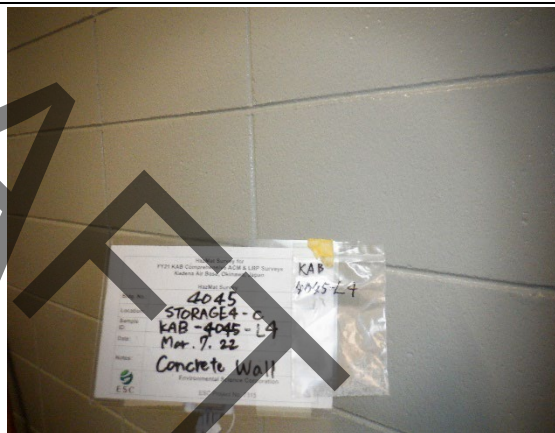
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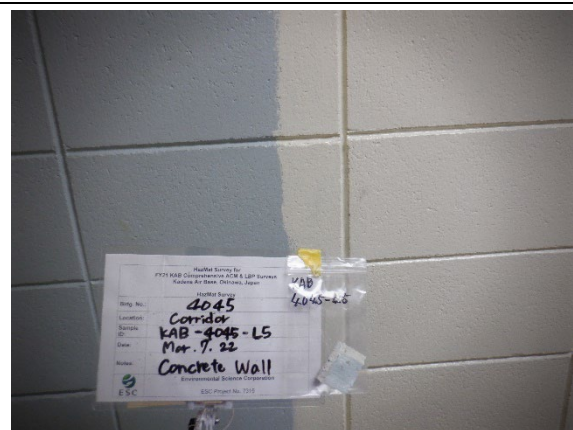
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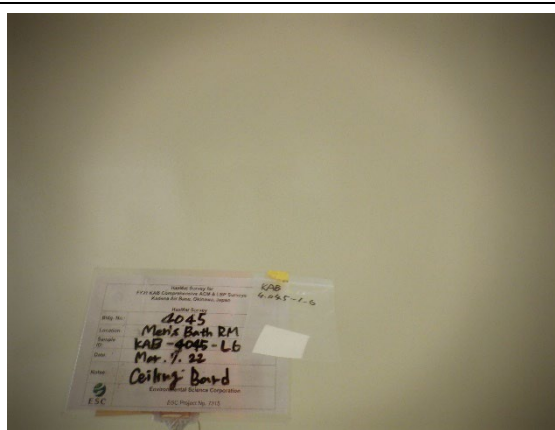
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KAB-4045-L4



KAB-4045-L5



KAB-4045-L6

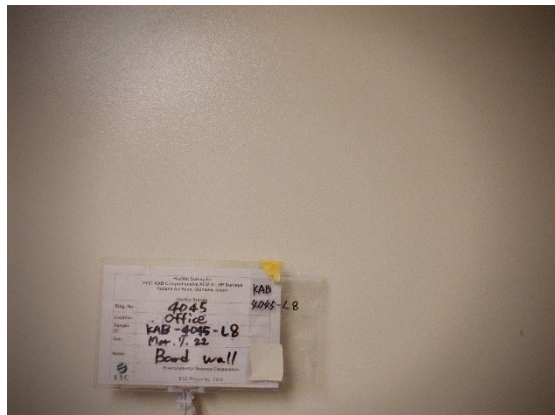


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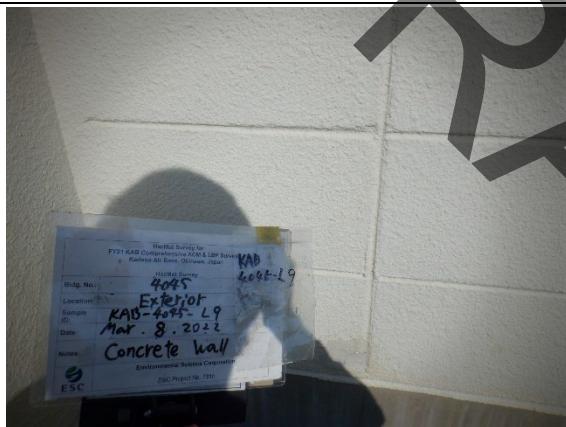
Lead Based Paint Sampling
Kadena Air Base, Bldg.4045



KAB-4045-L7



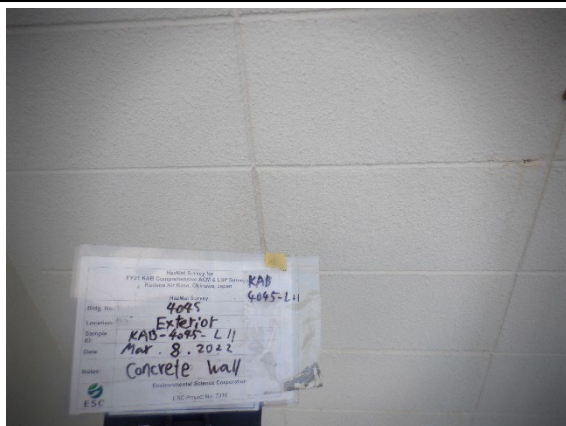
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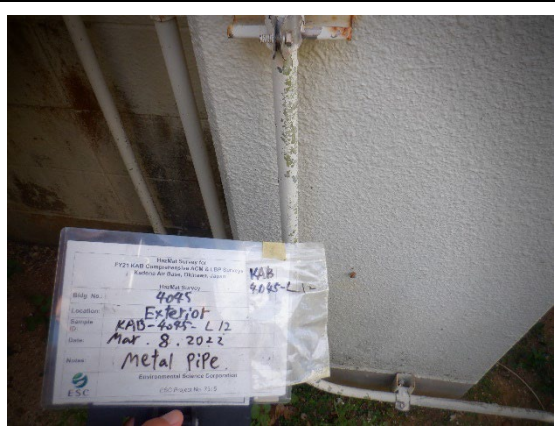
KAB-4045-L9



KAB-4045-L10



KAB-4045-L11



KAB-4045-L12

	Photo Record Sheet Lead Based Paint Sampling Kadena Air Base, Bldg.4045
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KAB-4045-L13	
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Intentionally Left Blank	Intentionally Left Blank

NOTE: The following report has been excerpted from the original version to provide the information pertinent to the execution of this project.

FY21 Comprehensive ACM & LBP Surveys
Kadena Air Base, Okinawa, Japan

Building 4046

Date: 9 December 2022
Prepared for: Kadena Air Base 718th CES/CENM
Prepared by: Environmental Science Corporation (ESC)

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2.0 ASBESTOS SAMPLING SUMMARY	1
2.1 ASBESTOS SAMPLING METHODS/ANALYSIS	2
3.0 LEAD BASED PAINT SAMPLING SUMMARY	2
3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE	2
4.0 REFERENCES	7

TABLES

TABLE 1 SUMMARY TABLE OF ASBESTOS AND LEAD PAINT SAMPLING	4
---	---

APPENDICES

A	ASBESTOS / LEAD PAINT SAMPLING LOCATIONS
B	ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
C	LBP SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS
D	PHOTOGRAPHIC RECORDS OF BULK SAMPLING
E	BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

1.0 SUMMARY

Environmental Science Corporation (ESC) conducted a hazardous material inspection at Building 4046, KAB, Okinawa, Japan. For materials suspected as possible Asbestos Containing Materials (ACM) and Lead-Based Paint (LBP), bulk sampling was conducted, and samples were sent for subsequent analysis at an accredited analytical laboratory.

The following tasks were conducted as part of the Hazardous Materials Survey:

- Evaluate Government-provided records for validity, conduct on-site sampling and analysis of materials/surfaces where Government-provided records were invalid, conduct new sampling and analysis for areas/surfaces where no records existed, report the evaluation results and new sampling laboratory analysis results in separate summary tables for Asbestos and Lead-Based Paint.

Summary data of all suspect asbestos and lead based paint samples collected from Government provided past records and/or within this survey are located in Table 1 below.

2.0 ASBESTOS SAMPLING SUMMARY

Materials assumed, reported, or confirmed to be ACM was not found at the site during this survey investigation. Out of twenty-eight (28) total asbestos samples collected for this survey, no samples were determined to be ACM.

The following additional caveats apply to this supplementary hazardous materials survey for ACM materials:

- Per the Japan Environmental Governing Standards (JEGS), any polarized light microscopy (PLM) sample result of “trace” is assumed to be $> 0.1\%$ asbestos and therefore ACM. For any “trace” sample re-analyzed using 1000-point counting and found to contain less than 0.1% asbestos, the associated homogenous area is considered non-ACM, as long as all other samples of that homogenous area are also less than 0.1% asbestos.
- The highest analytical result from all samples representative of each homogeneous area is used to recommend management of that material as a whole.
- Although not observed at the site during this survey, transite piping, used for older underground sewer and storm drain pipelines, shall be presumed ACM if encountered, unless verified non-ACM by a certified asbestos building inspector.
- Past Government provided ND records analyzed under a 1% laboratory Limit of Detection (LOD) shall also be treated as an invalid record per JEGS 2022 definition of ACM.

2.1 ASBESTOS SAMPLING METHODS/ANALYSIS

JEGS defines ACM as any material containing more than 0.1 percent asbestos by weight. All samples were provided to the laboratory accompanied with a copy of paragraph 5.8 from JEGS 2022 as analysis and reporting instructions. The laboratory was instructed to conduct 1000-point counting for all asbestos trace detections. Laboratory result reporting complies with the JEGS criteria.

3.0 LEAD BASED PAINT SAMPLING SUMMARY

The US Environmental Protection Agency (EPA) and JEGS define LBP as paint that contains equal to or greater than 0.5 percent lead by weight. Out of fourteen (14) total paint chip samples collected for this survey, no paint samples were determined to be lead-based paint.

Paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). The samples were delivered to SGS Forensic Laboratories, Carson, CA under chain of custody for analysis by EPA Method 3050B/7000B, Atomic Absorption Spectroscopy (EPA, 1996). Laboratory reports are included in Appendix C. Sample locations are shown in Appendix A.

3.1 LBP PAINT AS A GOJ SPECIALLY-CONTROLLED INDUSTRIAL WASTE

It should be noted that GOJ regulatory limit for lead as a specially-controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

Table 1 Summary Table of Asbestos and Lead Paint Sampling

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIABLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
ASBESTOS SAMPLES																
4046	102	H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Good	90 LM		Y	51553613	KAB-4046- A1	2022-10-07	ND	Valid	
4046	102	H2	Gypsum board wall	Off-white,smooth	N	Y	Good	175 SM		Y	51553614	KAB-4046- A2	2022-10-07	ND	Valid	
4046	102	H3	Concrete floor coating	Gray	Y	Y	Good	220 SM		Y	51553615	KAB-4046- A3	2022-10-07	ND	Valid	
4046	102	H4	Floor joint sealant	Gray mastic	N	Y	Good	N/A		Y	51553616	KAB-4046- A4	2022-10-07	ND	Valid	
4046	102	H5	Concrete block wall Coating	Off-white,smooth	N	Y	Good	175 SM		Y	51553617	KAB-4046- A5	2022-10-07	ND	Valid	
4046	102	H6	Door joint sealant	White mastic	N	Y	Good	15 LM		Y	51553618	KAB-4046- A6	2022-10-07	ND	Valid	
4046	101	H7	Pipe TSI(elbow)	Yellow fiberglass w/silver backing	N	N	Good	N/A		Y	51553619	KAB-4046- A7	2022-10-07	ND	Valid	
4046	101	H8	Pipe TSI	Yellow fiberglass w/silver backing	N	N	Good	N/A		Y	51553620	KAB-4046- A8	2022-10-07	ND	Valid	
4046	101	H9	ACU insulation	Black foam w/tan mastic	N	N	Good	N/A		Y	51553621	KAB-4046- A9	2022-10-07	ND	Valid	
4046	101	H3	Concrete floor coating	Gray	Y	Y	Good	220 SM		Y	51553622	KAB-4046- A10	2022-10-07	ND	Valid	
4046	101	H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Good	90 LM		Y	51553623	KAB-4046- A11	2022-10-07	ND	Valid	
4046	101	H2	Gypsum board wall	Off-white,smooth	N	Y	Good	175 SM		Y	51553624	KAB-4046- A12	2022-10-07	ND	Valid	
4046	101	H5	Concrete block wall Coating	Off-white,smooth	N	Y	Good	175 SM		Y	51553625	KAB-4046- A13	2022-10-07	ND	Valid	
4046	101	H4	Floor joint sealant	Gray mastic	N	Y	Good	N/A		Y	51553626	KAB-4046- A14	2022-10-07	ND	Valid	
4046	104	H4	Floor joint sealant	Gray mastic	N	Y	Good	N/A		Y	51553627	KAB-4046- A15	2022-10-07	ND	Valid	
4046	104	H3	Concrete floor coating	Gray	Y	Y	Good	220 SM		Y	51553628	KAB-4046- A16	2022-10-07	ND	Valid	
4046	104	H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Good	90 LM		Y	51553629	KAB-4046- A17	2022-10-07	ND	Valid	
4046	Toilet	H10	Ceramic wall tile	10x10 pale purple tile w/gray grout	Y	Y	Good	15 SM		Y	51553630	KAB-4046- A18	2022-10-07	ND	Valid	
4046	Toilet	H11	Ceramic floor tile	10x10 gray tile w/gray grout	Y	Y	Good	6 SM		Y	51553631	KAB-4046- A19	2022-10-07	ND	Valid	
4046	Toilet	H12	Sink joint sealant	White mastic	N	Y	Good	6 LM		Y	51553632	KAB-4046- A20	2022-10-07	ND	Valid	
4046	101	H13	Concrete wall Coating	Off-white,smooth	Y	Y	Good	20 SM		Y	51553633	KAB-4046- A21	2022-10-07	ND	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIABLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
4046	Toilet	H13	Concrete wall Coating	Off-white,smooth	Y	Y	Good	15 SM		Y	51553634	KAB-4046- A22	2022-10-07	ND	Valid	
4046	Exterior	H14	Floor joint sealant	Gray mastic	N	Y	Good	N/A		Y	51553635	KAB-4046- A23	2022-10-07	ND	Valid	
4046	Exterior	H15	Concrete wall Coating	Off-white,textured	Y	Y	Good	245 SM		Y	51553636	KAB-4046- A24	2022-10-07	ND	Valid	
4046	Exterior	H15	Concrete wall Coating	Off-white,textured	Y	Y	Good	245 SM		Y	51553637	KAB-4046- A25	2022-10-07	ND	Valid	
4046	Exterior	H16	Wall joint sealant	White mastic	N	Y	Good	60 LM		Y	51553638	KAB-4046- A26	2022-10-07	ND	Valid	
4046	Exterior	H15	Concrete wall Coating	Off-white,textured	Y	Y	Good	245 SM		Y	51553639	KAB-4046- A27	2022-10-07	ND	Valid	
4046	Exterior	H16	Wall joint sealant	White mastic	N	Y	Good	60 LM		Y	51553640	KAB-4046- A28	2022-10-07	ND	Valid	
LEAD PAINT SMAPLES																
4046	102	S1	Gypsum board wall	Off-white		Y	Used			Y	LM220565	KAB-4046- L1	2022-06-22	<0.006	Valid	
4046	102	S2	Metal door frame	Light brown		Y	Used			Y	LM220566	KAB-4046- L2	2022-06-22	0.017	Valid	
4046	101	S3	Metal pipe	Red		N	Used			Y	LM220567	KAB-4046- L3	2022-06-22	<0.02	Valid	
4046	101	S4	Metal pipe	Off-white		N	Used			Y	LM220568	KAB-4046- L4	2022-06-22	<0.02	Valid	
4046	101	S1	Gypsum board wall	Off-white		Y	Used			Y	LM220569	KAB-4046- L5	2022-06-22	0.014	Valid	
4046	101	S5	Concrete floor coating	Red		Y	Used			Y	LM220570	KAB-4046- L6	2022-06-22	0.009	Valid	
4046	104	S4	Metal pipe	Off-white		N	Used			Y	LM220571	KAB-4046- L7	2022-06-22	<0.01	Valid	
4046	Exterior	S6	Metal door frame	Off-white		Y	Used			Y	LM220572	KAB-4046- L8	2022-06-22	<0.006	Valid	
4046	Exterior	S7	Telephone box	Yellow		Y	Used			Y	LM220573	KAB-4046- L9	2022-06-22	<0.007	Valid	
4046	Exterior	S8	Concrete wall	Off-white		Y	Used			Y	LM220574	KAB-4046- L10	2022-06-22	<0.007	Valid	
4046	Exterior	S9	Metal pipe	Red over white		Y	Used			Y	LM220575	KAB-4046- L11	2022-06-22	<0.02	Valid	

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
BLDG NUMBER	ROOM OR FUNCTIONAL AREA	HOMOGENEOUS /SURFACE NUMBER	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIABLE	ACCESS	CONDITION	HA QTY	GOV T	C T R	LAB/ SAMPLE ID NO.	CTR COC SAMPLE ID NO.	LAB ANALYSIS DATE	RESULT	VALIDITY DETERMINATION	NOTES / INVALID OR UNKNOWN REASONING
4046	Exterior	S10	Metal flange	Ligth blue		Y	Used			Y	LM220576	KAB-4046- L12	2022-06-22	<0.03	Valid	
4046	Exterior	S8	Concrete wall	Off-white, textured		Y	Used			Y	LM220577	KAB-4046- L13	2022-06-22	<0.007	Valid	
4046	Exterior	S8	Concrete wall	Off-white, textured		Y	Used			Y	LM220578	KAB-4046- L14	2022-06-22	<0.006	Valid	

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4.0 REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.*

Government of Japan, 2001. *Japan Waste Management and Public Cleansing Law of 1970, updated 2001.*

United States Code 40 CFR Part 745. *Lead-Based Paint Poisoning Prevention in Certain Residential Structures.*

US EPA, 1996. *Method 3050B/Method No. 7420, Atomic Absorption Spectroscopy.*

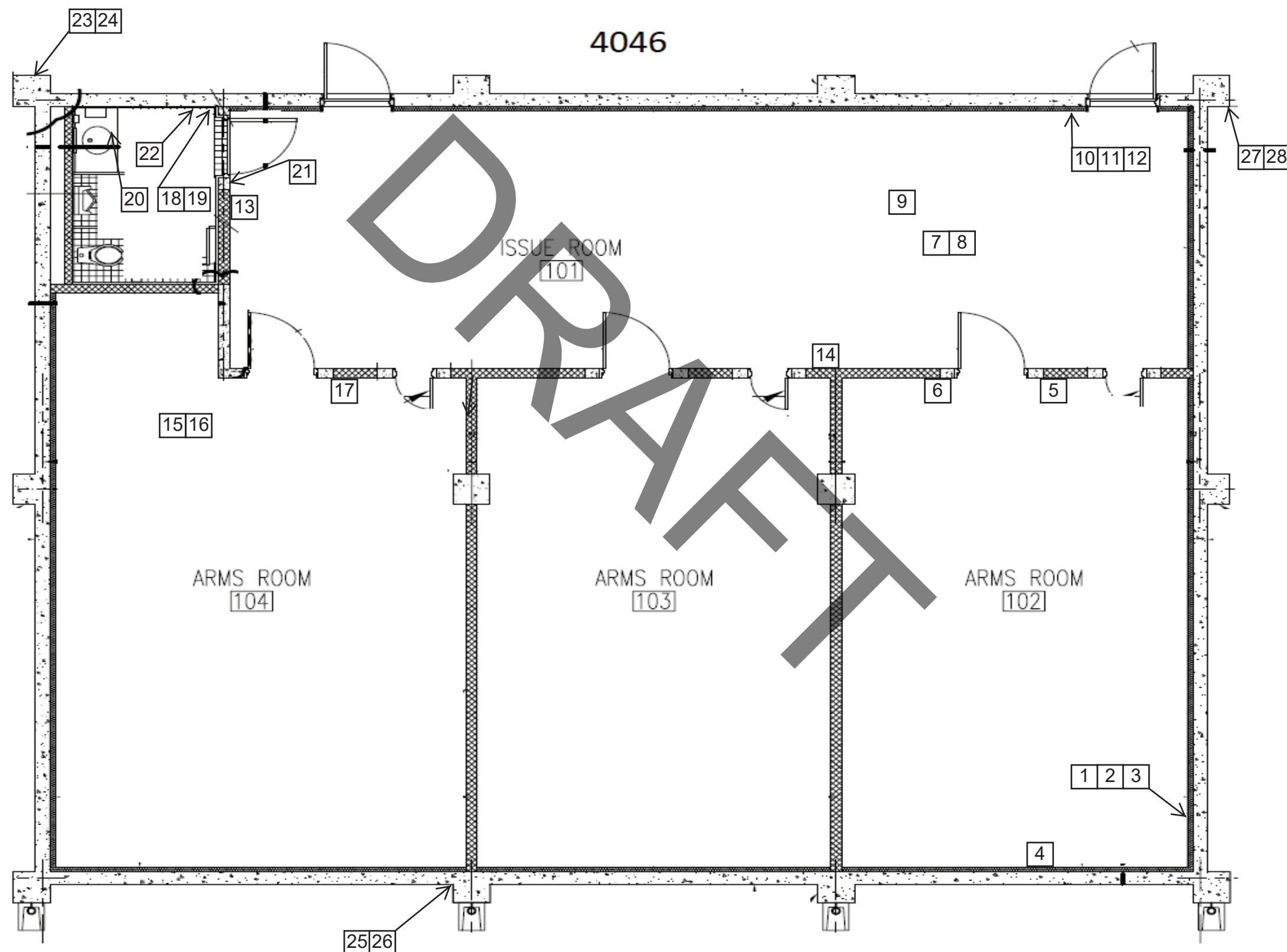
US EPA, 1993. *Method for the Determination of Asbestos in Bulk Building Materials*

USFJ, 2022. *Japan Environmental Governing Standards (JEGS).* April 2022.

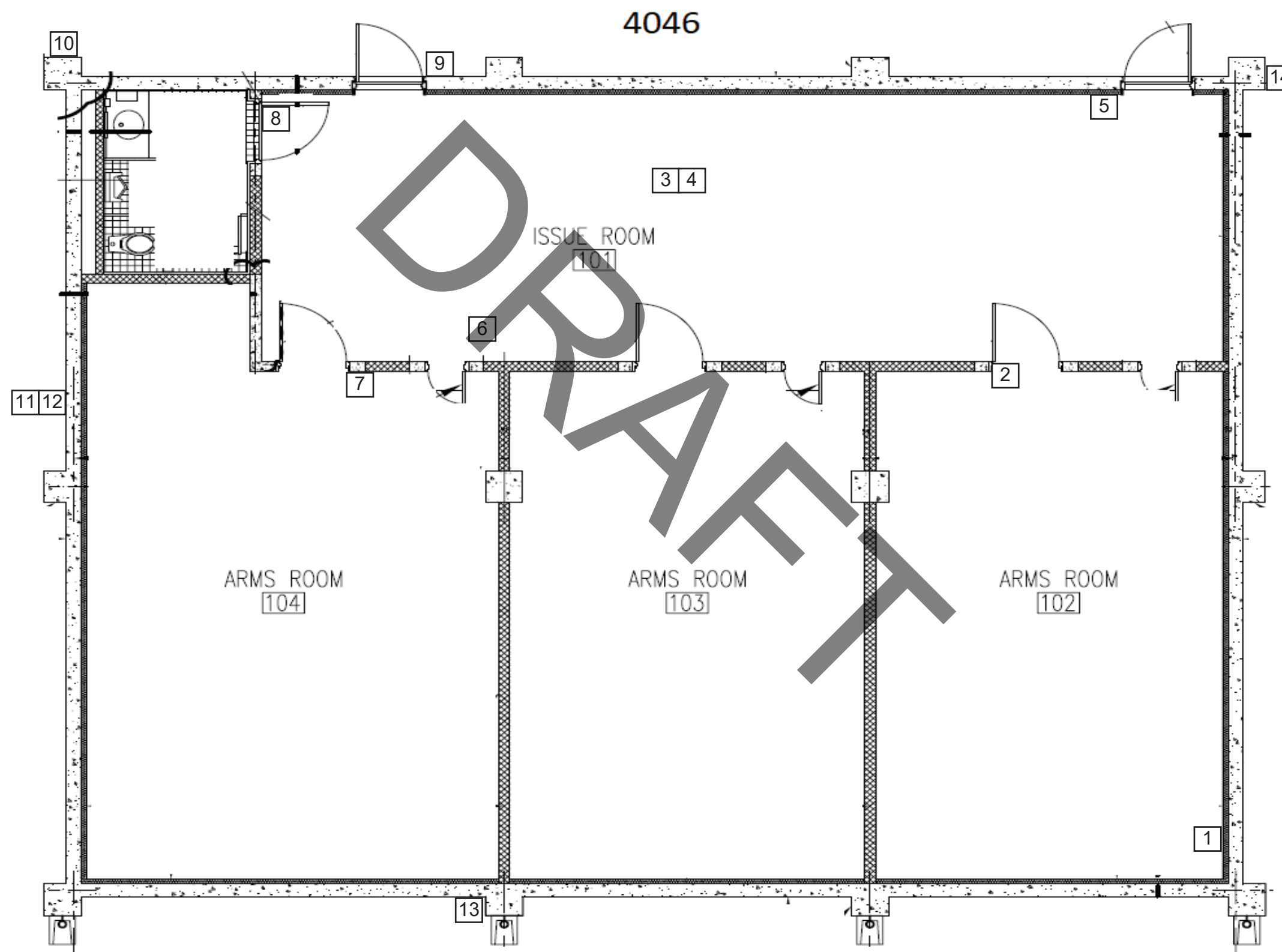
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APPENDIX A
ASBESTOS / LEAD PAINT SAMPLING LOCATIONS

ASBESTOS SAMPLES



LEAD PAINT SAMPLES



APPENDIX B
ACM SAMPLING CHAINS-OF-CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: ☐ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day	
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☒ PLM: ☒ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435		
Site Name: Kadena Air Base, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Bldg.4046		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project		
Comments: See attached sample logs.		☐ Metals Analysis Matrix: Method: Analytes:		
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4046-A1-A28		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				

Sampled By: Sean R Dudney		Date/Time: 4/20/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS	
Relinquished By: Sean R Dudney		Relinquished By:	Relinquished By:	
Date / Time: 26 April 2022		Date / Time:	Date / Time:	
Received By: Japan Post Office		Received By: <i>Chie M...</i>	Received By:	
Date / Time: 26 April 2022		Date / Time: 5/26/22 14:00 USPT	Date / Time:	
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No	Condition Acceptable? ☐ Yes ☐ No	

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
 Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE					ANALYSIS RECORD DATA										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMB	ROOM OR FUNCTION	HOMOGENEOU S/SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	TEST P C S	TEST P C S	CONDITIO N	G C O T	L A B/ S A M P L E I D	C T R	L A B A N A L Y S I S	R E S U L T	V A L I D I T Y D E T E R M I N A T I	N O T E S / I N V A L I D O R U N K N O W N R E A S O N I N G	
4046	102	H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Use				KAB- 4046-A1				
4046		H2	Gypsum board wall	Off-white, smooth	N	Y	Used				KAB- 4046-A2				
4046		H3	Concrete floor coating	Gray	Y	Y	Used				KAB- 4046-A3				
4046		H4	Floor joint sealant	Gray mastic	N	Y	Used				KAB- 4046-A4				
4046		H5	Concrete block wall	Off-white, smooth	N	Y	Use				KAB- 4046-A5				
4046		H6	Door joint sealant	White mastic	N	Y	Used				KAB- 4046-A6				
4046	101	H7	Pipe TSI(elbow)	Yellow fiberglass w/silver backing	N	N	Use				KAB- 4046-A7				
4046		H8	Pipe TSI	Yellow fiberglass w/silver backing	N	N	Use				KAB- 4046-A8				
4046		H9	ACU insulation	Black foam w/tan mastic	N	N	Use				KAB- 4046-A9				
4046		H3	Concrete floor coating	Gray	Y	Y	Used				KAB- 4046-A10				
4046		H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Use				KAB- 4046-A11				
4046		H2	Gypsum board wall	Off-white, smooth	N	Y	Used				KAB- 4046-A12				
4046		H5	Concrete block wall	Off-white, smooth	N	Y	Use				KAB- 4046-A13				
4046		H4	Floor joint sealant	Gray mastic	N	Y	Used				KAB- 4046-A14				
4046	104	H4	Floor joint sealant	Gray mastic	N	Y	Used				KAB- 4046-A15				
4046		H3	Concrete floor coating	Gray	Y	Y	Used				KAB- 4046-A16				
4046		H1	Vinyl cove base	10cm gray tile w/off white mastic	N	Y	Use				KAB- 4046-A17				
4046	Toilet	H10	Ceramic wall tile	10x10 pale purple tile w/gray grout	Y	Y	Use				KAB- 4046-A18				
4046		H11	Ceramic floor tile	10x10 gray tile w/gray grout	Y	Y	Use				KAB- 4046-A19				
4046		H12	Sink joint sealant	White mastic	N	Y	Used				KAB- 4046-A20				
4046	101	H13	Concrete wall	Off-white, smooth	Y	Y	Used				KAB- 4046-A21				
4046	Toilet	H13	Concrete wall	Off-white, smooth	Y	Y	Used				KAB- 4046-A22				
4046	Exterior	H14	Floor joint sealant	Gray mastic	N	Y	Used				KAB- 4046-A23				
4046		H15	Concrete wall	Off-white, textured	Y	Y	Used				KAB- 4046-A24				
4046		H15	Concrete wall	Off-white, textured	Y	Y	Used				KAB- 4046-A25				
4046		H16	Wall joint sealant	White mastic	N	Y	Used				KAB- 4046-A26				
4046		H15	Concrete wall	Off-white, textured	Y	Y	Used				KAB- 4046-A27				
4046		H16	Wall joint sealant	White mastic	N	Y	Used				KAB- 4046-A28				

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
NVLAP Lab Code: 101459-1

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: B333879
Date Received: 05/26/22
Date Analyzed: 10/07/22
Date Printed: 10/07/22
First Reported: 10/07/22

Job ID/Site: ESC.7315; Kadena air Base, Okinawa, Japan; Bldg. 4046

SGSFL Job ID: L1232-1
Total Samples Submitted: 28
Total Samples Analyzed: 28

Date(s) Collected: 04/20/2022

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4046-A1	51553613						
Layer: Grey Non-Fibrous Material			ND				
Layer: Paint			ND				
Layer: White Joint Compound			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A2	51553614						
Layer: White Drywall			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %) Fibrous Glass (Trace)							
KAB-4046-A3	51553615						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A4	51553616						
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A5	51553617						
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A6	51553618						
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation**Report Number:** B333879**Date Printed:** 10/07/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4046-A7	51553619						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)		Fibrous Glass (80 %)					
KAB-4046-A8	51553620						
Layer: Yellow Fibrous Material			ND				
Layer: White Woven Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)		Fibrous Glass (80 %)					
KAB-4046-A9	51553621						
Layer: Black Foam			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A10	51553622						
Layer: Grey Cementitious Material			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A11	51553623						
Layer: Grey Non-Fibrous Material			ND				
Layer: Off-White Mastic			ND				
Layer: Paint			ND				
Layer: White Skimcoat/Joint Compound			ND				
Layer: White Drywall			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A12	51553624						
Layer: White Drywall			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (20 %)		Fibrous Glass (Trace)					
KAB-4046-A13	51553625						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
KAB-4046-A14	51553626						
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333879

Date Printed: 10/07/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4046-A15	51553627						
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A16	51553628						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A17	51553629						
Layer: Grey Non-Fibrous Material			ND				
Layer: Tan Mastic			ND				
Layer: Paint			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A18	51553630						
Layer: Grey Ceramic Tile			ND				
Layer: Grey Mortar			ND				
Layer: Dark Grey Grout			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A19	51553631						
Layer: Grey Ceramic Tile			ND				
Layer: Dark Grey Grout			ND				
Layer: Tan Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A20	51553632						
Layer: Off-White Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A21	51553633						
Layer: Light Grey Plaster			ND				
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A22	51553634						
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B333879

Date Printed: 10/07/22

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
KAB-4046-A23	51553635						
Layer: Grey Non-Fibrous Material			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A24	51553636						
Layer: Grey Cementitious Material			ND				
Layer: White Joint Compound			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A25	51553637						
Layer: Light Grey Cementitious Material			ND				
Layer: Grey Cementitious Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A26	51553638						
Layer: White Non-Fibrous Material			ND				
Layer: Grey Foam			ND				
Layer: Tan Mastic			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A27	51553639						
Layer: Grey Cementitious Material			ND				
Layer: White Non-Fibrous Material			ND				
Layer: Paint			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							
KAB-4046-A28	51553640						
Layer: White Non-Fibrous Material with Debris			ND				
Layer: Grey Foam			ND				
Total Composite Values of Non-Asbestos Fibrous Components: Cellulose (Trace)							



Tiffani Ludd, Laboratory Supervisor, Carson Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

APPENDIX C
LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation - Okinawa ESC Building 1553-231 Ishikawa Iha, Uruma City, Okinawa 904-1115, Japan		Client No.: L1232	PO / Job#: ESC. 7315	Date: 26 April 2022			
Contact: Sean R Dudney		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day				
E-mail: sean.dudney@kankyoscience.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: Kadena Air Base, Okinawa, Japan		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Bldg. 4046		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments: See attached sample logs.		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project					
		☒ Metals Analysis Matrix: Paint Chip Method: Analytes: Lead					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. KAB-4046-L1-L14		See attached sample logs (Excel Sheet)	A P C				
2.			A P C				
3.			A P C				
4.			A P C				
5.			A P C				
6.			A P C				
7.			A P C				
8.			A P C				
9.			A P C				
10.			A P C				
Sampled By: Sean R Dudney		Date/Time: 4/20/22	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP EMS				
Relinquished By: Sean R Dudney		Relinquished By:		Relinquished By:			
Date / Time: 26 April 2022		Date / Time:		Date / Time:			
Received By: Japan Post Office		Received By: <i>Allen M...</i>		Received By:			
Date / Time: 26 April 2022		Date / Time: 5-26-22 14:00 USPT		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

HOMOGENEOUS MATERIAL OR SURFACE								ANALYSIS RECORD DATA							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BLDG NUMBE	ROOM OR FUNCTIONA	HOMOGENEOU S /SURFACE	MATERIAL NAME /APPLICATION	DESCRIPTION (Size, color, other)	FRIA	ACC	CONDITIO N	G O	C T	LAB/ SAMPLE ID	CTR COC	LAB ANALYSIS	RESULT	VALIDITY DETERMINATI	NOTES / INVALID OR UNKNOWN REASONING
4046	102	S1	Gypsum board wall	Off-white		Y	Use				KAB- 4046-L1				
4046		S2	Metal door frame	Light brown		Y	Used				KAB- 4046-L2				
4046	101	S3	Metal pipe	Red		N	Use				KAB- 4046-L3				
4046		S4	Metal pipe	Off-white		N	Use				KAB- 4046-L4				
4046		S1	Gypsum board wall	Off-white		Y	Use				KAB- 4046-L5				
4046		S5	Concrete floor coating	Red		Y	Used				KAB- 4046-L6				
4046	104	S4	Metal pipe	Off-white		N	Use				KAB- 4046-L7				
4046	Exterior	S6	Metal door frame	Off-white		Y	Used				KAB- 4046-L8				
4046		S7	Telephone box	Yellow		Y	Used				KAB- 4046-L9				
4046		S8	Concrete wall	Off-white		Y	Used				KAB- 4046- L10				
4046		S9	Metal pipe	Red over white		Y	Used				KAB- 4046- L11				
4046		S10	Metal flange	Ligth blue		Y	Used				KAB- 4046- L12				
4046		S8	Concrete wall	Off-white, textured		Y	Used				KAB- 4046- L13				
4046		S8	Concrete wall	Off-white, textured		Y	Used				KAB- 4046- L14				

Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Environmental Science Corporation
Sean Dudley
Itsuro Building 4F
3-7-2 Kanda-Nishikicho
Chiyoda-ku, Tokyo, JP 101-0054

Client ID: L1232
Report Number: M242185
Date Received: 05/26/22
Date Analyzed: 06/22/22
Date Printed: 06/22/22
First Reported: 06/22/22

Job ID / Site: ESC.7315; KAB Comprehensive Hazmat, Kadena Air Base, Okinawa, Japan,
Bldg. 4046

SGSFL Job ID: L1232-1

Date(s) Collected: 04/20/22

Total Samples Submitted: 14

Total Samples Analyzed: 14

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
KAB-4046-L1	LM220565	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4046-L2	LM220566	Pb	0.017	wt%	0.009	EPA 3050B/7000B
KAB-4046-L3	LM220567	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4046-L4	LM220568	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4046-L5	LM220569	Pb	0.014	wt%	0.007	EPA 3050B/7000B
KAB-4046-L6	LM220570	Pb	0.009	wt%	0.006	EPA 3050B/7000B
KAB-4046-L7	LM220571	Pb	< 0.01	wt%	0.01	EPA 3050B/7000B
KAB-4046-L8	LM220572	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B
KAB-4046-L9	LM220573	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4046-L10	LM220574	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4046-L11	LM220575	Pb	< 0.02	wt%	0.02	EPA 3050B/7000B
KAB-4046-L12	LM220576	Pb	< 0.03	wt%	0.03	EPA 3050B/7000B
KAB-4046-L13	LM220577	Pb	< 0.007	wt%	0.007	EPA 3050B/7000B
KAB-4046-L14	LM220578	Pb	< 0.006	wt%	0.006	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.



Kevin Poon, Laboratory Analyst, Hayward Laboratory

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Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

ASBESTOS SAMPLING PHOTOS

FY21 KAB Comprehensive ACM & LBP Surveys,

Kadena Air Base, Okinawa, Japan

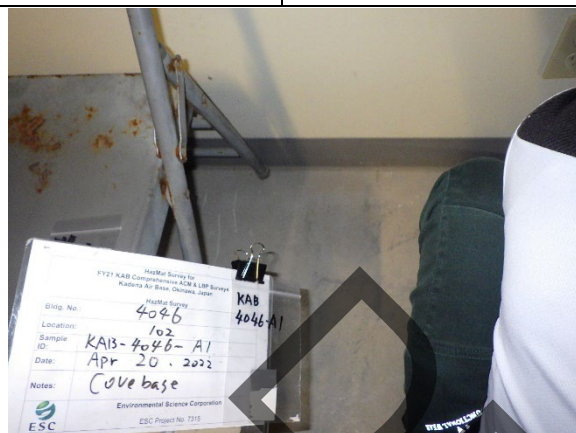
HazMat Survey

Bldg. 4046

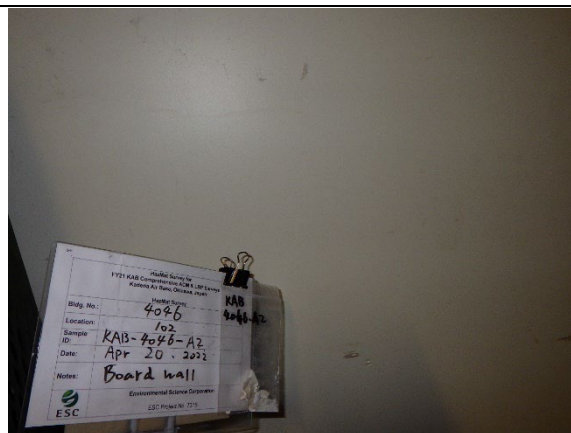


Photo Record Sheet

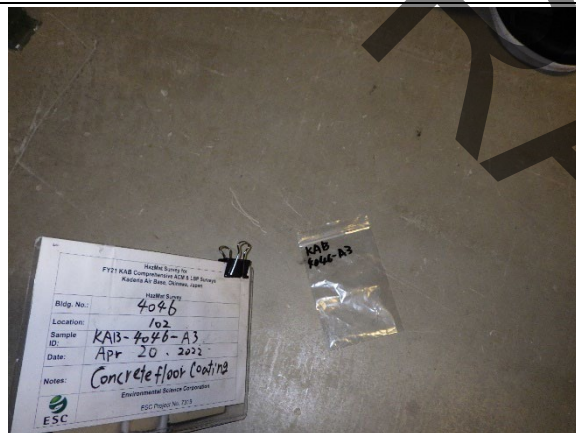
Asbestos Sampling
Kadena Air Base, Bldg. 4046



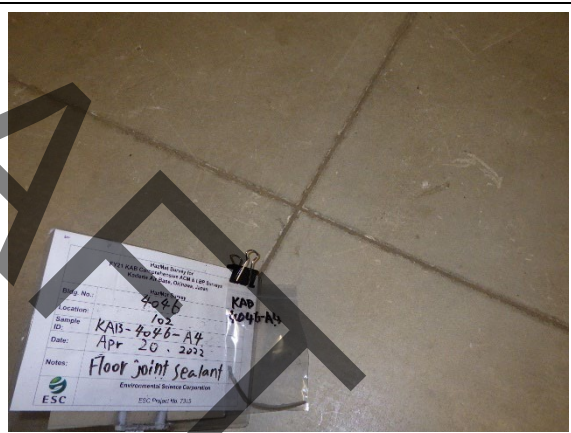
KAB-4046-A1



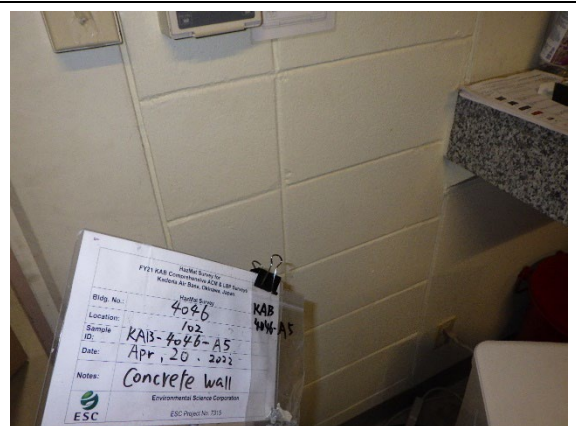
KAB-4046-A2



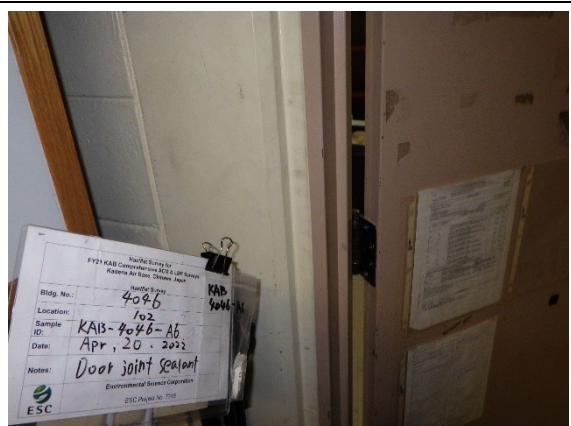
KAB-4046-A3



KAB-4046-A4



KAB-4046-A5



KAB-4046-A6